1. Characteristic of sound

1.1 Introduction

longitudinal wave motion in an elastic medium (air, water, solid)

Molecules in the medium (fluid particle) vibrate with low speed : $\sim 0.1 m/s$

propagate in a high speed ($c \approx 340$ m/s in air, 1500 m/s in water)

disturbs the fluid from its mean state P_0, ρ_0

perturbation (fluctation) $p'(\vec{x}, t)$, $\rho'(\vec{x}, t)$

total unsteady pressure and density : $P_0 + p'$, $\rho_0 + \rho'$

where $\left|\frac{p'}{P_0}\right| \ll 1$, $\left|\frac{\rho'}{\rho_0}\right| \ll 1$

$\Rightarrow$ Sound is linear motion $\leftarrow$ linear equation

- products of perturbation are negligible (high order term)
- response of acoustic field is linear

Threshold of ear: $10^{-10} \leq \frac{p'}{P_0} \leq 10^{-3}$

Means of measuring sound level

① Power level (PWL)

$$PWL = 10 \log_{10}\left(\frac{\text{Sound power output}}{10^{-12}\ \text{watts}}\right)$$

$$= 10 \log_{10}(\text{Sound power output}) + 120\ dB$$

- Sound power levels are usually small.

  eg: quiet human whisper of $10^{-10}\ W \Rightarrow PWL = 20\ dB$

  human shout of $10^{-5}\ W \Rightarrow PWL = 70\ dB$

- Enormous numerical range of PWL $\Rightarrow$ logarithmic scale + decibels

  eg: jet trasport at take-off of $10^{5}\ W \Rightarrow PWL = 170 dB$

② Sound pressure level (SPL)

$$SPL = 20 \log_{10}\left(\frac{\sqrt{\overline{p'^2}}}{2\times10^{-5}\ N/m^2}\right)$$

$p'^2_{rms} = \overline{p'^2}$

root mean square pressure fluctuation

$-$: mean value over a period

eg: $1\ atm = 10^5\ N/m^2 \Rightarrow SPL = 194\ dB$

threshold of hearing $= 2\times10^{-5}\ N/m^2 \Rightarrow SPL = 0\ dB$

threshold of pain $= 10^{-3}\ atm = 10^2\ N/m^2 \Rightarrow SPL \approx 134\ dB$

Example: two different sound:

A	B
ω_1	ω_2
phase shift σ	
SPL = 85dB	80dB

Solution: $p'(t) = A\cos\omega_1 t + B\cos(\omega_2 t + \sigma)$

↑ linear sound wave

$$p'^2 = A^2\cos^2(\omega_1 t) + 2AB\cos(\omega_1 t)\cos(\omega_2 t+\sigma) + B^2\cos^2(\omega_2 t+\sigma)$$
$$= A^2\cos^2(\omega_1 t) + B^2\cos^2(\omega_2 t+\sigma) + AB\left[\cos(\omega_1 t+\omega_2 t+\sigma) + \cos(\omega_1 t-\omega_2 t-\sigma)\right]$$

Averaging over t

$$\overline{[p']^2} = \tfrac{1}{2}A^2 + \tfrac{1}{2}B^2 + AB\overline{\left[\cos(\omega_1 t+\omega_2 t+\sigma) + \cos(\omega_1 t-\omega_2 t-\sigma)\right]}$$

if $\omega_1 \neq \omega_2$ ⇒ the third term is zero

⇒ $\overline{[p']^2} = \tfrac{1}{2}(A^2+B^2)$

if $\omega_1 = \omega_2$ ⇒ the third term is indepent of t

⇒ $\overline{p'^2} = \tfrac{1}{2}(A^2+B^2) + AB\cos\sigma$

for sound waves in phase : $\sigma = 0$ ⇒ $\overline{p'^2} = \tfrac{1}{2}(A+B)^2$

for identical sound waves : $A=B$, $\sigma=0$ ⇒ $\overline{p'^2} = 2A^2$

for a single wave : $p' = A\cos\omega_1 t$ ⇒ $\overline{p'^2} = \tfrac{1}{2}A^2$

$$85 = 20\log_{10}\left(\frac{A/\sqrt{2}}{2\times10^{-5}}\right) \Rightarrow A = 0.503\ N/m^2$$

$$80 = 20\log_{10}\left(\frac{B/\sqrt{2}}{2\times10^{-5}}\right) \Rightarrow B = 0.283\ N/m^2$$

if $\omega_1 \neq \omega_2$ ⇒ $20\log_{10}\left(\frac{\sqrt{A^2+B^2}/\sqrt{2}}{2\times10^{-5}}\right) = 86.2\,dB$

if $\omega_1 = \omega_2$, $\sigma = 0$ ⇒ $20\log_{10}\left(\frac{(A+B)/\sqrt{2}}{2\times10^{-5}}\right) = 88.9\,dB$

1.2 1-D wave equation

Sound wave virtually far from the source or in narrow pipe

Suppose the fluid is at rest $\Rightarrow \bar{u} = 0$

Mass equation

$$\frac{\partial(\rho_0+\rho')}{\partial t} + \frac{\partial(\rho_0+\rho')u}{\partial x} = 0$$

$$\boxed{\frac{\partial \rho'}{\partial t} + \rho_0 \frac{\partial u}{\partial x} = 0}$$

local mass conservation:

$$\frac{\partial}{\partial t}\rho + \operatorname{div}(\rho v) = 0$$

let $\rho = \rho_0 + \rho'(t)$

ρ_0: constant

$$\Downarrow$$

$$\frac{\partial \rho'}{\partial t} + \rho_0 \operatorname{div} v = 0$$

1-D

momentum equation

$$Re = \frac{2\pi c \lambda}{\nu} = \frac{\omega \lambda^2}{\nu} \sim 10^8 \Rightarrow \text{inviscid}$$

elastic fluid model, neglect body force

$$\operatorname{div} T + b = \rho \dot{v} \Rightarrow \rho \dot{v} = -\operatorname{grad} p$$

$$(\rho_0+\rho')\frac{\partial u}{\partial t} + (\rho_0+\rho')u\frac{\partial u}{\partial x} = -\frac{\partial p'}{\partial x}$$

$$\boxed{\rho_0 \frac{\partial u}{\partial t} + \frac{\partial p'}{\partial x} = 0}$$

irotational

$$\rho\left(v' + \operatorname{grad}\frac{v^2}{2} + \operatorname{curl} v \times v\right) = -\operatorname{grad} p$$

linearised

$$\rho_0 \frac{\partial v}{\partial t} = -\operatorname{grad} p'$$

1-D

$$\Rightarrow \quad \frac{\partial^2 \rho'}{\partial t^2} - \frac{\partial^2 p'}{\partial x^2} = 0$$

isentropic $p = p(\rho) \Rightarrow$

$$p = p_0 + p' = p_0 + \rho' \left.\frac{dp}{d\rho}\right|_{\rho=\rho_0}$$

$$\Rightarrow \quad p' = \rho' \left.\frac{dp}{d\rho}\right|_{\rho=\rho_0} = \rho' c^2$$

$$\Rightarrow \quad \boxed{\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x^2} = 0}$$

General solution : $p'(x,t) = f(x-ct) + g(x+ct)$

Harmonic waves

$$p'(x,t) = A e^{i\omega(t-\frac{x}{c})} + B e^{i\omega(t+\frac{x}{c})}$$

wavelength $\lambda = \frac{2\pi c}{\omega}$

wave number $k = \frac{\omega}{c}$

Evaluation of c

for an isentropic perfect gas

$$\frac{p}{\rho^\gamma} = \frac{p_0}{\rho_0^\gamma} = \text{const}$$

$$\Rightarrow \quad \frac{dp}{p} = \gamma \frac{d\rho}{\rho}$$

$$\Rightarrow \quad \boxed{c^2 = \left.\frac{dp}{d\rho}\right|_s = \gamma \frac{p}{\rho} = \gamma R T}$$

$R = 287 \ \ J/(kg \cdot K)$

The acoustic particle velocity

Consider a wave travelling to the right

$$p' = f(x-ct)$$

$$\Rightarrow \quad \rho' = \frac{p'}{c^2} = \frac{f(x-ct)}{c^2}$$

continue equation $\frac{\partial u}{\partial x} = -\frac{1}{\rho_0}\frac{\partial \rho'}{\partial t} = -\frac{1}{\rho_0}\frac{1}{c^2}\frac{df(X)}{dX}\frac{\partial X}{\partial t} = \frac{1}{\rho_0 c}\frac{\partial f(x-ct)}{\partial x}$

$$\Rightarrow \quad u(x,t) = \frac{1}{\rho_0 c} f(x-ct)$$

$$\Rightarrow \quad p' = \rho_0 c u$$

The acoustic impedance $Z = \frac{p'}{u} = \rho_0 c$

- $\rho_0 c$: real and dependent only on medium properties
- Characteristic impedance of the medium

In the same way, negative direction : $p' = -\rho_0 c u$

Acoustic velocity potential

Suppose that acoustic motion is irrotational

$\Rightarrow$ $u = \frac{\partial \phi}{\partial x}$

momentum equation $\rho_0 \frac{\partial u}{\partial t} + \frac{\partial p'}{\partial x} = 0$

$\Rightarrow$ $p' = -\rho_0 \frac{\partial \phi}{\partial t}$

$\vdots$

ϕ also satisfies the 1-D wave equation

Suppose that $\phi = e^{i\omega t - ikx}$

$\Rightarrow$ $p' = -i\omega \rho_0 \phi$ $\quad u = ik\phi$

1.3 Energy of 1-D acoustic waves

Multiply the mass conservation by $\rho' c^2$ ← $\frac{\partial \rho'}{\partial t} + \rho_0 \frac{\partial u}{\partial x} = 0$

$\Rightarrow$ $c^2 \rho' \frac{\partial \rho'}{\partial t} + \underbrace{\rho_0 c^2 \rho' \frac{\partial u}{\partial x}}_{\text{small}} = 0$

$\Rightarrow$ low energy level in sound wave

$\Rightarrow$ $\frac{c^2}{\rho_0} \frac{\partial (\rho'^2/2)}{\partial t} + p' \frac{\partial u}{\partial x} = 0$

momentum eq.

Multiplying linearised momentum equation by u

$$\rho_0 \frac{\partial u}{\partial t} + \frac{\partial p'}{\partial x} = 0$$

$$\Rightarrow \quad \frac{1}{2}\rho_0 \frac{\partial u^2}{\partial t} + u\frac{\partial p'}{\partial x} = 0$$

Adding up the two equations $\Rightarrow$

$$\frac{\partial}{\partial t}\left(\underbrace{\frac{1}{2}\rho_0 u^2}_{e_k} + \underbrace{\frac{c^2\rho'^2}{2\rho_0}}_{e_p}\right) + \frac{\partial}{\partial x}\underbrace{(p'u)}_{I} = 0$$

e_k —— kinetic energy density
e_p —— potential energy density
E —— total energy density
I —— Energy flux through unit surface area (acoustic intensity)

$$\Rightarrow \quad \frac{\partial E}{\partial t} = -\frac{\partial I}{\partial x}$$

$$\Rightarrow \quad \frac{\partial}{\partial t}\int_{x_0}^{x_0+l} AE\,dx = A\left[I(x_0,t) - I(x_0+l,t)\right]$$

The rate of change of acoustic energy inside a control volume V is equal to the flux of energy across the surface of V.

Sound intensity level : a meassure of the mean intensity $\bar{I}$

$$IL = 10\log_{10}\left(\frac{\bar{I}}{10^{-12}\ w/m^2}\right) \quad \text{in dB}$$

$$= 10\log_{10}\bar{I} + 120\ dB$$

For 1-D waves travelling to the right

$$\Rightarrow \quad e_k = \frac{1}{2}\rho_0 u^2 = \frac{c^2\rho'^2}{2\rho_0} = e_p \qquad \Rightarrow u^2 = \frac{c^2\rho'^2}{\rho_0^2} \Rightarrow u = \frac{\rho' c}{\rho_0}$$

also in 1-D sound wave

$$I = p'u = \rho' c^2 \cdot \frac{\rho' c}{\rho_0} = \frac{c^3\rho'^2}{\rho_0} = 2c\cdot\frac{c^2\rho'^2}{2\rho_0} = 2c\,e_p = c(e_k+e_p) = cE$$

$$\downarrow \quad \frac{\partial E}{\partial t} = -\frac{\partial I}{\partial x}$$

$$\frac{\partial E}{\partial t} + c\frac{\partial E}{\partial x} = 0 \qquad (\text{right})$$

$$\Rightarrow \quad \frac{\partial E}{\partial t} - c\frac{\partial E}{\partial x} = 0 \qquad \text{(left)}$$

Example 1.2 Two sound waves of the same frequency ω travelling in opposite directions. Determine the mean acoustic energy density and energy flux.

Solution:

$$\phi = A e^{i\omega(t - x/c)} + B e^{i\omega(t + x/c)}$$

$$\Rightarrow \quad p' = -i\omega\rho_0\phi = \rho_0\omega\left\{A\sin[\omega(t-x/c)] + B\sin[\omega(t+x/c)]\right.$$

$$\Rightarrow \quad p'^2 =$$

$$\Rightarrow \quad \overline{p'^2} = (\rho_0\omega)^2\left(\tfrac{1}{2}A^2 + \tfrac{1}{2}B^2 + AB\cos\frac{2\omega x}{c}\right)$$

$$\Rightarrow \quad \overline{\rho'^2} = \frac{(\rho_0\omega)^2}{c^4}\left(\tfrac{1}{2}A^2 + \tfrac{1}{2}B^2 + AB\cos\frac{2\omega x}{c}\right)$$

$$\Rightarrow \quad \overline{u^2} = \left(\frac{\omega}{c}\right)^2\left(\tfrac{1}{2}A^2 + \tfrac{1}{2}B^2 - AB\cos\frac{2\omega x}{c}\right)$$

$$\Rightarrow \quad \bar{E} = \tfrac{1}{2}\rho_0\overline{u^2} + \frac{c^2}{2\rho_0}\overline{\rho'^2} = \frac{\rho_0\omega^2}{2c^2}(A^2+B^2) \qquad \bar{E} \text{ dependent of position}$$

$$\bar{I} = \overline{p'u} = \frac{\rho_0\omega^2}{2c}(A^2 - B^2)$$

If $A = B \Rightarrow \bar{I} = 0$

Example 1.3 Suppose that the sound radiated from a speaker. At a distance of 20m the SPL is 110dB

Solution:

$$SPL = 20\log_{10}\left(\frac{p'_{rms}}{2\times10^{-5}\,N/m^2}\right) = 110 \quad \Rightarrow p'_{rms} = 6.32 \ N/m^2$$

$$\Rightarrow \quad u_{rms} = \frac{p'_{rms}}{\rho_0 c} = 1.55\times10^{-2}\ m/s$$

$$\Rightarrow \quad \bar{I} = \overline{p'u} = \frac{\overline{p'^2}}{\rho_0 c} = p'_{rms}u_{rms} = 9.8\times10^{-2}\ W/m^2$$

$$\Rightarrow \quad 4\pi\cdot 20^2\cdot\bar{I} = 492.6\ W$$

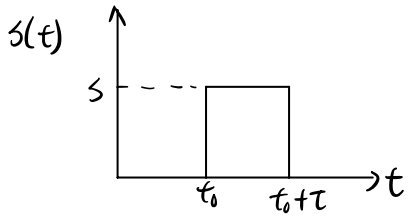
1.4 Sound spectra

$$S(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\omega)\, e^{i\omega t}\, d\omega$$

$$F(\omega) = \int_{-\infty}^{+\infty} S(t)\, e^{-i\omega t}\, dt$$

The sound is made up of harmonic waves with amplitudes $F(\omega)\,\delta\omega/2\pi$ in the frequency band $[\omega, \omega+\delta\omega]$

Example 1.4 The harmonic decomposition of a single pulse.



$$s(t) = \begin{cases} S, & t_0 \le t \le t_0+\tau \\ 0, & \text{other} \end{cases}$$

$$\Rightarrow \quad F(\omega) = S\int_{t_0}^{t_0+\tau} e^{-i\omega t}dt = S\tau e^{-i\omega(t_0+\tau/2)}\frac{\sin\omega\tau/2}{\omega\tau/2}$$

If $S = \frac{I}{\tau}$, $\tau \to 0$ $\Rightarrow$ $\int s(t)\,dt = \frac{\tau I}{\tau} = I$

$\delta(t)$ function

$$\delta(t) \xrightarrow{\mathcal{F}} 1$$

property of δ function

$$\int_{-\infty}^{+\infty} g(t)\,\delta(t-t_0)\,dt = g(t_0)$$

$\Rightarrow$ for the impulse signal $s(t) = I\delta(t-t_0)$

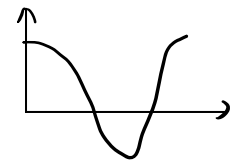
$\Rightarrow$ $F(\omega) = I\int_{-\infty}^{+\infty} e^{-i\omega t}\delta(t-t_0)\,dt = I e^{-i\omega t_0}$ $\Rightarrow$ $\int_{-\infty}^{+\infty} e^{-i\omega t}\delta(t-t_0)\,dt = e^{-i\omega t_0}$

$\Rightarrow$ $\delta(t-t_0) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{i\omega(t-t_0)}\,d\omega$

$\Rightarrow$ $\int_{-\infty}^{+\infty} e^{ixy}\,dx = 2\pi\,\delta(y)$

Suppose a harmonic signal $s(t) = Ie^{i\omega_0 t}$

$\Rightarrow$ $F(\omega) = I\int_{-\infty}^{+\infty} e^{i(\omega_0-\omega)t}dt = I\,2\pi\,\delta(\omega_0-\omega)$



- Spectrum of a sound may be discrete
- Pure tones : spikes in spectrum

- Most real noise has continous spectrum

Summary of Chapter 1

Sound pressure level $SPL = 20 \log_{10}\left(\frac{\sqrt{\overline{p'^2}}}{2\times10^{-5}\ N/m^2}\right)$ in dB

1-D wave equation and general solution

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x^2} = 0$$

$$\Rightarrow \quad p'(x,t) = f(x-ct) + g(x+ct)$$

The speed of sound

$$c^2 = \left.\frac{dp}{d\rho}\right|_s = \gamma R T$$

The acoustic impedance $Z = \frac{p'}{u} = \rho_0 c$

The acoustic potential velocity $p' = -\rho_0 \frac{\partial \phi}{\partial t}$ $u = \frac{\partial \phi}{\partial x}$

The acoustic energy density $E = \underbrace{\frac{1}{2}\rho_0 u^2}_{e_k} + \underbrace{\frac{c^2 \rho'^2}{2\rho_0}}_{e_p}$

The acoustic energy flux (acoustic intensity)

$$I = p'u = \underbrace{c\,(e_k + e_p)}_{\text{in 1-D waves}}$$

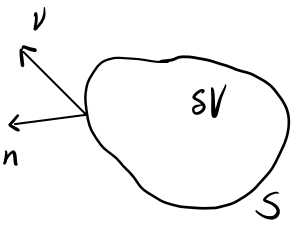
Fourier transform

$$s(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\omega)\, e^{i\omega t}\, d\omega$$

$$F(\omega) = \int_{-\infty}^{+\infty} s(t)\, e^{-i\omega t}\, dt$$

δ function

2. 3-D sound wave



conservation of mass

$$SV\frac{\partial \rho'}{\partial t} = -\int_S \rho_0 v\cdot n\, dS$$

↓ Gauss's theorem

$$SV\frac{\partial \rho'}{\partial t} = -\iiint_{SV} \rho_0 \, div\, v\, dV$$

↓

$$\frac{\partial \rho'}{\partial t} + \rho_0 \, div\, v = 0$$

or $\boxed{\frac{\partial \rho'}{\partial t} + \rho_0 \nabla\cdot v = 0}$ ← $\frac{\partial}{\partial t}$

or $\frac{\partial \rho'}{\partial t} + \rho_0 \frac{\partial v_i}{\partial x_i} = 0$

momentum equation

$\boxed{\rho_0 \frac{\partial v}{\partial t} + \nabla p' = 0}$ ← $\nabla\cdot$

or $\rho_0 \frac{\partial v_i}{\partial t} + \frac{\partial p'}{\partial x_i} = 0$

$\nabla^2 p' = \nabla\cdot(\nabla p')$

$$\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0$$

↓

$$\boxed{\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0}$$

$\nabla\times$ → $\rho_0 \frac{\partial}{\partial t}(\nabla\times v) = 0$ because $\nabla\times\nabla p' = 0$

p — scalar v — vector

$\nabla\times(\nabla p) = 0$

$\nabla\cdot(\nabla\times v) = 0$

标量场梯度的旋度为零

向量场旋度的散度为零

证明详见连续介质力学第三次作业

For an initial state of rest

$\nabla \times v = 0$ ⟵ irrotational flow

↓

exist a velocity potential $\phi(x, t)$,

$v = \nabla \phi$

↓ momentum equation $\rho_0 \frac{\partial v}{\partial t} + \nabla p' = 0$

$\nabla \left(\rho_0 \frac{\partial \phi}{\partial t} + p' \right) = 0$

↓

$p' = -\rho_0 \frac{\partial \phi}{\partial t}$

$\Rightarrow$ $\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = 0$ velocity potential also satisfy wave equation

In the limit of i) incompressible flow

ii) very slow motion

iii) flow near a singularity or small source

$\Rightarrow$ $\nabla^2 \phi = 0$

Near a singularity ⟶ $\phi \sim r^{-\alpha}$ $(\alpha > 0,\ r \to 0)$

$\Rightarrow$ $\nabla^2 \phi \sim r^{-\alpha-2}$, $\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} \sim \frac{\omega^2}{c^2} r^{-\alpha} = O\left(\frac{\omega^2}{c^2} r^2 \nabla^2 \phi\right)$

i) $c \to \infty$ $\Rightarrow$ incompressible ii) $\omega \to 0$ $\Rightarrow$ slow iii) $r \to 0$ $\Rightarrow$ near

- In steady aerodynamics $p + \frac{1}{2}\rho v^2 = \text{constant}$ (nonlinear term)
- In unsteady aerodynamics $p + \frac{1}{2}\rho v^2 + \rho_0 \frac{\partial \phi}{\partial t} = \text{constant}$

?

- In acoustic of a still medium $p' + \rho_0 \frac{\partial \phi}{\partial t} = \text{constant}$

  unsteady linear term

2.2 Energy of 3-D acoustic waves

$$\frac{1}{c^2}\frac{\partial p'}{\partial t} + \rho_0 \nabla \cdot v = 0 \quad \times \frac{p'}{\rho_0}$$

$$\rho_0 \frac{\partial v}{\partial t} + \nabla p' = 0 \quad \times v_0$$

$$\frac{p'}{\rho_0 c^2}\frac{\partial p'}{\partial t} + p' \nabla \cdot v + \rho_0 v \cdot \frac{\partial v}{\partial t} + v \cdot \nabla p' = 0$$

$$\downarrow$$

$$\frac{\partial}{\partial t}\left(\underbrace{\frac{1}{2}\rho_0 v^2}_{e_k} + \underbrace{\frac{1}{2\rho_0 c^2} p'^2}_{e_p}\right) + \nabla \cdot \underbrace{(p' v)}_{I} = 0$$

$$\downarrow$$

声能量密度的变化 = 声强通量

$$\frac{\partial}{\partial t}(e_p + e_k) = -\nabla \cdot I = -\frac{\partial I_x}{\partial x_i}$$

$$\downarrow \int_V$$

integral form

$$\frac{\partial}{\partial t}\int_V (e_p + e_k)\, dV = -\int_S I_n \, dS$$

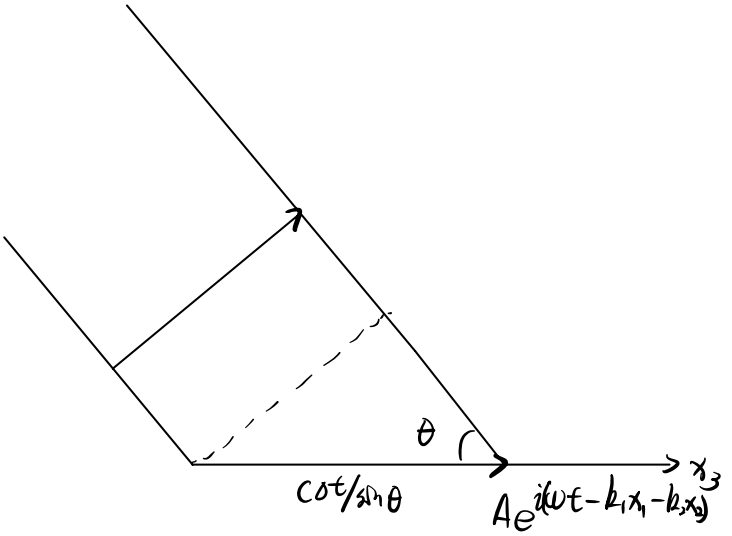
- Energy decreases because of its transport out across the

surface S

- Energy increase ⟵ negative divergence of $\mathbf{I}$

convergence

2.3 3-D plane wave field



$|k|^2 = \frac{\omega^2}{c^2}$ phase speed $c = \frac{\omega}{|k|}$

In the surface of $x_3 = 0$

$$p'(x,t) = Ae^{i(\omega t - k_1 x_1 - k_2 x_2)}$$

$$\frac{\omega}{(k_1^2+k_2^2)^{\frac{1}{2}}} = \frac{\omega}{|k|\sin\theta} = \frac{c}{\sin\theta} > c$$

(supersonic surface phase speed)

if $(k_1^2+k_2^2)^{\frac{1}{2}} > \frac{\omega}{c}$ ⇒ subsonic surface

Evanescent waves

Consider a disturbance travelling subsonically in the plane $x_3 = 0$

$$p'(x_1, x_2, 0, t) = A e^{i(\omega t - k_1 x_1 - k_2 x_2)} \quad \text{with} \quad (k_1^2 + k_2^2)^{\frac{1}{2}} > \frac{\omega}{c}$$

Such a disturbance is possible by a sound wave incident through a different medium with a lower c, or a vibrating mechanical surface. (wave speed $<< c$)

In $x_3 > 0$, try the solution form

$$p'(x, t) = f(x_3)\, e^{i(\omega t - k_1 x_1 - k_2 x_2)}$$

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0$$

Wave equation imphes an ODE

$$\frac{d^2 f}{dx_3^2} - (k_1^2 + k_2^2 - \frac{\omega^2}{c^2}) f = 0$$

$$\Rightarrow \quad f(x_3) = A e^{-\gamma x_3} + B e^{\gamma x_3} \qquad \text{where} \quad \gamma = (k_1^2 + k_2^2 - \frac{\omega^2}{c^2})^{\frac{1}{2}}$$

The solution must be finite for large $x_3 > 0$ $\longrightarrow$ $B = 0$

Hence $$p'(x, t) = A e^{i(\omega t - k_1 x_1 - k_2 x_2) - \gamma x_3}$$

- Evanescent wave: a wave that propagates in the surface $x_3 = 0$ and decays exponentially away from $x_3 = 0$
- Effect confined to the vicinity of the surface

影响仅局限于表面附近

2.4 Spherical wave field

$$p' = p'(r, t) \qquad v = v(r, t)$$

r: radial distance

Only radial velocity $u(r, t)$

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0$$

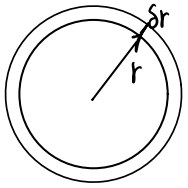
$$\nabla^2 p' = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial p'}{\partial r}\right) + \frac{1}{r^2}\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial p'}{\partial\theta}\right) + \frac{1}{r^2\sin\theta}\frac{\partial^2 p'}{\partial\phi^2}$$

neglect

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial p'}{\partial r}\right) = 0$$

Alternative derivation

Consider a spherical shell of radius r and thickness δr



Linearised mass equation

$$4\pi r^2 \delta r\frac{\partial \rho'}{\partial t} = -\rho_0\delta(4\pi r^2 u)$$

$$\Rightarrow \quad \frac{r^2}{c^2}\frac{\partial p'}{\partial t} = -\rho_0\frac{\partial}{\partial r}(r^2 u) \qquad \frac{\partial}{\partial t}$$

Linearised momentum conservation

$$\rho_0\frac{\partial u}{\partial t} = -\frac{\partial p'}{\partial r} \qquad r^2\frac{\partial}{\partial r}$$

$$\frac{r^2}{c^2}\frac{\partial^2 p'}{\partial t^2} = \frac{\partial}{\partial r}\left(r^2\frac{\partial p'}{\partial r}\right)$$

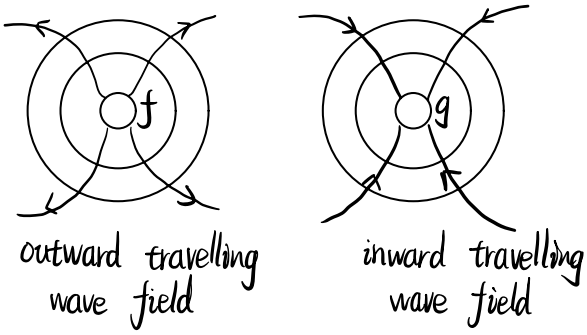
$\because$ $\frac{1}{r}\frac{\partial}{\partial r}\left(r^2\frac{\partial p'}{\partial r}\right) = 2\frac{\partial p'}{\partial r} + r\frac{\partial^2 p'}{\partial r^2}$

$\frac{\partial^2}{\partial r^2}(rp') = \frac{\partial}{\partial r}\left(p' + r\frac{\partial p'}{\partial r}\right) = 2\frac{\partial p'}{\partial r} + r\frac{\partial^2 p'}{\partial r^2}$

$\therefore$ $\frac{1}{c^2}\frac{\partial^2}{\partial t^2}(rp') - \frac{\partial^2}{\partial r^2}(rp') = 0$

$\Rightarrow$ $rp' = f(r-ct) + g(r+ct)$

$\Rightarrow$ $p' = \frac{f(r-ct)}{r} + \frac{g(r+ct)}{r}$



outward travelling wave field → causal solution of a source near origin

inward travelling wave field → excluded by the causality constraint

Sommerfeld's radiation condition:

$$\lim_{r\to\infty} r\left(\frac{\partial p'}{\partial t} + c\frac{\partial p'}{\partial r}\right) = 0 \quad ?$$

In open space, only outward travelling waves could have any real existence

Pressure and velocity in a sperical wave

$$p'(r,t) = \frac{A}{r} e^{i\omega\left(t-\frac{r}{c}\right)}$$

↓ momentum equation $\rho_0 \frac{\partial u}{\partial t} = -\frac{\partial p'}{\partial r}$

$$i\omega\rho_0 u = \left(\frac{A}{r}\frac{i\omega}{c} + \frac{A}{r^2}\right) e^{i\omega\left(t-\frac{r}{c}\right)}$$

far field / near field

$$\Rightarrow \quad u(r,t) = \frac{A}{r\rho_0 c}\left(1 + \frac{c}{i\omega r}\right) e^{i\omega\left(t-\frac{r}{c}\right)}$$

$$= \frac{1}{\rho_0 c}\sqrt{1+\left(\frac{c}{\omega r}\right)^2}\, e^{-i\Phi}\cdot\frac{A}{r} e^{i\omega\left(t-\frac{r}{c}\right)}$$

the phase angle $\Phi = \tan^{-1}\frac{c}{\omega r}$

$$\Rightarrow \quad u(r,t) = \frac{\sqrt{1-(c/\omega r)^2}}{\rho_0 c}\, p'\left(r, t-\frac{\phi}{\omega}\right)$$

- In general, p', u not in phase, the impedance $p'/u < \rho_0 c$
- In near field $\left(r \ll \frac{c}{\omega}\right)$, $u \approx \frac{p'}{i\omega r \rho_0}$

  $\Rightarrow$ p' leads u by 90°, $p' \approx \rho_0 r \frac{\partial u}{\partial t}$
- As $r \uparrow$, phase lead $\downarrow$ $\quad p'/u \to \rho_0 c$
- In far field $\left(r \gg \frac{c}{\omega}\right)$ $\quad u = \frac{p'}{\rho_0 c}$

  p', u nearly in phase, plane wave impedance $\rho_0 c$ recovered

spherical wave $\longrightarrow$ locally plane wave

(At large r, the wave fronts are virtually plane)

Example 2.1 The amplitude of p' at a distance 1m from a source in air is 1 Nm^{-2} and the frequency of the wave is 100 Hz. Calculate the amplitude of particle velocity at 0.1m and 1m from the source if the waves are (a) plane (b) spherically symmetric

Solution:

(a) $p'(x,t) = Ae^{i\omega(t-\frac{|x|}{c})}$ and $u = \frac{p'}{\rho_0 c}$

$$\Rightarrow \quad \hat{u} = \frac{A}{\rho_0 c} = 2.5\times10^{-3}\ m/s$$

(b) $p'(r,t) = \frac{A}{r} e^{i\omega(t-\frac{r}{c})}$ $\qquad u(r,t) = \frac{\sqrt{1+(c/\omega r)^2}}{\rho_0 c}\, p'(r, t-\phi/\omega)$

$$\Rightarrow \quad \hat{u} = \frac{A}{r}\frac{\sqrt{1+(c/\omega r)^2}}{\rho_0 c} = \begin{cases} 0.13 \ m/s & \text{at } r = 0.1m \\ 2.8\times10^{-3}\ m/s & \text{at } r = 1m \end{cases}$$

Example 2.2 A spherical wave has a frequency of 200 Hz and an SPL of 100 dB 1m away from the source. Calculate the acoustic energy flow rate through a spherical surface 1m and 0.1m away from the source, and confirm they are equal.

Hint: The energy flow rate $\int_S \bar{I}_n \, dS$ where $\bar{I}_n = \overline{p'u}$

Solution:

$$SPL = 100\ dB = 20 \log_{10} \frac{p'_{rms}}{2\times10^{-5}\ N/m^2} \quad \text{at} \quad r = 1m$$

$$\Rightarrow \quad p'_{rms} = 2\ N/m^2 \quad \text{at} \quad r = 1m$$

Since $p' \propto \frac{1}{r}$, $p'_{rms} = 20\ N/m^2$ at $r = 0.1\ m$

$$\bar{I}_n = \overline{p'u} = \overline{p(r,t)\, p'(r, t - \bar{\phi}/\omega)} \, \frac{\sqrt{1+(c/\omega r)^2}}{\rho_0 c}$$

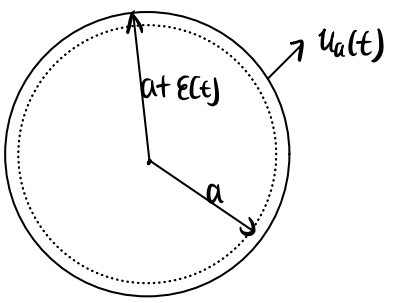
$$= p'^2_{rms} \cos\bar{\phi} \, \frac{\sqrt{1+(c/\omega r)^2}}{\rho_0 c} = \frac{p'^2_{rms}}{\rho_0 c}$$

$\therefore$ Energy flow rate $= 4\pi r^2 \frac{p'^2_{rms}}{\rho_0 c} = 0.12\ W$ for both surfaces.

2.5 Simple sound sources.

脉动球源

2.5.1 Sound of a pulsating sphere



A highly idealised source: a spherical shell of varible radius.

B.C. $u(t) = u_a(t)$ on $r = a + \varepsilon(t)$

The outward travelling centred wave by surface vibration

$$p'(r,t) = \frac{f(t - r/c)}{r}$$

write $p'(r,t) = \hat{p}(r)\, e^{i\omega t}$

complex amplitude

$$\Rightarrow \quad \hat{p}(r) = \frac{A}{r} e^{-ikr}$$

$$\rho_0 \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x}$$

And $$i\omega\rho_0 \hat{u}(r) = A\left(\frac{1}{r^2} + \frac{ik}{r}\right) e^{-ikr}$$

linearised B.C. $$\hat{u}(a+\varepsilon) = \hat{u}(a) + \underbrace{\varepsilon \frac{d\hat{u}}{dr}}_{\text{2nd order}}$$

$\Rightarrow$ B.C.: $u(t) = u_a(t)$ on $r = a$

$$\hat{u}_a = \frac{A}{i\omega\rho_0 a^2}(1+ika)\, e^{-ika}$$

$$\Rightarrow \quad A = \frac{i\omega\rho_0 a^2 \hat{u}_a e^{ika}}{1+ika}$$

$$\Rightarrow \quad \hat{p}(r) = \frac{a}{r}\frac{ika}{1+ika}\rho_0 c \hat{u}_a e^{-ik(r-a)}$$

Surface (radiation) impedance $$Z = \left.\frac{\hat{p}}{\hat{u}}\right|_{r=a} = \frac{ika}{1+ika}\rho_0 c$$

The Helmholtz number $$ka = \frac{2\pi a}{\lambda}$$ ⟶ compactness ratio

i) $ka \gg 1$ (non-compact) 非紧致声源

($kr \gg 1$)

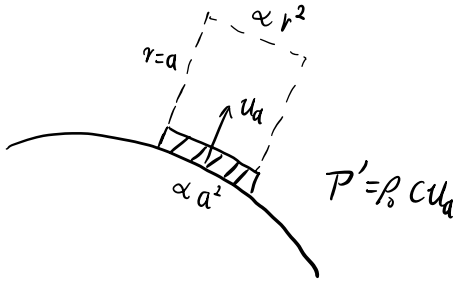
$$\hat{p}(r) = \frac{a}{r}\rho_0 c \hat{u}_a e^{-ik(r-a)}$$

i.e. $$p'(r,t) = \frac{a}{r}\rho_0 c\, u_a\left(t - \frac{r-a}{c}\right)$$

$Z = \rho_0 c$ on surface $r=a$ ⟶ $p' = \rho_0 c u$ everywhere as in a plane sound wave

Alternatively, ray theory for high-freq sound waves.

- locally plane wave $p' = \rho_0 c u$



- amplitude variation $\frac{\overline{p'^2} A}{\rho_0 c} = \text{constant}$

On the surface $p'(a) = \rho_0 c u_a$ and $A \propto a^2$

$$\Rightarrow \frac{\overline{p'^2} A}{\rho_0 c} \propto \rho_0 c u_a^2 a^2$$

$$\Rightarrow p'(r) \propto \frac{a}{r} \rho_0 c u_a$$

ii) $ka \ll 1$ (compact) (緊致声源)

The expansion $e^{ika} = 1 + ika + \frac{1}{2}(ika)^2$

$$\hat{p}(r) = \frac{a}{r} ika \rho_0 c \hat{u}_a e^{-ikr} [1 + o(ka)] = \frac{i\omega a^2}{r} \rho_0 \hat{u}_a e^{-ikr}$$

i.e. $p'(r,t) = \frac{a^2}{r} \rho_0 \frac{\partial u_a(t - r/c)}{\partial t}$

? • p' independent of c (and medium compressibility)

• p' and u have phase difference of $\pi/2$ on $r=a$

$$\Rightarrow \overline{p'u} = 0 \quad \text{on} \quad r=a$$

$V(t)$ = volume of pulsating sphere, $\dot{V}(t) = dV/dt = 4\pi a^2 u_a(t)$

$$\Rightarrow p'(r,t) = \frac{\rho_0 \dot{V}(t - r/c)}{4\pi r}$$ → monopole source

- will see later this remains ture for compact body of arbitrary shape
- In the compact limit, $p'(r,t)$ is not depend on the body's shape but only on the $\frac{d^2}{dt^2}$ of its volume

Comparing the two cases of $ka \gg 1$ and $ka \ll 1$

$$\left| \frac{\hat{p}(r)|_{ka \ll 1}}{\hat{p}(r)|_{ka \gg 1}} \right| = ka \ll 1$$

- For the same $\hat{u}_a$, a small fraction ka of p' in the non-compact case is generated by a compact source
- Small acoustic radiators are very inefficient (低效) and their response increase with freq.
- Large sources are efficient with virtually 1-D acoustic waves and no freq. dependence. (恒值没有频率依赖性)

Example 2.3 A spherical loudspeaker of radius 30 cm vibrates with simple harmonic motion of peak-to-peak amplitude of 2mm. Determine on expression of the acoustic power radiated by the loudspeaker as a function of frequency. How much power is radiated at frequencies of 10, 100 and 1000 Hz?

Solution:

The radiated sound field

$$\hat{p}(r) = \frac{a}{r} \frac{ika}{1+ika} \rho_0 c \hat{u}_a e^{-ik(r-a)}$$

with $\hat{u}_a = i\omega\varepsilon$ where $\varepsilon = 1m$

Hence the radiated power

$$P = \frac{1}{2} \frac{|\hat{p}|^2}{\rho_0 c} 4\pi r^2$$

from $\overline{p'^2}$ surface of sphere

$$|\hat{p}| = \frac{a}{r}\frac{ka}{\sqrt{1+(wa/c)^2}}\rho_0 c w \varepsilon = \frac{a^2}{r}\frac{\rho_0 w^2 \varepsilon}{\sqrt{1+(wa/c)^2}}$$

$$\Rightarrow \quad P = \frac{2\pi \rho_0 \varepsilon^2 a^4 w^4}{c\left[1+(wa/c)\right]^2} = \begin{cases} 2.8\ mW & \text{at } 10\ Hz \\ 21.4\ W & \text{at } 100\ Hz \\ 8.8\ kW & \text{at } 1000\ Hz \end{cases}$$

Example 2.4 A boundary vibrates in air at a frequency of 100 Hz with peak-to-peak displacement of 2×10^{-3} m. Determine the SPL generated by that vibration at a distance 10 m from the boundary.

(a) the boundary is a plane surface with uniform displacement.

(b) the boundary is a sphere, radius 5 m

What are the SPLs if the vibration occurs at 10 kHz

Solution:

(a) $p' = \rho_0 cu \Rightarrow |\hat{p}| = \rho_0 c w \varepsilon$

with $\hat{u} = iw\varepsilon$ where $\varepsilon = 1\ mm$

$$\Rightarrow SPL = 20\log_{10}\frac{p'_{rms}}{2\times10^{-5}\ N/m^2} = 20\log_{10}\frac{|\hat{p}|}{\sqrt{2}\times2\times10^{-5}\ N/m^2}$$

$$= \begin{cases} 139\ dB & \text{at } 100\ Hz \\ 179\ dB & \text{at } 10\ kHz \end{cases}$$

(b) For the sphere

$$\hat{p}(r) = \frac{a}{r}\frac{ika}{1+ika}\rho_0 c \hat{u}_a e^{-ik(r-a)}$$

and $$|\hat{p}| = \frac{a^2}{r}\frac{\rho_0 w^2 \varepsilon}{\sqrt{1+(w/c)^2}}$$

$ka \ll 1$

$$\Rightarrow \quad SPL = 20\log_{10}\left(\frac{|\hat{p}|}{\sqrt{2}\times 2\times10^{-5}\,N/m^2}\right) = \begin{cases} 72.4\ dB & \text{at } 100\,Hz \\ 133\ dB & \text{at } 10kHz \end{cases}$$

$ka \gg 1$

$ka \approx 0.1 \rightarrow \Delta SPL \approx 20dB$

However, $\hat{u}_a \sim w \rightarrow \Delta SPL \approx 40dB$

振荡球源

2.5.2 Sound of an oscillating rigid sphere

Other solutions of the wave equation

If ϕ is a solution $\quad \frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2} - \nabla^2\phi = 0$

$$\Rightarrow \quad \frac{\partial}{\partial x_1}\left(\frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2} - \nabla^2\phi\right) = 0$$

swap order $\quad \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\left(\frac{\partial\phi}{\partial x_1}\right) - \left(\nabla^2\frac{\partial\phi}{\partial x_1}\right) = 0$

$$\Rightarrow \quad p'(x,t) = \frac{\partial}{\partial x_1}\left[\frac{f(r-ct)}{r}\right] \text{ is a solution}$$

The chain rule of differentiation

$$p'(x,t) = \frac{\partial r}{\partial x_1}\Big|_{x_2,x_3}\frac{\partial}{\partial r}\left[\frac{f(r-ct)}{r}\right]$$

since $\quad r = |x| = (x_1^2+x_2^2+x_3^2)^{\frac{1}{2}}$

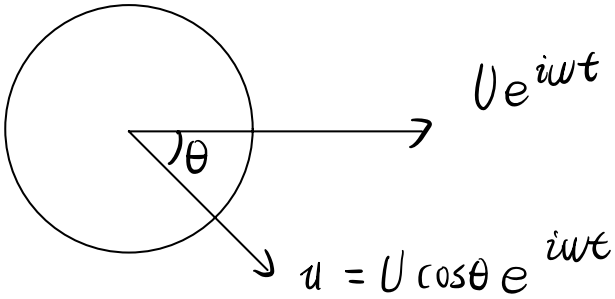
$$\Rightarrow \frac{\partial r}{\partial x_1} = 2x_1\frac{1}{2}(x_1^2+x_2^2+x_3^2)^{-\frac{1}{2}} = \frac{x_1}{r} = \cos\theta$$

with θ the angle between x_1 and r

$$\Rightarrow \quad p'(x,t) = \cos\theta\frac{\partial}{\partial r}\left[\frac{f(r-ct)}{r}\right]$$

harmonic form

$$\Rightarrow \quad p'(x,t) = A\cos\theta\frac{\partial}{\partial r}\left[\frac{e^{iw(t-\frac{r}{c})}}{r}\right]$$



Apply the radial momentum equation on the surface

轴向的动量矩方程

$$\rho_0 \frac{\partial u}{\partial t} = i\omega\rho_0 u = -\frac{\partial p'}{\partial r} = -A\cos\theta \frac{\partial^2}{\partial r^2}\left[\frac{e^{i\omega(t-\frac{r}{c})}}{r}\right]$$

$$\Rightarrow \quad u = -\frac{A\cos\theta}{i\omega\rho_0}\left(\frac{2}{r^3} + \frac{2ik}{r^2} - \frac{k^2}{r}\right)e^{i\omega(t-r/c)}$$

To find A, use $u = U\cos\theta\, e^{i\omega t}$ on $r = a$

$$\Rightarrow \quad A = -\frac{i\omega\rho_0 U a^3 e^{ika}}{2(1+ika) - (ka)^2}$$

$$\Rightarrow \quad p'(r,\theta,t) = -\frac{i\omega\rho_0 U a^3\cos\theta}{2(1+ika) - (ka)^2}\frac{\partial}{\partial r}\left\{\frac{e^{i[\omega t - k(r-a)]}}{r}\right\}$$

$$= \frac{i\omega\rho_0 U a^3\cos\theta}{2(1+ika) - (ka)^2}\left(\frac{1}{r^2} + \frac{ik}{r}\right)e^{i[\omega t - k(r-a)]}$$

i) $ka \gg 1$ (non-compact, $kr \gg 1$)

$$p'(r,\theta,t) = \frac{i\omega\rho_0 U a^3\cos\theta}{2(1+ika) - (ka)^2} k^2\left(\frac{1}{k^2r^2} + \frac{i}{kr}\right)e^{i[\omega t - k(r-a)]}$$

这一项相对于$(ka)^2$是小量

$$= \rho_0 c U\cos\theta \frac{a}{r} e^{i[\omega t - k(r-a)]}$$

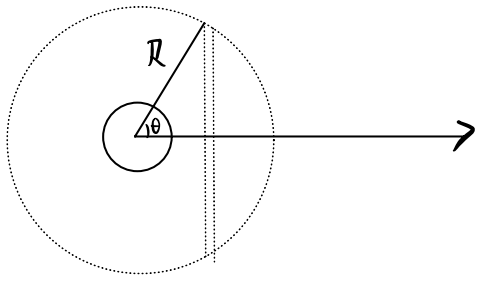
$$= \frac{a}{r} p'\left(a,\theta,t - \frac{r-a}{c}\right)$$

Radiated Sound power

$$P = \int \overline{p'u}\, dS = \int \frac{\overline{p'^2}}{\rho_0 c}\, dS = \rho_0 c U^2 \frac{a^2}{R^2} \cdot \frac{1}{2} \int \cos^2\theta\, dS$$

↑ large sphere

$dA = 2\pi (R\sin\theta)(R\, d\theta)$



$$= \rho_0 c U^2 \frac{a^2}{R^2} \frac{1}{2} \int_0^\pi \cos^2\theta \cdot 2\pi R^2 \sin\theta\, d\theta$$

$$= \frac{2}{3}\pi \rho_0 c U^2 a^2$$

↘ 与半径 a，速度 U 有关

$$\text{Drag} = -\int_0^\pi p' \cos\theta\, dS = -\int_0^\pi \rho_0 c U \cos^2\theta\, 2\pi a^2 \sin\theta\, d\theta\, e^{i\omega t}$$

$$= -\frac{4}{3}\pi \rho_0 c U a^2 e^{i\omega t}$$

rate of work done $= \overline{DU} = \frac{2}{3}\pi \rho_0 U^2 a^2 =$ radiated power

↙ both contain $e^{i\omega t}$

ii) $ka \ll 1$ (compact)

$$p'(r,\theta,t) = \frac{i\omega \rho_0 U a^3 \cos\theta}{2(1+ika) - (ka)^2} k^2 \left(\frac{1}{k^2 r^2} + \frac{i}{kr}\right) e^{i[\omega t - k(r-a)]}$$

$$= \frac{i\omega \rho_0 U a^3 \cos\theta}{2(1+ika)-(ka)^2} \frac{ik}{r}\left(-\frac{i}{kr}+1\right) e^{i(\omega t - kr + ka)}$$

$$= -\frac{1}{2}(ka)^2 \rho_0 c\, U \cos\theta \frac{a}{r} e^{i(\omega t - kr)} \left(1-\frac{i}{kr}\right)$$

$1-\frac{i}{kr} \approx -\frac{i}{kr}$

- In the near field $kr \ll 1$

$$p'(r,\theta,t) = \frac{1}{2}\rho_0 i\omega U \cos\theta \frac{a^3}{r^2} e^{i(\omega t - kr)}$$

$\Rightarrow$ p' independent of c (and medium compressibility)

p' and u have a phase difference of $\pi/2$ $\Rightarrow$ $\overline{p'u} = 0$

- In the far field $kr \gg 1$

$$p'(r,\theta,t) = -\frac{1}{2}(ka)^2 \rho_0 c U \cos\theta \frac{a}{r} e^{i(\omega t - kr)}$$

radiated sound power

$$\int \overline{p'u}\, dS = \int \frac{\overline{p'^2}}{\rho_0 c} \cdot 2\pi r^2 \sin\theta\, d\theta = \frac{1}{4}(ka)^4 \rho_0 c U^2 \frac{a^2}{r^2} \cdot \frac{1}{2} \cdot 2\pi r^2 \cdot \frac{2}{3}$$

$$= \frac{1}{6}\pi \rho_0 c U^2 a^2 (ka)^4$$

Comparing the two cases of $ka \gg 1$ and $ka \ll 1$ ($kr \gg 1$)

$$\left| \frac{P|_{ka\ll 1}}{P|_{ka\gg 1}} \right| = \frac{(ka)^4}{4} < 1$$

A compact sphere radiates sound very inefficiently

Compare for pulsating sphere $(ka \ll 1)$

$$p'(r,t) = \frac{a}{r} ika\, \rho_0 c\, \hat{u}_a e^{i(\omega t - kr)}$$

$$\text{Power} = \frac{\overline{p'^2}}{\rho_0 c} \cdot 4\pi r^2 = 2\pi \rho_0 c\, |\hat{u}_a|^2 a^2 (ka)^2$$

紧致的 pulsating sphere 辐射声音比 oscillating sphere

Compact pulsating sphere or bubble is more effective at generating sound than oscillating sphere by an order of $(2(ka)^{-2})$

True in general: Surface motion that leads to volume change is more effective acoustic source than vibration at constant volume

Example 2.5: A sphere of radius a oscillates parallel to the x-axis with speed $U = Ue^{i\omega t}$, where U is small. Calculate

(i) the drag the field exerts on the sphere

(ii) the rate of which work is done on the fluid by the sphere

(iii) the sound power ratiated to infinity

Comment on the limiting forms of these expressions for high $(\omega a/c \gg 1)$ and low $(\omega a/c \ll 1)$ frequencies.

Solution:

$$p'(r,\theta,t) = \frac{i\omega\rho_0 U a^3\cos\theta}{2(1+ika)-(ka)^2}\left(\frac{1}{r^2}+\frac{ik}{r}\right)e^{i[\omega t-k(r-a)]}$$

on $r=a$ $\quad p'(a,\theta,t) = \frac{ika(1+ika)}{2(1+ika)-(ka)^2}\rho_0 c U\cos\theta\, e^{i\omega t}$

(i) The force the fluid exerts on the sphere is in x-direction

$$F_x = -\int_0^\pi p'(a,\theta,t)\cos\theta\cdot 2\pi a\sin\theta\, a\, d\theta$$

$$= -\frac{ika(1+ika)}{2(1+ika)-(ka)^2}\frac{4}{3}\pi a^2\rho_0 c U e^{i\omega t}$$

(ii) The rate at which work is done on the fluid by sphere

$$\overline{F_x U} = -\frac{1}{2}\mathrm{Re}(F_x U^*)$$ (U^*: 共轭)

$$= \mathrm{Re}\left[\frac{ika(1+ika)}{1+ika-(ka)^2/2}\right]\frac{1}{3}\pi a^2\rho_0 c\,|U|^2$$

$$= \frac{(ka)^4}{1+(ka)^4/4} \cdot \frac{1}{6}\pi a^2 \rho_0 c |U|^2$$

Time average of $Ae^{i\omega t}$, $Be^{i\omega t}$, use real parts only

$$\mathrm{Re}(Ae^{i\omega t}) \cdot \mathrm{Re}(Be^{i\omega t}) = \left(\frac{Ae^{i\omega t} + A^* e^{-i\omega t}}{2}\right) \cdot \left(\frac{Be^{i\omega t} + B^* e^{-i\omega t}}{2}\right)$$

$$= \frac{1}{4}(ABe^{2i\omega t} + A^*B^* e^{-2i\omega t} + A^*B + AB^*)$$

$$\Rightarrow \overline{\mathrm{Re}(Ae^{i\omega t}) \cdot \mathrm{Re}(Be^{i\omega t})} = \frac{1}{4}(0+0+A^*B+AB^*) = \frac{1}{2}\mathrm{Re}(AB^*)$$

What if there is a phase difference of $\frac{\pi}{2}$

$$AB^* \sim A(iA)^* = -i|A|^2$$

$$\Rightarrow \frac{1}{2}(\mathrm{Re}\, AB^*) = 0$$

(iii) For large r

$$p'(r,\theta,t) = -\frac{k^2 a^2}{2(1+ika)-(ka)^2} \cdot \rho_0 c U \frac{a}{r}\cos\theta\, e^{i[\omega t - k(r-a)]}$$

The sound power radiated to infinity $= \int \frac{\overline{p'^2}}{\rho_0 c} 2\pi r^2 \sin\theta\, d\theta$

$$= \frac{(ka)^4}{1+(ka)^4/4} \cdot \frac{1}{6}\pi a^2 \rho_0 c |U|^2$$

$$= \text{result in (ii)}$$

- for $ka \gg 1$, $F_x = -\frac{4}{3}\pi a^2 \rho_0 c U e^{i\omega t}$

$\Rightarrow$ Drag is in phase with velocity

$\overline{F_x U} = \frac{2}{3}\pi \rho_0 c |U|^2 a^2 \simeq$ the power radiated to infinity

- for $ka \ll 1$

$$F_x = -ika\frac{2}{3}\pi a^2 \rho_0 c U e^{i\omega t} = -\frac{2}{3}\pi a^3 \rho_0 \cdot \underbrace{i\omega U e^{i\omega t}}_{\dot{U}}$$

  - —— The largest term in the drag is in phase with $\dot{U}$ and describes the force the sphere must exert on the fluid to overcome its inertia
  - —— No net work done against the force $-\frac{1}{2}\mathrm{Re}(F_x U^*) = 0$
  - —— A very much smaller component of $-F_x$ in phase with the velocity that does work on the fluid at a rate $\frac{1}{6}\pi \rho_0 c |U|^2 a^2 (ka)^4$
  - —— This power is radiated to infinity is a factor of $(ka)^4/4$ smaller than the power radiated by the large vibrating sphere.
    - ↳ a measure of the very small amount of power that is radiated into the sound field by the rigid motion of a compact sphere.

Example 2.6

How much power is radiated in the various harmonic sound fields whose pressure are $p'(x,t) = P_0 l^{n+1} \frac{\partial^n}{\partial x_1^n}\{r^{-1}\cos[\omega(t-r/c)]\}$

↓ source strength

l is an length scale

Solution:

$$p'(x,t) = Re\left[P_0 (-l)^{n+1} \frac{\partial^n}{\partial x_1^n} \left(\frac{e^{i\omega(t-r/c)}}{r} \right) \right]$$

Now $r = (x_1^2 + x_2^2 + x_3^2)^{\frac{1}{2}}$ and $\left.\frac{\partial r}{\partial x_1}\right|_{x_2, x_3} = \frac{x_1}{r} = \cos\theta$

For large r, keep the term of order $\frac{1}{r}$

$$\Rightarrow \quad p'(x,t) = Re\left[P_0 (-l)^{n+1} \cos^n\theta \, (ik)^n \frac{e^{i\omega(t-r/c)}}{r} \right]$$

$$\Rightarrow \quad \overline{p'^2} = P_0^2 (-l)^{2n+2} \cos^{2n}\theta \, k^{2n} \frac{1}{2r^2}$$

$$\Rightarrow \quad \text{Radiated sound power} \quad \int_0^\pi \overline{p'u} \, dS = \int_0^\pi \frac{\overline{p'^2}}{\rho_0 c} 2\pi r \sin\theta \, r \, d\theta$$

$$= \frac{2\pi P_0^2 l^2}{\rho_0 c (2n+1)} (kl)^{2n}$$

Summary of Chapter 2

The 3-D wave equation $\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0$

The velocity potential $\phi(x,t)$

$$\frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = 0$$

$$p' = -\rho_0 \frac{\partial \phi}{\partial t} \quad , \quad v = \nabla \phi$$

vector intensity $I = p' v$

General form for a spherically symmetric sound field

$$p'(r,t) = \frac{f(r-ct)}{r} + \frac{g(r+ct)}{r}$$

↓ radial momentum equation

radial velocity

Far from source : $u = \frac{p'}{\rho_0 c}$

sound power $P = \int_S \frac{\overline{p'^2}}{\rho_0 c} dS$ where $dS = 2\pi r^2 \sin\theta \, d\theta$

How to use the symmetric solution of the wave equation

$$\phi(r,t) = \frac{f(t-r/c)}{r} + \frac{g(t+r/c)}{r}$$

and the solution with $\cos\theta$ dependence

$$\phi(r,t) = \cos\theta \frac{\partial}{\partial r}\left[\frac{f(t-r/c)}{r} + \frac{g(t+r/c)}{r}\right]$$

applying appropriate initial and boundary conditions to determine the functions f and g

You should have an appreciation that

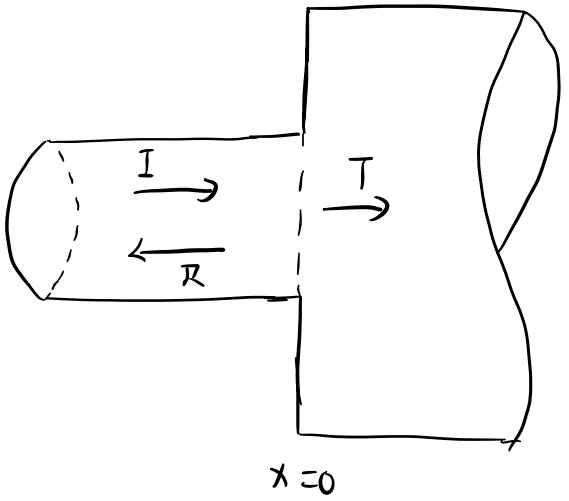
- compact acoustic source ($ka \ll 1$) are inefficient at generating sound, with motion of a rigid body being even more inefficient than motion involving a volume change.
- high frequency surface motion ($ka \gg 1$) has the relation $p' = \rho_0 cu$ as in a plane wave, with sound propagating away from its source like rays.

3 Waves in pipes and resonators

3.1 Plane waves

If sound wavelength $\lambda \gg$ pipe diameter d

i.e. low-freq sound wave $\longrightarrow$ 1-D plane wave



Incident wave $\quad p_i' = I e^{i\omega(t - x/c)}$

reflected wave $\quad p_r' = R e^{i\omega(t + x/c)}$

transmitted wave $\quad p_t' = T e^{i\omega(t - x/c)}$

$$\Rightarrow \quad p' = \begin{cases} I e^{i\omega(t - x/c)} + R e^{i\omega(t + x/c)}, & x < 0 \\ T e^{i\omega(t - x/c)}, & x > 0 \end{cases}$$

mass flux in = mass flux out $\Rightarrow \quad \rho_0 A_1 u_1 = \rho_0 A_2 u_2$

$$\Downarrow$$

$$\frac{A_1}{\rho_0 c}(I - R) = \frac{A_2}{\rho_0 c} T$$

energy flux in = energy flux out $\Rightarrow \quad A_1 p_1' u_1 = A_2 p_2' u_2$

$$\Downarrow$$

$$p_1' = p_2'$$

$$\Downarrow$$

$$I + R = T$$

$$\Rightarrow \quad R = \frac{A_1 - A_2}{A_1 + A_2} I \quad , \quad T = \frac{2A_1}{A_1 + A_2} I$$

- For an open pipe end, $A_2 = \infty \longrightarrow R = -I$, $T = 0$
- Plane wave cannot escape from open pipes

Transmission loss

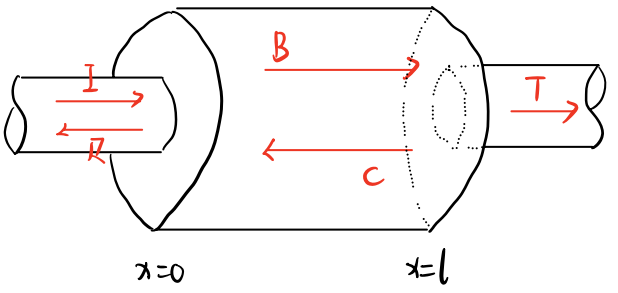
$$TL = 10 \log_{10} \left(\frac{\text{incident power}}{\text{transmitted power}} \right)$$

In this case, $TL = 10 \log_{10} \left(\frac{A_1 I^2}{A_2 T^2} \right) = 10 \log_{10} \left(\frac{(A_1 + A_2)^2}{4 A_1 A_2} \right)$

- TL symmetric in A_1 and A_2
- TL the same whether sound incident from A_1 or A_2

3.1.1 A single expansion chamber "silencer"

$$p' = \begin{cases} I e^{i\omega(t - \frac{x}{c})} + R e^{i\omega(t + \frac{x}{c})} & , x < 0 \\ B e^{i\omega(t - \frac{x}{c})} + C e^{i\omega(t + \frac{x}{c})} & , 0 < x < l \\ T e^{i\omega(t - \frac{x}{c})} & , x > l \end{cases}$$



B.C.

$$\begin{cases} A_1 (I - R) = A_2 (B - C) \\ I + R = B + C \end{cases}$$

$$\begin{cases} A_1 T e^{-i\omega \frac{l}{c}} = A_2 (B e^{-i\omega \frac{l}{c}} - C e^{i\omega \frac{l}{c}}) \\ T e^{-i\omega \frac{l}{c}} = B e^{-i\omega \frac{l}{c}} + C e^{i\omega \frac{l}{c}} \end{cases}$$

$$\Rightarrow \quad \boxed{R = \frac{\left(\frac{A_1}{A_2} - \frac{A_2}{A_1} \right) i \sin \frac{\omega l}{c} I}{2 \cos \frac{\omega l}{c} + \left(\frac{A_1}{A_2} + \frac{A_2}{A_1} \right) i \sin \frac{\omega l}{c}}}$$

$$T = \frac{2I e^{i\omega l/c}}{2\cos\frac{\omega l}{c} + \left(\frac{A_1}{A_2} + \frac{A_2}{A_1}\right) i \sin\frac{\omega l}{c}}$$

$$\Rightarrow \quad |R|^2 + |T|^2 = |I|^2$$

- Acoustic energy is conserved
- This "silencer" does not reduce sound energy
- To produce acoustic energy reduction ⟵ sound absorbing material

$$TL = 10\log_{10}\left(\frac{|I|^2}{|T|^2}\right) = 10\log_{10}\left[1 + \frac{1}{4}\left(\frac{A_1}{A_2} - \frac{A_2}{A_1}\right)^2 \sin^2\frac{\omega l}{c}\right]$$

$\Rightarrow$ max TL when $\sin\frac{\omega l}{c} = \pm 1 \Rightarrow \omega = \frac{\pi c}{2l}, \frac{3\pi c}{2l} \cdots$

$\Rightarrow$ $\frac{1}{4}$ wavelength resonance, $l = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \cdots$

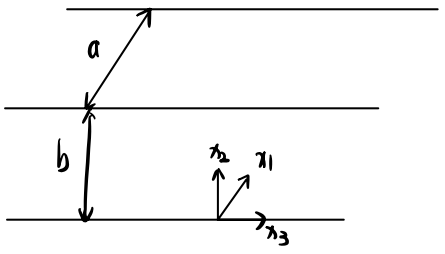
$TL = 0$ when $\sin\frac{\omega l}{c} = 0 \Rightarrow \omega = \frac{\pi c}{l}, \frac{2\pi c}{l}, \cdots$

$\Rightarrow$ $\frac{1}{2}$ wavelength resonance, $l = \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \cdots$

- The silencer should be tuned max. TL at dominant frequency of the coming sound.

3.2 High order modes

3.2.1 Ducts of retangular cross-section



Example 3.1 Show that acoustic disturbances of frequency ω have the form

$$p'(x,t) = \sum_{m,n} \cos\left(\frac{m\pi x_1}{a}\right)\cos\left(\frac{n\pi x_2}{b}\right)\cdot\left[A_{mn}e^{i(\omega t - k_{mn}x_3)} + B e^{i(\omega t + k_{mn}x_3)}\right]$$

$\omega < \pi c\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)^{\frac{1}{2}}$ vary exponentially along the duct

Find the frequency below which only the plane wave mode ($m=n=0$) propagates.

Solution:

$$\nabla^2 p' - \frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} = 0$$

分离变量法

seek a seperable solution $p'(x_i, t) = f(x_1)\,g(x_2)\,h(x_3)\,e^{i\omega t}$

$$\Rightarrow \quad \frac{\ddot{f}}{f} = -\frac{\ddot{g}}{g} - \frac{\ddot{h}}{h} - \frac{\omega^2}{c^2} \quad \overset{\text{set}}{=\!=} \text{constant} = -\alpha_1^2$$

$$\Rightarrow \quad \frac{\ddot{f}}{f} = -\alpha_1^2 \quad \text{has solution} \quad f(x_1) = A_1\cos(\alpha_1 x_1) + B_1\sin(\alpha_1 x_1)$$

To satisfy B.C. $\frac{df}{dx_1} = 0$ on $x_1 = 0 \Rightarrow B_1 = 0$

for $\frac{df}{dx_1} = 0$ on $x_1 = a \Rightarrow \alpha_1 = \frac{m\pi}{a}$ (m is interger)

$$\Rightarrow \quad f(x_1) = A_1\cos\left(\frac{m\pi x_1}{a}\right) \quad \text{(standing wave solution)}$$

Similar reasoning gives $g(x_2) = A_2\cos\left(\frac{n\pi x_2}{b}\right)$

Then from the wave equation

$$\frac{d^2h}{dx_3^2} + \left[\frac{\omega^2}{c^2} - \pi^2\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)\right]h = 0$$

$$\Rightarrow \quad h(x_3) = A_{mn}e^{-ik_{mn}x_3} + Be^{ik_{mn}x_3} \quad \text{(travelling waves)}$$

The general solution is a superposition of these modes.

$$h(x_3) = \sum_{m,n}\left(A_{mn}e^{-ik_{mn}x_3} + B_{mn}e^{ik_{mn}x_3}\right)$$

$$\text{with} \quad k_{mn} = \left[\frac{\omega^2}{c^2} - \pi^2\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)\right]^{\frac{1}{2}}$$

① If $\omega > \pi c\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)^{\frac{1}{2}}$, k_{mn} is real $\longrightarrow$ "cut-on" modes

Travelling waves along the x_3 axis at the speed $\omega/k_{mn} \geq c$

② If $\omega < \pi c \left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)^{\frac{1}{2}}$, $k_{mn} = \pm i\gamma_{mn}$

h must be finite when $x_3 \to \pm\infty$

$\Rightarrow$ $k_{mn} = -i\gamma_{mn}$ for $+x$ and $-x$ propagates waves

$\Rightarrow$ $h(x_3) = \sum_{m,n} \left(A_{mn} e^{-\gamma_{mn} x_3} + B_{mn} e^{\gamma_{mn} x_3} \right)$

In case of $x_3 \to \pm\infty$, h reduces to $\sum_{m,n} A_{mn} e^{-\gamma_{mn} x_3}$ or $\sum_{m,n} B_{mn} e^{\gamma_{mn} x_3}$

- p' and u_3 are 90° out of phase using momentum eq.

  $\Rightarrow$ mean acoustic intensity $\overline{p' u_3} = 0$

- Evanescent disturances vary exponentially in amplitude along the duct.
- For $\omega < \pi c \cdot \min(\frac{1}{a}, \frac{1}{b})$, all modes except $m=n=0$ satisfy

$$\omega < \pi c \left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)^{\frac{1}{2}}$$

without loss of generality, assume $a > b$

$\Rightarrow$ when $2a < \lambda$ (i.e $\omega < \frac{\pi c}{a}$), only the plane wave ($m=n=0$) propagates, and other mode "cut-off"

- The plane wave can always propagates, while higher order modes only propagate if at least $\omega > \frac{\pi c}{a}$

Example 3.2 Determine the "cut-on" frequency (i.e. the lowest frequency at which a non-plannar wave propagates) for a duct of rectangular cross-section of 0.3m by 0.5m. At half of this frequency, what is the decay rate in dB per metre of the slowest decaying non-planar wave mode?

Solution:

$$\omega > \pi c \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$

for $m=1, n=0$ (if $a = 0.5\,m$, $b = 0.3m$)

'cut-on' frequency $\omega = \frac{\pi c}{0.5}$

At half this frequency, the velocity potential and p' vary with x_3 like

$$e^{-\gamma_{mn} x_3} \quad \text{where} \quad \gamma_{mn} = \left[\pi^2\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right) - \frac{\omega^2}{c^2}\right]^{\frac{1}{2}} \quad \text{with} \quad \omega = \pi c$$

The slowest decaying mode is for $m=1, n=0$

$$\Rightarrow \quad \gamma_{10} = \sqrt{(2\pi)^2 - \pi^2} = \sqrt{3}\pi$$

$$\Rightarrow \quad p'_{rms}(x=1m) = p'_{rms}(x=0)\, e^{-\sqrt{3}\pi}$$

In dB $\quad dB(x=1m) = dB(x=0m) + 20 \log_{10} e^{-\sqrt{3}\pi}$

$$\Rightarrow \text{ attenuation of } 47.3\,dB$$

This illustrates how beneficial it can be to ensure that a source in a duct generates modes that are cut-off.

3.2.2 Ducts of circular cross-section

Consider p' with azimuthal mode number n in a cylindical duct of radius a

$$p'(r, x_3, \theta, t) = f(r)\, g(x_3)\, e^{i(\omega t - n\theta)}$$

↓ in θ direction satisfy periodical condition

↓ wave equation

柱坐标形式的波动方程：

$$\frac{\partial^2 p'}{\partial x^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p'}{\partial r}\right) + \frac{1}{r^2}\frac{\partial p'}{\partial \theta^2} - \frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} = 0$$

$$\frac{1}{rf}\frac{d}{dr}[r f'(r)] - \frac{n^2}{r^2} + k^2 = -\frac{\ddot{g}(x_3)}{g} = \text{constant} = k_x^2$$

The solution to $\frac{\ddot{g}(x_3)}{g} = -k_x^2$

$$\Rightarrow \quad g(x_3) = Ae^{-ik_x x_3} + Be^{ik_x x_3}$$

The equation for $f(r)$ is

$$\frac{1}{rf}\frac{d}{dr}[rf(r)] - \frac{n^2}{r^2} + k^2 - k_x^2 = 0 \qquad \text{(Bessel equation)}$$

$$\Rightarrow \quad J_n(\alpha r) \quad \text{with} \quad \alpha^2 = k^2 - k_x^2$$

B.C. rigid duct wall $\frac{\partial p'}{\partial r} = 0$ on $r=a \Rightarrow J_n'(\alpha a) = 0$

Z_{mn} = the mth zero of $J_n' \Rightarrow \alpha = \frac{Z_{mn}}{a} \Rightarrow k_x^2 = k^2 - \frac{Z_{mn}^2}{a^2}$

	n = 0	n = 1	n = 2
m = 1	0.000	1.841	3.054
m = 2	3.832	5.331	6.706
m = 3	7.016	8.536	9.969

$$\Rightarrow p'(r, x_3, \theta, t) = J_n(Z_{mn} r/a)\left(A_{mn} e^{-ik_{mn}x_3} + Be^{ik_{mn}x_3}\right) e^{i(\omega t - n\theta)}$$

where $k_{mn} = \sqrt{k^2 - Z_{mn}^2/a^2}$

- Plane waves $(m=1, n=0) \longrightarrow Z_{mn} = 0$, $k_{mn} = k$

  $\longrightarrow$ no variations with r or θ
- The smallest nonzero value of $Z_{mn} = 1.841$ for $m=1, n=1$

  $\longrightarrow$ The first mode with azimuthal dependence $e^{-i\theta}$
- $k < 1.841/a$, k_{mn} purely imaginary for all modes expect the

plane wave $\Rightarrow$ cut off mode with exponential decay

B.C. duct linear with surface impedance $\frac{p_s'}{u_n} = \rho_0 c z$

$$u_n \longleftarrow \rho_0 \frac{\partial u_r}{\partial t} = -\frac{\partial p'}{\partial r}$$

$$\rho_0 i\omega u_r = -\alpha \dot{J}_n(\alpha r)\left(A_{mn} e^{-ik_x x_3} + B_{mn} e^{ik_x x_3}\right) e^{i(\omega t - n\theta)}$$

$$\downarrow$$

$$p' = J_n(\alpha r)\left(A_{mn} e^{-ik_x x_3} + B_{mn} e^{ik_x x_3}\right) e^{i(\omega t - n\theta)}$$

Substitude in the B.C. $\Rightarrow$ $J_n(\alpha a) = -\frac{1}{\rho_0 i\omega} \alpha \dot{J}_n(\alpha a) \rho_0 c z$

Let $\alpha a = z_{mn}$, the roots of $-ika\, J_n(z_{mn}) = z_{mn}\, \dot{J}_n(z_{mn}) z$ how

Then the solution is as before

$$p'(r, x_3, \theta, t) = J_n(z_{mn} r/a)\left(A_{mn} e^{-ik_{mn} x_3} + B_{mn} e^{ik_{mn} x_3}\right) e^{i(\omega t - n\theta)}$$

where $k_{mn} = \sqrt{k^2 - z_{mn}^2/a^2}$

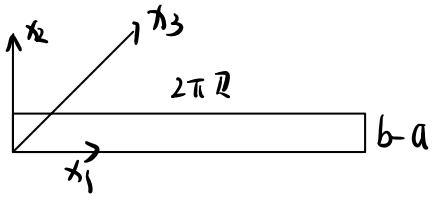
- The roots z_{mn} are in general complex $\longleftarrow$ numerial iteration
- The complex k_{mn} $\longrightarrow$ propagation and attenuation of p'

3.2.3 Ducts of thin annulus

Consider a rigid walled annulus duct of inner radius a and outer radius b

For a thin annulus $\frac{b}{a} \approx 1$

$\Downarrow$



where $R = \frac{a+b}{2}$

把周向展平

The periodic condition

$$p'(r, x_3, \theta+2\pi, t) = p'(r, x_3, \theta, t)$$

Solution $p'(x,t) = f(x_1)g(x_2)h(x_3)e^{i\omega t}$

↓ wave equation

$$f(x_1) = A_1\cos(k_1 x_1) + B_1 \sin(k_1 x_1)$$

↓ $f(x_1) = f(x_1 + 2\pi R)$

$$k_1 2\pi R = 2\pi n \quad \Rightarrow \quad k_1 = \frac{n}{R}$$

The rigid walls a distance b-a apart $\Rightarrow$ $U_2(x_2=0) \Rightarrow \sin[k_2(b-a)] = 0$

$$k_2 = \frac{m\pi}{b-a}$$

$$g(x_2) = A_2\cos(k_2 x_2) \quad \text{with} \quad k_2 = \frac{m\pi}{b-a}$$

$$\Rightarrow \quad h(x_3) = A_{mn} e^{-ik_{mn}x_3} + B_{mn} e^{ik_{mn}x_3}$$

where $$k_{mn} = \left[\frac{\omega^2}{c^2} - \frac{n^2}{R^2} - \left(\frac{m\pi}{b-a}\right)^2\right]^{\frac{1}{2}}$$

- If $\frac{\omega}{c} < \frac{n}{R}$, k_{mn}^2 will be negative for all m

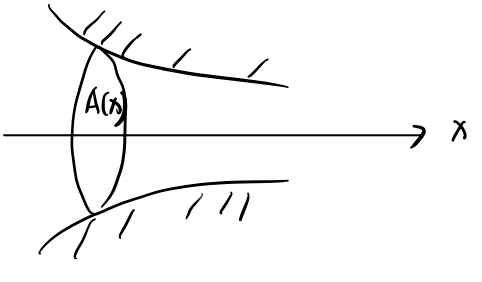
  ie. all circumferential modes of number n are "cut-off" when $n > Rk$

  or $\omega < \frac{nc}{R}$

这里是因为 $R \gg b-a$

3.3 Pipes of varying cross-section

Consider a rigid with cross-sectional area A(x)



If $d \ll \lambda$ and A varying slowly with x

continue equation $A \frac{\partial \rho'}{\partial t} = -\rho_0 \frac{\partial}{\partial x}(uA)$

momentum equation $\rho_0 \frac{\partial u}{\partial t} = -\frac{\partial p'}{\partial x}$

$\frac{\partial}{\partial t}$

$\cdot A, \frac{\partial}{\partial x}$ 先两边乘 A 再求散度

$$\frac{A}{c^2}\frac{\partial^2 p'}{\partial t^2} = \frac{\partial}{\partial x}\left(A\frac{\partial p'}{\partial x}\right)$$

Simple solutions for some geometries

For example, for an 'exponential horn' $A(x) = A_0 e^{\alpha x}$

$A_0 = A(0)$

α: flare constant

$\frac{A^2}{c^2}\frac{\partial^2 p'}{\partial t^2} = \frac{\partial}{\partial x}\left(A\frac{\partial p'}{\partial x}\right)$

这一项表示，朝喇叭开口方向传播时，恒定会 decay

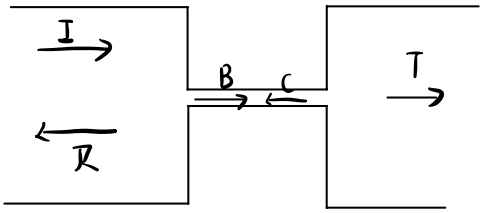
$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t} = \frac{\partial^2 p'}{\partial x^2} + \alpha\frac{\partial p'}{\partial x}$$

$$\Rightarrow \quad p'(x,t) = e^{-\alpha x/2}\left[Ae^{i(\omega t - kx)} + Be^{i(\omega t + kx)}\right]$$

where $k = \sqrt{\omega^2/c^2 - \alpha^2/4}$

- $\omega > \alpha c/2$: waves propagate with attenuating p' (but ont energy flux $p'uA$)
- $\omega < \alpha c/2$: 'cut-off' modes with mean intensity $\overline{p'u} = 0$

3.4 Organ pipes



In sec 3.1.1, $\mathcal{R}=0$ when $\sin\frac{\omega l}{c}=0$

$\Rightarrow$ $T=I$ (perfect sound transmission)

- v for constriction $(A_1>A_2)$ or expansion $(A_1<A_2)$
- large acoustic activity within the narrow tube when $A_1 \gg A_2$

$$\cdots \quad B=\frac{\left(1+\frac{A_1}{A_2}\right)Ie^{i\omega l/c}}{2\cos\frac{\omega l}{c}+\left(\frac{A_1}{A_2}+\frac{A_2}{A_1}\right)i\sin\frac{\omega l}{c}}$$

$$C=\frac{\left(1-\frac{A_1}{A_2}\right)Ie^{i\omega l/c}}{2\cos\frac{\omega l}{c}+\left(\frac{A_1}{A_2}+\frac{A_2}{A_1}\right)i\sin\frac{\omega l}{c}}$$

In the limit of large constraction

$$\frac{A_1}{A_2}\to\infty \quad\Rightarrow\quad |B|^2=|C|^2=\frac{|I|^2\frac{A_1^2}{A_2^2}}{4\cos^2(\omega l/c)+\frac{A_1^2}{A_2^2}\sin^2(\omega l/c)}\approx\frac{|I|^2}{\sin^2(\omega l/c)}$$

- When $\omega l/c=n\pi$

$$\lambda=2\pi c/\omega=2l/n \qquad |B/I|^2\longrightarrow\infty$$

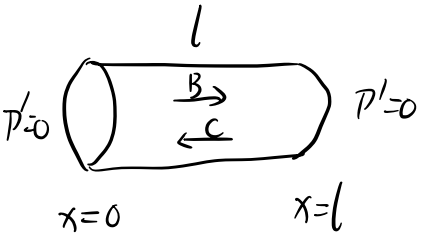
- Resonator $\longrightarrow$ large amplitude response to excitation at some pure tones (resonance frequency)

- For an expansion chamber ($A_1 < A_2$) $\Rightarrow |B/I|^2 < 1$

  $\Rightarrow$ no amplification of pressure waves with the resonator

The case of $A_1 \gg A_2$, $\omega l/c = n\pi \longrightarrow |B| \gg |I|$

$\Rightarrow$ organ pipe resonator ($p \approx p_{atm}$, $p'=0$ at two open ends)



For 1-D waves

$$p' = Be^{i\omega(t-\frac{x}{c})} + Ce^{i\omega(t+\frac{x}{c})}$$

B.C. | $p'=0$ at $x=0 \Rightarrow B=-C$

$p'=0$ at $x=l \Rightarrow Be^{-i\omega l/c} + Ce^{i\omega l/c} = 0$

open end condition

$$\sin(\omega l/c) = 0 \Rightarrow \omega l/c = n\pi \quad n=1, 2\cdots$$

The organ pipe frequencies at $\omega = \frac{\pi c}{l}, \frac{2\pi c}{l}, \frac{3\pi c}{l} \cdots$

wavelengths of these modes $\lambda = 2\pi c/\omega = 2l, l, \frac{2}{3}l$

$$l = \frac{1}{2}\lambda, \lambda, \frac{3}{2}\lambda$$

$n=1 \Rightarrow \omega = \pi c/l$

putting $B=-C$ back to $p' = Be^{i\omega(t-\frac{x}{c})} + Ce^{i\omega(t+\frac{x}{c})}$

$$\Rightarrow p' = -2iBe^{i\omega t}\sin(\pi x/l)$$

$$\Rightarrow u = \frac{2B}{\rho_0 c} e^{i\omega t}\cos(\pi x/l)$$

- p' and u 90° out of phase $\Rightarrow$ $\overline{p'u} = 0$

  $\Rightarrow$ acoustic energy conserved in organ pipe)

The particle displacement

$$\eta = \int u\,dt = -\frac{2iB}{\rho_0 c\omega} e^{i\omega t} \cos\left(\frac{\pi x}{l}\right)$$

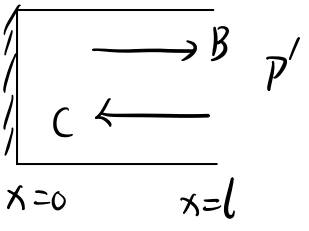
Hence relate p' and η

$$\frac{p'}{\rho_0 c^2} = \frac{\pi \eta}{l} \tan\left(\frac{\pi x}{l}\right)$$

- If $\tan\left(\frac{\pi x}{l}\right) > 0$ (i.e. $0 < x < l/2$) $\longrightarrow$ p' and η in phase
- If $\tan\left(\frac{\pi x}{l}\right) < 0$ (i.e. $l/2 < x < l$) $\longrightarrow$ p' and η out of phase

  (antiphase $\delta\varphi = \pi$)

Example 3.3 Alternative end conditions



$\frac{1}{4}$ wavelength

3.5 The Rijke tube

Rijke tube: a vertically held pipe open at both ends with a heated grid

air heated by the grid $\longrightarrow$ hot air rises $\uparrow$

η of sound wave superimposed on this natural convection

- $\eta > 0 \uparrow \Rightarrow$ more colder air below pass the grid, heat transfer $q \uparrow$
- $\eta < 0 \downarrow \Rightarrow$ relatively little q

$\Rightarrow$ q_{max} occurs at $\eta > 0$

At the grid $x = x_G$

- If $x_G < l/2$, η, p' in phase $\longrightarrow$ q, p' in phase, energy fed into p'
  $\Rightarrow$ fundamental pipe resonance excited
- If $x_G > l/2$, η, p' and q, p' out of phase $\longrightarrow$ organ pipe mode not excited

Lord Rayleigh in "Theory of Sound" (1894) explained:

If heat $\rightarrow$ sound wave $q > 0$, p'?

observed heat addition

- When $T' > 0 \longrightarrow p' \uparrow$ energy fed into sound wave
- When $T' < 0 \longrightarrow p' \downarrow$ sound wave loses energy

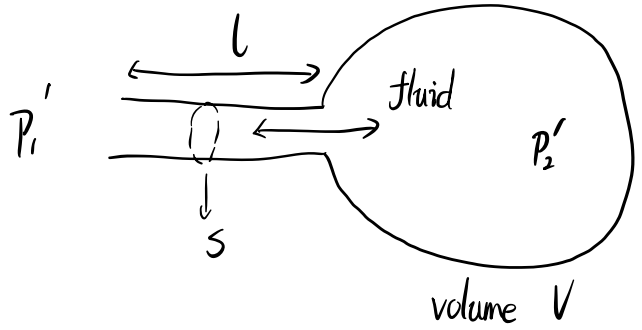
adiabatic compression in sound wave $\longrightarrow$ p' and T' in phase

Hence heat addition ($q>0$) causes

- $p' \uparrow$ for $+p'$ (q, p' in phase, $x_G < l/2$)
- $p' \downarrow$ for $-p'$ (q, p' out of phase, $x_G > l/2$)

3.6 The Helmholtz resonator

Consider a Helmholtz resonator with a short neck of length l and cross-sectional area S, and a large bulb of volume V



p' at the end of neck $\rightarrow$ mass flow rate Q (kg/s)

$$p_1' \Longleftrightarrow Q \ ?$$

Mass flow Q into the bulb V $\rightarrow$ $\rho \uparrow$

$$V\frac{\partial \rho'}{\partial t} = Q \quad \Rightarrow \quad \rho' = \frac{Q}{i\omega V}$$ for harmonic waves

$\rho' \Rightarrow p_2' \uparrow$ $\quad p_2' = c^2\rho' = \frac{c^2 Q}{i\omega V}$

assume that isentropic

Momentum balance in the neck

$$(p_1' - p_2')S = \rho_0 sl\frac{\partial u}{\partial t}$$ ← take the control volume of neck as a mass cube

□

$\because \quad Q = \rho_0 S u \quad \Rightarrow \quad p_1' - p_2' = \frac{i\omega l Q}{S}$

Substitude p_2' in $p_2' = c^2 \rho' = \frac{c^2 Q}{i\omega V}$

$$p_1' = \left(\frac{c^2}{i\omega V} + \frac{i\omega l}{S} \right) Q = \left(\omega^2 - \frac{c^2 S}{Vl} \right) \frac{il Q}{\omega S}$$

At $\omega = \sqrt{\frac{c^2 S}{Vl}} \longrightarrow$ resonance : very small $p_1' \longrightarrow$ large Q

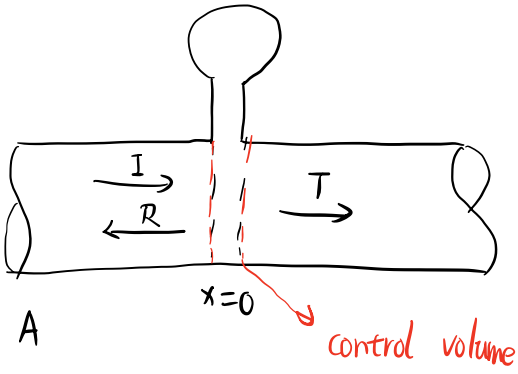
- For a typical beer bottle $V = 5 \times 10^{-4}\ m^3$, $S = 2 \times 10^{-4}\ m^2$

  $l = 5 \times 10^{-2}\ m$

  $\Rightarrow \quad \omega = 1000\ rad/s \approx 150\ Hz$

- Higher notes $\omega \uparrow \leftarrow V \downarrow$ partly filled bottles.

Use as efficient muttler



$$p' = \begin{cases} I e^{i\omega(t - x/c)} + R e^{i\omega(t + x/c)} & , x < 0 \\ T e^{i\omega(t - x/c)} & , x > 0 \end{cases}$$

Continuity of p' at the neck $x = 0$

$$p_1' = (I + R) e^{i\omega t} = T e^{i\omega t}$$

Continuity of mass flux in to the control volume

$$\rho_0 \frac{A}{\rho_0 c}(1-R)e^{i\omega t} = Q + \rho_0 \frac{AT}{\rho_0 c} e^{i\omega t}$$

$$\Rightarrow P_1' = \left(\frac{c^2}{i\omega V} + \frac{i\omega l}{S}\right) Q$$

$$\Rightarrow \quad T = \frac{I}{1 + \frac{1}{2A}\left(\frac{c}{i\omega V} + \frac{i\omega l}{cS}\right)^{-1}}$$

Transmission loss

$$TL = 10 \log_{10}\left(\frac{|I|^2}{|T|^2}\right) = 10 \log_{10}\left[1 + \frac{1}{4A^2}\left(\frac{c}{\omega V} - \frac{\omega l}{cS}\right)^{-2}\right]$$

- Significant sound attenuation only at frequencies close to $\omega = \sqrt{\frac{c^2 S}{Vl}}$

Summary of chapter 3

You should know:

- Only plane wave propagate in a duct whose dimensions are small in comparison with the wavelength $2\pi c/\omega$
- How to determine the wave amp and resonance freq. in a pipe by applying appropriate B.C.s
- Transmission loss $TL = 10 \log_{10}\left(\frac{\text{incident power}}{\text{transmitted power}}\right)$
- How to use separable solution to determine p' of higher order modes in a duct

You should have an appreciation that:

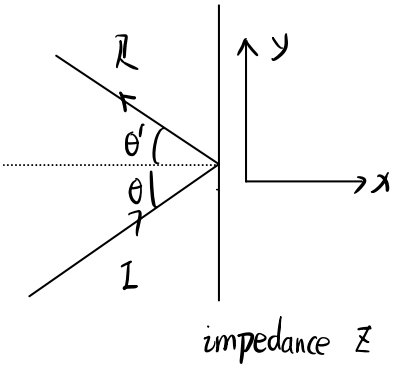
- If $w <$ "cut-off" freq. , k_{mn} purely imaginary

  $\longrightarrow$ "cut-off" modes with

- At the resonance frequencies of resonator, small p' can encite large amplitude response of pressure perturbation and mean flow

4. Sound wave at interfaces

4.1 Waves incident on a flexible surface



Consider a plane harmonic wave incident at an angle θ

$$p_i'(x, y, t) = I e^{i\omega(t - x\cos\theta/c - y\sin\theta/c)}$$

$$p_r'(x, y, t) = R e^{i\omega(t + x\cos\theta'/c - y\sin\theta'/c)}$$

The surface pressure at $x = 0$

$$p'(0, y, t) = I e^{i\omega(t - y\sin\theta/c)} + R e^{i\omega(t - y\sin\theta'/c)}$$

Continuity of velocity at the surface $x=0$

$$u(0, y, t) = \frac{1}{\rho_0 c}\left[I\cos\theta\, e^{i\omega(t - y\sin\theta/c)} - R\cos\theta'\, e^{i\omega(t - y\sin\theta'/c)}\right]$$

impedance $p' = Zu$

- For a rigid or very heavy surface : $u_n = 0 \Rightarrow Z \to \infty$
- For a hmp, light surface $p' = 0$, hence $Z = 0$
- In general, Z is a complex

Subsitude $p'(0, y, t)$ and $u(0, y, t)$ into

$$I e^{i\omega(t - y\sin\theta/c)} + R e^{i\omega(t - y\sin\theta'/c)}$$

$$= \frac{Z}{\rho_0 c}\left[I\cos\theta\, e^{i\omega(t - y\sin\theta/c)} - R\cos\theta'\, e^{i\omega(t - y\sin\theta'/c)}\right]$$

$\Rightarrow$ $\boxed{\theta = \theta'}$ ← If and only if ⋯,

反射角 = 入射角

Hence the solution

$$\mathcal{R} = \frac{\mathcal{Z} - \rho_0 c/\cos\theta}{\mathcal{Z} + \rho_0 c/\cos\theta}\, \mathcal{I}$$

- $\rho_0 c/\cos\theta$: "oblique incidence" wave impedance

  $\rho_0 c/\cos\theta \longrightarrow \infty$ for tangential incidence ($\cos\theta \to 0$)

- For a rigid or heavy surface $\mathcal{Z} \to \infty$, $\mathcal{R} \approx \mathcal{I}$ (where $\cos\theta \neq 0$)

  $\Rightarrow$ SPL ↑ 6 dB (which means SPL increase 6 dB due to total reflection)

- For a limp, light surface $\mathcal{Z} = 0$, or grazing incidence $\cos\theta = 0$, $\mathcal{R} = -\mathcal{I}$

  the direct and reflected wave cancel exactly → no sound

- If $\mathcal{Z} = \rho_0 c/\cos\theta$ (surface impedance = plane-wave impedance), $\mathcal{R} = 0$

  → perfectly transparent wall to the second wave.

Example 4.1 Consider a plane wave obliquely incident on a wall of mechanical impedance $\mathcal{Z}$. Determine the total time-averaged net energy-flux in the wall normal direction. Show that the averaged rate of working by the wall per unit area is $\frac{1}{2}\mathrm{Re}(\mathcal{Z})\,|u_n|^2$. How does this relate to the net energy flux?

Solution :

Rate of working by wall $= p'u = \mathcal{Z}u^2$

$$\Rightarrow \quad \overline{\mathcal{Z}u^2} = \frac{1}{2}\mathrm{Re}\left(\mathcal{Z}uu^*\right) = \frac{1}{2}\mathrm{Re}(\mathcal{Z})\,|u|^2$$

Net energy flux $= p'u$ on $y = 0$

$$p'(x,0,t) = \mathcal{I}e^{i\omega(t - x\cos\theta/c)} + \mathcal{R}e^{i\omega(t + x\cos\theta/c)}$$

$$u(x,0,t) = \frac{1}{\rho_0 c}\left[I\cos\theta\, e^{i\omega(t-x\cos\theta/c)} - R\cos\theta\, e^{i\omega(t+x\cos\theta/c)}\right]$$

$$\Rightarrow \quad \overline{Pu} = \frac{1}{2}\mathrm{Re}(p'u^*) = \frac{1}{2}\mathrm{Re}\left\{\left[Ie^{i\omega(t-x\cos\theta/c)} + Re^{i\omega(t+x\cos\theta/c)}\right]\cdot \frac{1}{\rho_0 c}\left[I^*\cos\theta\, e^{i\omega(-t+x\cos\theta/c)} - R^*\cos\theta\, e^{i\omega(-t-x\cos\theta/c)}\right]\right\}$$

$$= \frac{1}{2\rho_0 c}\mathrm{Re}\left\{|I|^2\cos\theta - |R|^2\cos\theta - IR^*\cos\theta\, e^{-2i\omega x\cos\theta/c} + I^*R\cos\theta\, e^{2i\omega x\cos\theta/c}\right\}$$

$$= \frac{1}{2\rho_0 c}\mathrm{Re}\left\{\cos\theta\,(|I|^2-|R|^2) - \cos\theta\underbrace{\left[IR^*e^{-2i\omega x\cos\theta/c} - \left[(IR^*)e^{-2i\omega x\cos\theta/c}\right]^*\right]}_{\text{purely imaginary}}\right\}$$

$$= \frac{\cos\theta}{2\rho_0 c}(|I|^2 - |R|^2)$$

The net energy flux = the rate of working by the wall

Sound absorbers

Sound absorbent material ⟶ reduce acoustic energy

↙ surface impedance

Practical sound absorbers: porous material (多孔材料), perforated (穿孔板) or honeycomb panel (蜂窝板)

for example, acoustic liners (声衬) in ducts.

The absorption coefficient $\alpha = 1 - \left|\frac{R}{I}\right|^2 - \left|\frac{T}{I}\right|^2$

If the sound absorber is backed by a rigid wall, $u_n = 0$ at the wall, $T = 0$

$$\alpha = 1 - \left|\frac{R}{I}\right|^2 = 1 - \left|\frac{Z - \rho_0 c/\cos\theta}{Z + \rho_0 c/\cos\theta}\right|^2$$

- If Z is purely imaginary, $|R/I| = 1 \Rightarrow \alpha = 0$
- $\alpha = \alpha(\omega, \theta)$ ↘ 表示吸声系数与频率 ω, 入射角 θ 有关

Example 4.2 A plane sound wave of SPL 160 dB, frequency 1 kHz is normally incident on an acoustic linear whose surface impedance is $\rho_0 c\,(0.11 + 0.25i)$. Determine the SPL of the reflected wave. What fraction of incident energy in a plane wave is absorbed?

Solution:

Take the surface to be at $x = 0$

$$\frac{p'}{u} = \rho_0 c \frac{I+R}{I-R} = Z$$

$$\Rightarrow \left|\frac{R}{I}\right| = \left|\frac{Z-\rho_0 c}{Z+\rho_0 c}\right| = 0.8125$$

$\Rightarrow$ The SPL of the reflected wave

$$SPL = 160 + 20\log_{10}\left(\left|\frac{R}{I}\right|\right) = 158.2 \text{ dB}$$

the SPL of reflected wave is

$$SPL = 20\log_{10}\left(\frac{|R|}{2\times10^{-5}}\right) \text{ dB}$$

$$= 20\log_{10}\left(\frac{|R|}{|I|}\right) + 20\log_{10}\left(\frac{|I|}{2\times10^{-5}}\right) \text{ dB}$$

$$\text{energy flux} = \overline{p'u} = \frac{1}{2}\mathrm{Re}(p'u^*) = \frac{1}{2\rho_0 c}\mathrm{Re}\left[(Ie^{-ikx} + Re^{ikx})(I^*e^{ikx} - R^*e^{-ikx})\right]$$

$$\downarrow \quad (RI^*e^{ikx})^* = R^*Ie^{-2ikx} \longrightarrow RI^*e^{2ikx} - IR^*e^{-2ikx}$$

purely imaginary

$$\text{energy flux} = \frac{1}{2\rho_0 c}\left(|I|^2 - |R|^2\right)$$

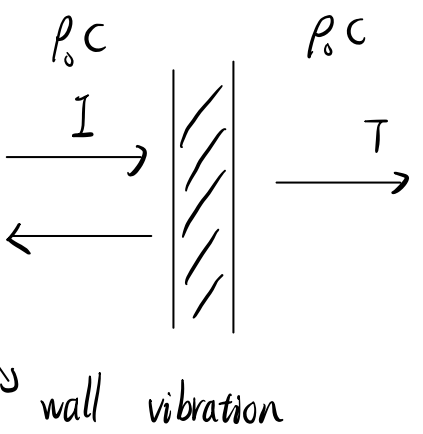
$RI^* - IR^*$ 是纯虚数

Incident energy flux: $\frac{|I|^2}{2\rho_0 c}$ Reflected energy flux: $\frac{|R|^2}{2\rho_0 c}$

$\Rightarrow$ Fraction of incident energy absorbed

$$\alpha = 1 - \left|\frac{R}{I}\right|^2 = 1 - 0.8125^2 = 0.34$$

4.2 Sound propagation through vibrating walls



Consider a thin wall at $x=0$ in a homogeneous fluid with ρ_0, c

A plane wave normally incident on the wall

$$\begin{cases} p_1'(x,t) = Ie^{i\omega(t-x/c)} + Re^{i\omega(t+x/c)}, & x<0 \\ p_2'(x,t) = Te^{i\omega(t-x/c)}, & x>0 \end{cases}$$

$\Delta p'$ across the wall $\longrightarrow$ wall vibrates at $u = Ve^{i\omega t}$

$$\underbrace{(I+R-T)e^{i\omega t}}_{\text{net force}} = \underbrace{m}_{\text{mass per unit aera}} \frac{\partial u}{\partial t} \quad (\text{식})$$

$$\Rightarrow \quad I+R-T = mi\omega V$$

Continuity of velocity at $x=0$

$$V = \frac{I-R}{\rho_0 c} = \frac{T}{\rho_0 c}$$

$$\Rightarrow \quad R = \frac{i\omega m I}{2\rho_0 c + i\omega m} \qquad T = \frac{2\rho_0 c I}{2\rho_0 c + i\omega m}$$

$$V = \frac{2I}{2\rho_0 c + i\omega m}$$

- $\dfrac{p'}{u} = i\omega m + \rho_0 c$: surface impedance with fluid loading

The surface pressure must drive both the wall motion and the

fluid on the right

- Energy conservation : $|R|^2 + |T|^2 = |I|^2$ (i.e. $\overline{p'u}$ = constant)

  net force $\sim i\omega V$ out of phase

Energy transmission coefficient

$$\left|\frac{T}{I}\right|^2 = \frac{4\rho_0^2 c^2}{4\rho_0^2 c^2 + \omega^2 m^2}$$

- For massive walls or high-freq waves, $\omega m \gg \rho_0 c$, $|T/I| \approx 0$, $|R/I| \approx 1$

  → very little high-freq waves transmitted through the wall

  (or when $\rho_0 \times 2\pi c/\omega \ll m$ i.e. mass of air within one λ of the wall per unit area $\ll$ wall mass per unit area)

- For low-freq. waves $\omega m \ll \rho_0 c$, $|T/I| \approx 1$, $|R/I| \approx 0$

Example 4.3 Typical values for wall attenuation

wall mass $m = 50\ kg/m^2$ $\rho_0 = 1.2\ kg/m^3$ $c = 340\ m/s$

($\rho_0 c = 408\ kg \cdot m^{-2} \cdot s^{-1}$)

Solution :

$$TL = 10 \log_{10}\left(\left(\frac{T}{I}\right)^2\right) = 10 \log_{10}\left(\frac{4\rho_0^2 c^2}{4\rho_0^2 c^2 + \omega^2 m^2}\right)$$

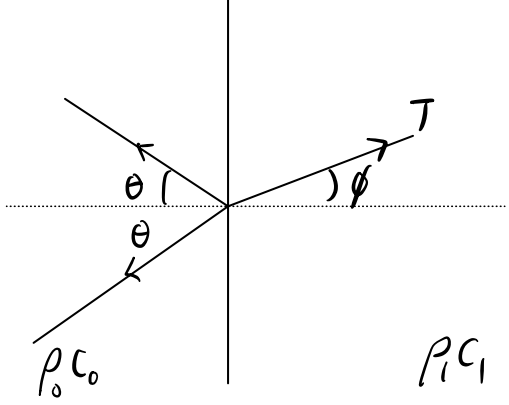
For $10\ Hz$ $\Rightarrow$ $TL = 12\ dB$

For $1\ kHz$ $\Rightarrow$ $TL = 50\ dB$

4.3 Sound refraction at interfaces of fluid discontinuity

In $x<0$

$$p_1'(x,y,t) = I e^{i\omega(t - x\cos\theta/c_0 - y\sin\theta/c_0)} + R e^{i\omega(t + x\cos\theta/c_0 - y\sin\theta/c_0)}$$



In $x>0$

$$p_2'(x,y,t) = T e^{i\omega(t - x\cos\phi/c_1 - y\sin\phi/c_1)}$$

Continuous p' across the interface at $x=0$

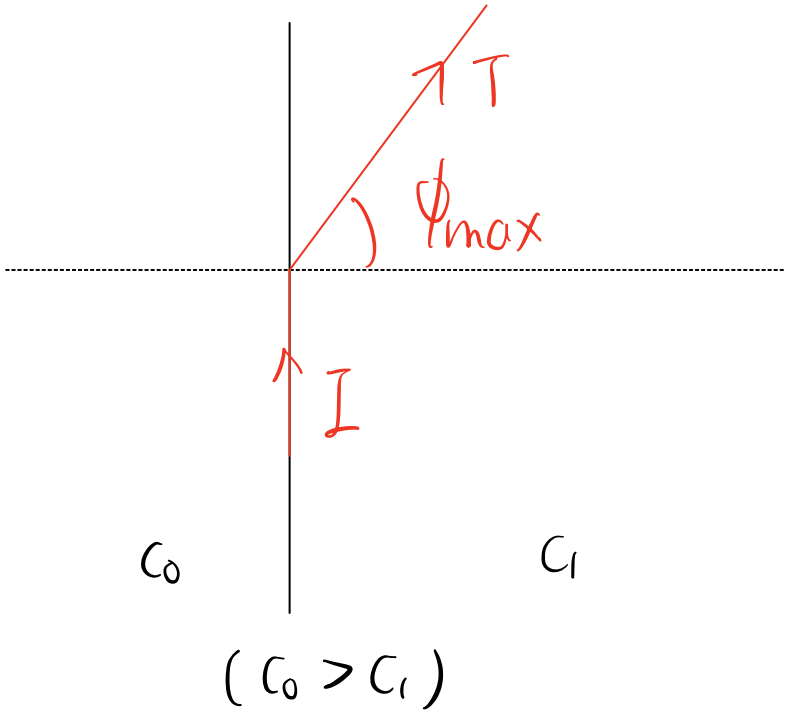
$$I e^{-i\omega y\sin\theta/c_0} + R e^{-i\omega y\sin\theta/c_0} = T e^{-i\omega y\sin\phi/c_1}$$

For this eq. to be true all along the interface (i.e. for all y)

$\Rightarrow$ Snell's law ✡✡✡

$$\frac{\sin\theta}{c_0} = \frac{\sin\phi}{c_1}$$

- If $c_0 > c_1$, then $\theta > \phi$, transmitted waves bent towards the normal $\Rightarrow$ $\phi_{max} = \sin^{-1}(c_1/c_0)$ for grazing incident wave $(\theta = \pi/2)$



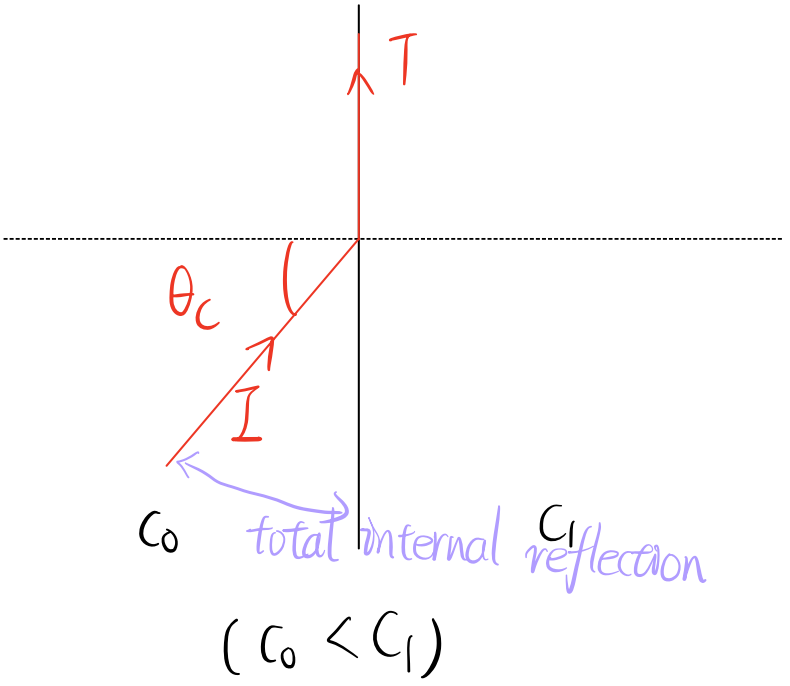
$(c_0 > c_1)$

- If $C_0 < C_1$, then $\theta < \phi$, transmitted wave bent away from the normal

$$\sin\phi = \sin\theta \cdot \frac{C_1}{C_0} \leq 1 \Rightarrow \text{critical angle } \theta_c = \sin^{-1}\left(\frac{C_0}{C_1}\right)$$

$\Rightarrow$ grazing transmitted wave ($\phi = \pi/2$)

For $\theta > \theta_c$, no transmitted wave $\Rightarrow$ total internal reflection



($C_0 < C_1$)

Continous p' at $x = 0$ $\Rightarrow$ $I + R = T$

Continous normal particle velocity $\Rightarrow$ $\frac{I}{\rho_0 c}\cos\theta - \frac{R}{\rho_0 c}\cos\theta = \frac{T}{\rho_1 c_1}\cos\phi$

The solution is

$$R = \frac{\frac{\rho_1 c_1}{\cos\phi} - \frac{\rho_0 c_0}{\cos\theta}}{\frac{\rho_1 c_1}{\cos\phi} + \frac{\rho_1 c_0}{\cos\theta}} I \qquad T = \frac{\frac{2\rho_1 c_1}{\cos\phi}}{\frac{\rho_1 c_1}{\cos\phi} + \frac{\rho_1 c_0}{\cos\theta}} I$$

- the ratio between the two oblique incidence wave impedances
  ↘ R, T
- Energy conservation: $\frac{|R|^2\cos\theta}{\rho_0 c_0} + \frac{|T|^2\cos\phi}{\rho_1 c_1} = \frac{|I|^2\cos\theta}{\rho_0 c_0}$ (i.e. $\overline{p' u_n} = \text{constant}$)

4.4 Evanescent wave

Total internal reflection $\longrightarrow$ no transmitted wave

Subsonic surface speed $\quad c_0/\sin\theta < c_1$

Transmitted acoustic pressure

$$p'(x, y, t) = Te^{i(\omega t - k_x x - k_y y)}$$

Continuous frequency and wave speed on $x=0$

$$k_y = \frac{\omega \sin\theta}{c_0} > \frac{\omega}{c_1}$$

Substitute p_t' in the 2D wave equation

$$\frac{1}{c^2}\frac{\partial^2 p_t'}{\partial t^2} = \frac{\partial^2 p_t'}{\partial x^2} + \frac{\partial^2 p_t'}{\partial y^2}$$

$$\Rightarrow \quad -\frac{\omega^2}{c_1^2} T = (-k_x^2 - k_y^2) T$$

Suppose that $T \neq 0$

$$k_x = \pm i\sqrt{k_y^2 - \frac{\omega^2}{c_1^2}} = \pm i\gamma$$

- Wavenumber in the x-direction k_x purely imaginary
- Positive root rejected as non-physical $\longleftarrow$ p_t' must be finite for large $+x$

$$\Rightarrow \quad p_t'(x, y, t) = Te^{i\omega t - \gamma x - ik_y y} \qquad x > 0$$

- Surface wave speed $\omega/k_y = c_0/\sin\theta < c_1 \longrightarrow$ Subsonic surface wave
- Evanescent wave: transmitted wave propagates subsonically along the interface but decays exponentially away from the surface.

- For example: intense motion on sea surface rapidly decays with depth, waves on ship submarine hull can not be heard by observers.

Solve the complete sound field in the case of evanescent wave

From the momentum $\Rightarrow$ normal particle velocity

$i\omega\rho_1 u_x = -\frac{\partial p'}{\partial x}$ $\quad\Rightarrow\quad u_x = \frac{\gamma}{i\omega\rho_1} T e^{i\omega t - \gamma x - ik_y y}$

continous normal velocity at $x=0$

$$\frac{I}{\rho_0 c_0}\cos\theta - \frac{R}{\rho_0 c_0}\cos\theta = -i\frac{\gamma T}{\rho_1 \omega}$$

continous p' at $x=0$, $\quad I+R=T$

$$\Rightarrow \quad \frac{R}{I} = \frac{\frac{\cos\theta}{\rho_0 c_0} + \frac{i\gamma}{\rho_1\omega}}{\frac{\cos\theta}{\rho_0 c_0} - \frac{i\gamma}{\rho_1\omega}} \qquad \frac{T}{I} = \frac{\frac{2\cos\theta}{\rho_0 c_0}}{\frac{\cos\theta}{\rho_0 c_0} - \frac{i\gamma}{\rho_1\omega}}$$

- Reflection coefficient $|R/I| = 1 \Rightarrow$ total internal reflection
- p_t' and u_x always out of phase $\Rightarrow$ $\overline{p_t' u_x} = 0$

  i.e. no energy transported away from the surface

Example 4.4 The time-average energy flux in an evanescent wave is precisely zero.

Solution:

$$\phi(x,y,t) = e^{i\omega t - ik_y y - \gamma x}$$

$$\Rightarrow \quad u(x,y,t) = \nabla\phi = (-\gamma, -ik_y)\, e^{i\omega t - ik_y y - \gamma x}$$

$$p'(x,y,t) = -\rho_1 \frac{\partial \phi}{\partial t} = -i\rho_1 \omega\, e^{i\omega t - ik_y y - \gamma x}$$

$$I_x = \overline{p' u_x} = \frac{1}{2} \mathrm{Re}\,(p' u_x^*)$$

$$= \frac{1}{2} \mathrm{Re}\left[(-i\rho_1 \omega e^{i\omega t - ik_y y - \gamma x})(-\gamma e^{i\omega t - ik_y y - \gamma x}) \right]$$

$$= \frac{1}{2} \mathrm{Re}\,(i\rho_0 \omega \gamma e^{-2\gamma x})$$

$$= 0$$

In y-direction

$$I_y = \overline{p' u_y} = \frac{1}{2} \mathrm{Re}\,(p' u_y^*)$$

$$= \frac{1}{2} \mathrm{Re}\left[(-i\rho_1 \omega\, e^{i\omega t - ik_y y - \gamma x})(-ik_y e^{i\omega t - ik_y y - \gamma x}) \right]$$

$$= \frac{1}{2} \mathrm{Re}\,(-\rho_1 \omega k_y e^{-2\gamma x})$$

$$\neq 0$$

$\Rightarrow$ Time-average energy flux $I_x = 0$, $I_y \neq 0$

Summary of Chapter 4

You should know

How to determine the wave amplitudes (reflected and transmitted) and energy flux by applying appropriate B.C.s at the interface (rigid, limp, surface impedance, etc)

The absorption coefficient $\alpha = 1 - \left|\frac{R}{I}\right|^2 - \left|\frac{T}{I}\right|^2$

Energy transmission coefficient $\left|\frac{T}{I}\right|^2 = \frac{4\rho_0^2 c^2}{4\rho_0^2 c^2 + \omega^2 m^2}$

Snell's law $\frac{\sin\theta}{c_0} = \frac{\sin\phi}{c_1}$

If $c_0 < c_1$, total reflection occur when $\theta > \theta_c = \sin^{-1}(c_0/c_1)$

In the case of evanescent wave, transmitted wave decays exponentially away from the interface with subsonic surface wave and zero energy flux $\overline{p'u} = 0$

5 Ray Theory

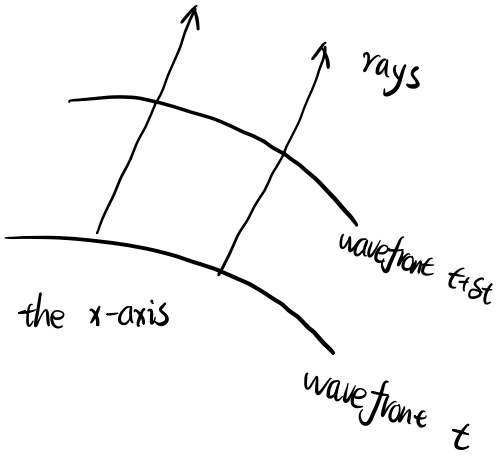
5.1 The ray theory equations

Consider situations in which c varies continously, e.g. in oceans or atmosphere

Ray theory: a mathematical technique for high-freq sound wave

- L: the typical length scale over which medium properties vary (e.g. ρ, c)

- If $\lambda \ll L$ (i.e. high-freq waves) $\rightarrow$ locally plane wave with direction of propagation and amplitude slowly varying as it propagates.
- Rays : curves always normal to the wavefronts and along the propagation direction



Direction variation

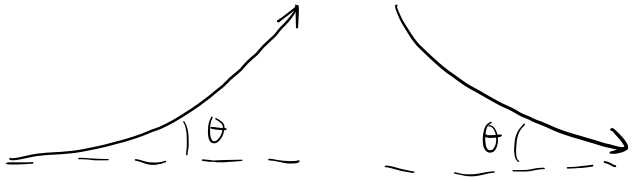
Suppose that $c = c(x)$,

θ : the angle between the ray and the x-axis

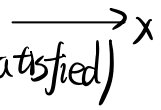
Snell's law $\rightarrow$ continuous medium

$$\frac{\sin\theta}{c(x)} = \text{constant}$$

- $c \uparrow \rightarrow \theta \uparrow \Rightarrow$ a ray travelling into region of $\uparrow c$ is bent away from the direction of $c \uparrow \Rightarrow$ eventually be reflected back
- $c \downarrow \rightarrow \theta \uparrow \Rightarrow$ a ray travelling into a region of $c \downarrow$ is bent towards the x-axis



- No sound travels into regions where $c > c_0/\sin\theta_0$, c_0, θ_0 at any point along the ray ($c \leq c_0/\sin\theta$ always satisfied)



Let the ray path be $y = y(x) \Rightarrow y' = \frac{dy}{dx} = \tan\theta$

$$\because \sin^2\theta = \frac{\tan^2\theta}{1+\tan^2\theta} \Rightarrow \frac{y'}{c(x)\,(1+y^2)^{1/2}} = \text{constant}$$

Example 5.1 Linear sound speed variation $c(x) = \alpha x + \beta$, trace the

ray path which starts from $(0,0)$ at angle θ_0

Solution:

B.C. : At $(0,0)$, $\theta = \theta_0$, $c(0) = \beta$

From the Snell's law

$$\frac{y'}{(1+y'^2)^{1/2}} \cdot \frac{1}{\alpha x + \beta} = \frac{\sin\theta_0}{\beta}$$

$$\Rightarrow \quad y'^2 = \frac{\sin^2\theta_0 (\alpha x + \beta)^2}{\beta^2 - \sin^2\theta_0 (\alpha x + \beta)^2}$$

$$\Rightarrow \quad y' = \frac{dy}{dx} = \pm \frac{\sin\theta_0 (\alpha x + \beta)}{\sqrt{\beta^2 - \sin^2\theta_0 (\alpha x + \beta)^2}}$$

$$\Rightarrow \quad y = \mp \frac{1}{\alpha \sin\theta_0} \sqrt{\beta^2 - \sin^2\theta_0 (\alpha x + \beta)^2} + C$$

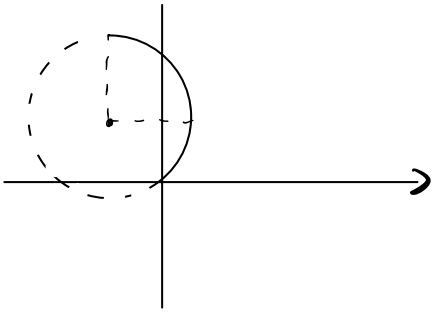
At $x=0$, $y=0$ $\Rightarrow$ $C = \pm \frac{\beta\cos\theta_0}{\alpha\sin\theta_0}$

$$\Rightarrow \quad y = \frac{1}{\alpha \sin\theta_0} \left(\mp \sqrt{\beta^2 - \sin^2\theta_0 (\alpha x + \beta)^2} \pm \beta\cos\theta_0 \right)$$

Rearrange and square

$$\left(x + \frac{\beta}{\alpha}\right)^2 + \left(y - \frac{\beta}{\alpha\tan\theta_0}\right)^2 = \frac{\beta^2}{\alpha^2 \sin^2\theta_0}$$

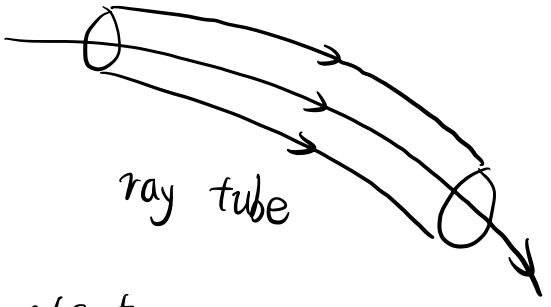
- radius $\beta/(\alpha\sin\theta_0)$ centre $(-\beta/\alpha, \cot\theta_0 \beta/\alpha)$
- $x_{max} = (\csc\theta_0 - 1)\beta/\alpha$
- All rays (except $\theta_0 = 0$) are eventually reflected backwards



Amplitude variation

Consider a ray tube formed by adjacent rays

- Ray tube always parallel to the direction of propagation



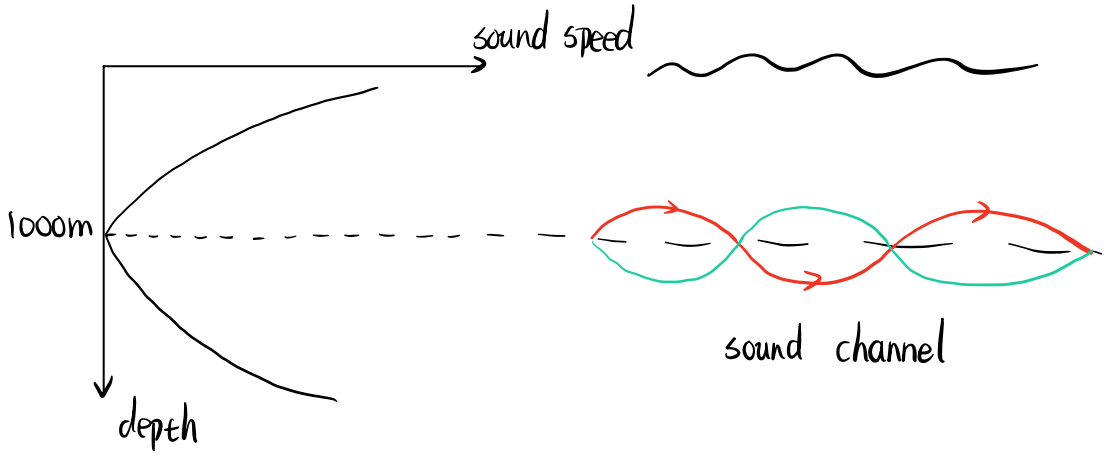
- Acoustic energy conserved → $\overline{p'u}\,A = \text{constant}$
- Locally plane wave ($\lambda \ll L$)

$$\frac{\overline{p'^2}\,A}{\rho_0 c} = \text{constant}$$

- Diverging rays : $A\uparrow \rightarrow p'\downarrow$
- Converging rays : $A\downarrow \rightarrow p'\uparrow$ (rays focused)

5.2 Underwater sound propagation

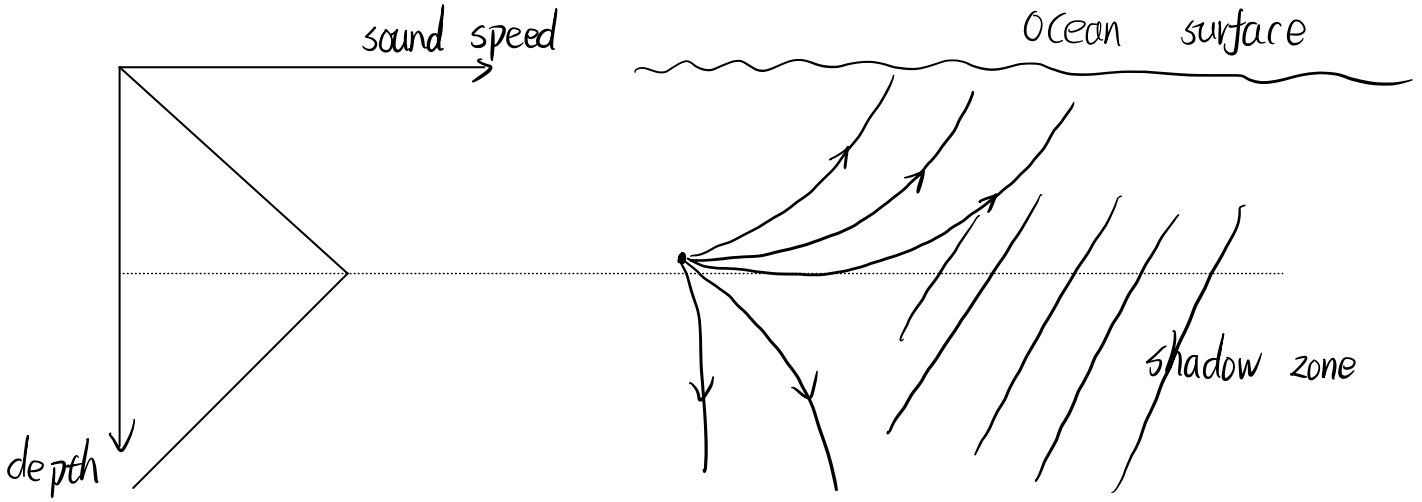
- From ocean surface $T\downarrow$ with $d\uparrow \Rightarrow c\downarrow$
- At $d \approx 1000m$ $T \sim$ constant (isothermal)
- $d > 1000m$, $p\uparrow$ with $d\uparrow \Rightarrow c\uparrow$



- sound channel : rays trapped within certain levels in the ocean
- sound can only spread out in a circle, rather than over a sphere (homogeneous medium) ⟶ sound decays more slowly.

An abnormal profile of c with d

- As d↑, c↑ then ↓ hnearly ⟶ c_{max}
- In a fjord where cold water enters the sea at a high level.



For a sound source near the position of c_{max}

- Ray ↑ → c↓

⇒ bent towards depth direction
i.e. reflected upwards

- Ray ↓ → c↓ → refracted downwards more steeply
- shadow zone (silence zone) : No rays penetrate

5.3 Sound propagating

During the day time :

- $h \uparrow \Rightarrow T \downarrow \Rightarrow c \downarrow$
- Rays are bent up $\Rightarrow$ Sound passes over a distant observer

At night (ground cools more quickly than air)

- $h \uparrow \Rightarrow T \uparrow \Rightarrow c \uparrow$
- Rays are bent back to the ground $\Rightarrow$ sound is heard more easily

$\Rightarrow$ Sound travels further at night.

Uniform medium

medium uniform $\Longrightarrow$ rays are straight lines

Example 5.2 Sound source in a uniform medium in 2D or 3D space

2D

Ray 'tube' formed by adjacent rays $\delta\theta$ apart

ray tube aera $\quad A = \delta\theta \cdot r$

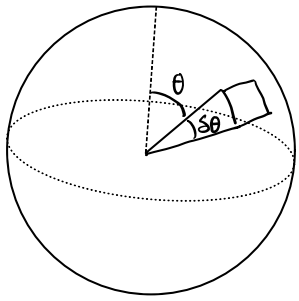
$$\Rightarrow \frac{\overline{p'^2}}{\rho_0 c} r\,\delta\theta = \text{constant}$$

$$\Rightarrow \quad p'_{rms} \propto \frac{1}{\sqrt{r}}$$

3D

Need to consider a small solid angle

$$A = r\delta\theta \cdot r\sin\theta\,\delta\phi = r^2 \sin\theta\,\delta\theta\,\delta\phi$$



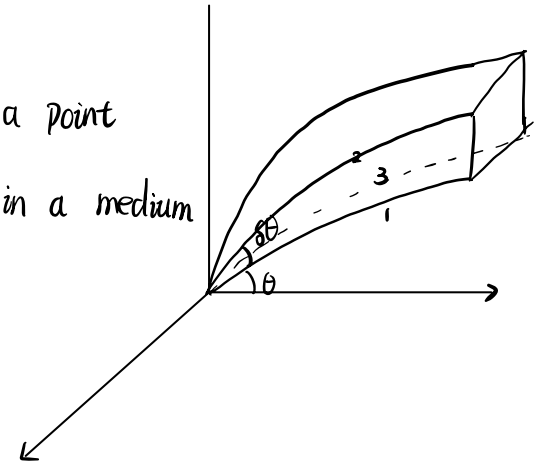
$$\Rightarrow \quad \frac{\overline{p'^2}}{\rho_0 c} r^2 \sin\theta\,\delta\theta\,\delta\phi = \text{constant}$$

$$\Rightarrow \quad p'_{rms} \propto \frac{1}{r}$$

(Recall for spherical wave $p' = \frac{f(r-ct)}{r}$)

Nonuniform medium

Consider a ray tube launched by a point source at the origin and travelling in a medium that varies only in the x-axis



Rays 1 and 2 make angles with the x-axis of θ and $\theta+d\theta$

Ray 3 makes angle $\delta\phi$ to ray 1 (rays 1 and 3 are symmetriced about the x-axis)

When the rays have travelled a distance X along the axis:

Rays 1 and 2 are apart by a, while the distance b between rays 1 and 3 along the z-axis is the same as in uniform medium

$$b = X\,\delta\phi \longrightarrow a\ ?$$

Example 5.3 Derive the pressure variation above a source in a linearly stratified atmosphere with $c(x) = \alpha x + \beta$, start from (0,0) with θ_0 to the x-axis

Solution:

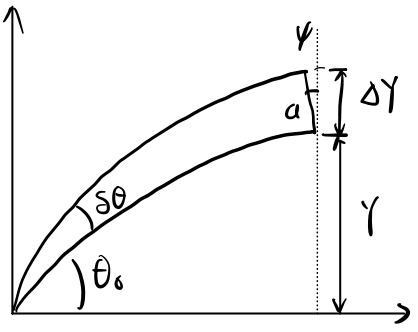
From Example 5.1 $\Rightarrow$ rays are in circle

$$\left(X+\frac{\beta}{\alpha}\right)^2+\left(Y-\frac{\beta}{\alpha}\cot\theta_0\right)^2=\frac{\beta^2}{\alpha^2}\csc^2\theta_0$$

$$\frac{\partial}{\partial\theta_0}\downarrow$$

$$\frac{d\cot\theta}{d\theta}=-\csc^2\theta$$

$$\frac{d\csc\theta}{d\theta}=-\csc\theta\cot\theta$$



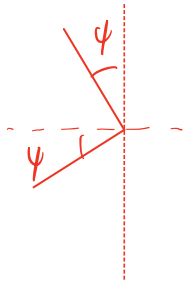
$$\frac{dY}{d\theta_0}\left(Y-\frac{\beta}{\alpha}\cot\theta_0\right)+\frac{\beta}{\alpha}Y\csc^2\theta_0=0$$

$$\downarrow$$

$$\frac{dY}{d\theta_0}=\frac{\beta Y\csc^2\theta_0}{\beta\cot\theta_0-\alpha Y}$$

$$\downarrow$$

$$\Delta Y=\frac{dY}{d\theta_0}\delta\theta$$



Snell's Law $\quad\frac{\sin\psi}{c(x)}=\frac{\sin\theta_0}{c(0)}\quad\Rightarrow\quad\psi=\sin^{-1}\left[\left(1+\frac{\alpha}{\beta}X\right)\sin\theta_0\right]$

$$\Rightarrow\quad a=\frac{dY}{d\theta_0}\delta\theta\cos\psi\qquad\leftarrow\quad a=\Delta Y\cos\psi$$

The ray tube area $\quad ab=\frac{dY}{d\theta_0}X\delta\theta\delta\phi\cos\psi$

The variation of P'_{rms} to be worked out from energy conservation

$$\frac{\overline{P'^2}}{\rho_0 c}\frac{dY}{d\theta_0}X\delta\theta\delta\phi\cos\phi=\text{constant}$$

Consider a sphere centred at the source with small radius ε and constant c

within it. When the ray tube formed by $(\delta\theta, \delta\phi)$ reaches the sphere surface:

it has travelled a small distance $\varepsilon\cos\theta_0$ along the x-axis.

the distance between rays 1 and 3 along the z-axis, $b = \varepsilon\cos\theta_0\delta\phi$

The energy flux per unit area on the surface is $W/4\pi\varepsilon^2$

$\Rightarrow$ The energy flux through the ray tube

$$\frac{W}{4\pi\varepsilon^2}\,\varepsilon\,\delta\theta\cdot\varepsilon\cos\theta_0\,\delta\phi = \frac{W\cos\theta_0\,\delta\theta\,\delta\phi}{4\pi}$$

Hence $\delta\theta\,\delta\phi$ will be cancelled.

Summary of Chapter 5

You should know:

For continuous media, high-frequency sound waves ($\lambda \ll L$) can be regarded as rays whose direction of propagation and amplitude slowly vary as they propagates.

A ray tube is parallel to the direction of propagation, within which acoustic energy is conserved.

The propagation direction of a ray follows Snell's law $\frac{\sin\theta}{c(x)} = \text{constant}$

Within a ray tube, the amplitude of sound wave varies as $\frac{\overline{p'^2}A}{\rho_0 c} = \text{constant}$

How to determine the ray path using the profile of linear sound speed variation underwater and in the atmosphere

6. Sources of sound

Question: What sound field can occupy all of space?

(i) in the absence of boundaries

(ii) no incoming field from ∞

1D field $f(x, \pm ct)$ $\quad x \longleftarrow$ from $-\infty$ to ∞

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{f(t-r/c)}{r} = 0$$

$\Rightarrow$ $\dfrac{f(t-r/c)}{r}$ $\rightarrow$ a solution of the homogeneous wave equation

$\nabla^2 p' = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial p'}{\partial r}\right)$ meaningless at $r=0$

i.e. $\dfrac{f(t-r/c)}{r}$ has a singularity at the origin

In section 2.1 : near a singularity $(r \to 0)$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{f(t-r/c)}{r} = 0 \quad \Rightarrow \quad \text{Laplace equation}$$

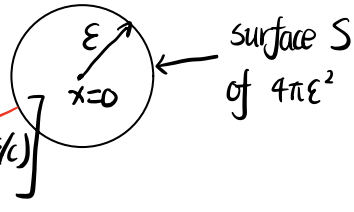
Integral over a vanishing small volume V surrounding the origin $x=0$ is finite.

$$\int_V \nabla^2\left[\frac{f(t-r/c)}{r}\right] dV = \int_S \frac{\partial}{\partial n}\left[\frac{f(t-r/c)}{r}\right] dS$$

divergence theorem

$$= 4\pi\varepsilon^2\left[-\frac{f(t-\varepsilon/c)}{\varepsilon^2} + \frac{1}{\varepsilon}\frac{\partial f}{\partial r}(t-\varepsilon/c)\right]$$

$$= -4\pi f(t) \quad \text{as } \varepsilon \to 0$$



3D δ-function $\delta(x)$: integral over a vanishing V surrounding $x=0$ is unity.

$$\int_V \delta(x)\, dV = \begin{cases} 1 & , \text{ if } V \text{ includes } x=0 \\ 0 & , \text{ otherwise} \end{cases}$$

$$\int_V \left[\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{f(t-r/c)}{r}\right] dV \approx 4\pi f(t)$$

$$\int_V 4\pi f(t)\,\delta(x)\, dV$$

$\times 4\pi f(t)$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{f(t-r/c)}{r} = 4\pi f(t)\,\delta(x)$$

- A building block ⟶ the solution of practical problems of sound generation
- Homogeneous ⟵ only way is to set $f(t)=0$
- Silence is the only homogeneous sound field in unbounded space!

Sound p': weak disturbances that conform with the wave eq.

$$\underbrace{\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0}_{\text{source } q(t)}$$

The only solution to this homogeneous eq. in unbounded space is $p'=0$

↳ no sound $p'=0$ $\Longleftrightarrow$ no source q ?

Arbitrary definition

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = q(x,t)$$

- $q=0$ in the sound field ⟵ p' satisfies the homogeneous wave eq.

- $q \neq 0$ in the source region

✗ sound field $\longrightarrow$ sound source (non-unique)

由声场可以推出声源，但结果并不唯一

Consider the sound fields $p'+q$ and $p'+f(q)$

- $p' = p'+q = p'+f(q) \quad \Leftarrow \quad q=0,\ f(0)=0$ in the sound field
- sources of the three sound fields are different.

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = q$$

$$\frac{1}{c^2}\frac{\partial^2 (p'+q)}{\partial t^2} - \nabla^2 (p'+q) = q + \frac{1}{c^2}\frac{\partial^2 q}{\partial t^2} - \nabla^2 q$$

$$\frac{1}{c^2}\frac{\partial^2 [p'+f(q)]}{\partial t^2} - \nabla^2 [p'+f(q)] = q + \frac{1}{c^2}\frac{\partial^2 f(q)}{\partial t^2} - \nabla^2 f(q)$$

✗ source field q $\rightarrow$ sound field (unique)

write the source in Eq $\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = q(x,t)$ as

$$q(x,t) = \int_V q(y,t)\,\delta(x-y)\,d^3y$$

Apply $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{f(t-r/c)}{r} = 4\pi f(t)\,\delta(x)$

$$\Rightarrow \quad \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{q(y,t-r/c)}{r} = 4\pi\, q(y,t)\,\delta(x-y)$$

$r \rightarrow |x-y|$

Integrate y

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\int_V \frac{q(y,t-|x-y|/c)}{|x-y|}\,d^3y = 4\pi\, q(x,t)$$

$$\Rightarrow \quad p'(x,t) = \int_V \frac{q(y, t-|x-y|/c)}{4\pi|x-y|}\, d^3y$$

在x处的观察者得到的来自y处声源辐射的声压

- Source element at y $\xrightarrow{\text{radiating sound}}$ observer at x
- Sound arriving at observer t $\longleftarrow$ emitted from source at time $\underbrace{t-|x-y|/c}_{\text{retarded time}}$
- When the source region V is compact, and $|y| \ll |x|$

$$p'(x,t) = \frac{Q(t-|x|/c)}{4\pi|x|}$$

where $Q(t) = \int_V q(y,t)\, d^3y$

6.2 The monopole source

6.2.1 Injection of mass at a point (质量的注入)

- an idealised source
- the building block $\longrightarrow$ sound generation in a range of applications

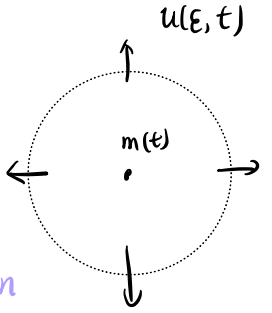
Suppose mass injected at the origin at a rate of $m(t)$

$\longrightarrow$ spherically symmetric sound field

velocity potential $\quad \phi(r,t) = \frac{f(t-r/c)}{r}$

$$\Rightarrow \quad u = \frac{\partial \phi}{\partial r} = -\frac{f(t-r/c)}{r^2} - \frac{\dot{f}(t-r/c)}{rc}$$

Consider a small sphere of radius ε



Conservation of mass flux

$4\pi\varepsilon^2 \rho_0 u(\varepsilon,t) = m(t)$ → rate of mass addition

mass flux through a sphere

$$4\pi\rho_0\left[-f(t-\varepsilon/c) - \frac{\varepsilon}{c}\dot{f}(t-\varepsilon/c)\right] \approx -4\pi\rho_0 f(t)$$

$$\longrightarrow \quad 4\pi\varepsilon^2\rho_0\left[-\frac{f(t-\varepsilon/c)}{\varepsilon^2}-\frac{\dot{f}(t-\varepsilon/c)}{\varepsilon c}\right]=m(t) \quad \text{as } \varepsilon\to 0$$

$$\Rightarrow \quad f(t)=-\frac{m(t)}{4\pi\rho_0}$$

$$\Longrightarrow \quad p'(t)=-\rho_0\frac{\partial\phi}{\partial t}=-\rho_0\frac{\dot{f}(t-r/c)}{r}=\frac{\dot{m}(t-r/c)}{4\pi r}$$

monopole source strength 质量注入率的变化率

monopole point source

The inhomogeneous wave equation

Conservation of mass

$$\frac{\partial\rho'}{\partial t}+\rho_0\nabla\cdot v=m(t)\,\delta(x) \qquad \frac{\partial}{\partial t}$$

Momentum equation

$$\rho_0\frac{\partial v}{\partial t}+\nabla p'=0 \qquad \nabla\cdot$$

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2}-\nabla^2 p'=\dot{m}(t)\,\delta(x)$$

↓ Source at $x=0$

The solution is

$$p'(r,t)=\frac{\dot{m}(t-r/c)}{4\pi r}$$

In general, the solution to

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2}-\nabla^2\right)p'=Q(t)\,\delta(x)$$

is $\quad p'(x,t)=\dfrac{Q(t-r/c)}{4\pi r}, \quad r=|x|$

For a source of strength $Q(y,t)$ at y, observer at x

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2}-\nabla_x^2\right)p'=Q(y,t)\,\delta(x-y)$$

The solution is

$$p'(x,t) = \frac{Q(y, t - |x-y|/c)}{4\pi |x-y|}$$

6.2.2 Monopole Source distribution

Definition

If $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = q(x,t)$

and $\int q(x,t)\, dV = Q(t) \neq 0$

$\Rightarrow$ $q(x,t)$: a monopole or simple source distribution

Derivation of solution

Decompose $q(x,t)$ into a superposition of point sources

$$q(x,t) = \int q(y,t)\, \delta(x-y)\, d^3y$$

- $\delta(x-y) = 0$ if $x \neq y$
- $\int \delta(x-y)\, d^3y = 1$

Superposition of individual sound fields $\longrightarrow$ total sound field

$$\Rightarrow \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla_x^2\right) \frac{q(y, t-|x-y|/c)}{4\pi|x-y|} = q(y,t)\,\delta(x-y)$$

Integrate over y

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla_x^2\right) \int \frac{q(y, t-|x-y|/c)}{4\pi|x-y|}\, d^3y = \int q(y,t)\,\delta(x-y)\, d^3y = q(x,t)$$

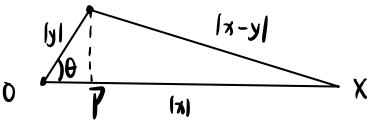
The solution to $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = q(x,t)$ is

$$p'(x,t) = \int \frac{q(y, t-|x-y|/c)}{4\pi|x-y|}\, d^3y$$

重要的是从源项到声压场的推导过程

The far-field form

Choose the origin to be within the source region



If $|x| \gg |y|$,

$$|x-y| \approx \overline{Ox} - OP = |x| - |y|\cos\theta = |x| - \frac{x\cdot y}{|x|}$$

Alternatively

$$|x-y| = [(x-y)\cdot(x-y)]^{1/2} = [|x|^2 - 2x\cdot y + |y|^2]^{1/2}$$

$$= |x|\left[1 - \frac{2x\cdot y}{|x|^2} + \frac{|y|^2}{|x|^2}\right]^{1/2}$$

$$= |x|\left[1 - \frac{1}{2}\frac{2x\cdot y}{|x|^2} + O\left(\frac{|y|}{|x|}\right)^2\right]$$

$$= |x| - \frac{x\cdot y}{|x|}$$

$(1+\varepsilon)^n = 1 + n\varepsilon + \varepsilon^2$

$$\frac{1}{|x-y|} \approx \frac{1}{|x|}\left(1 - \frac{x\cdot y}{|x|}\right)^{-1}$$

$(1+\varepsilon)^n = 1 + n\varepsilon + \varepsilon^2,\ \varepsilon \to 0$

$$= \frac{1}{|x|}\left[1 + \frac{x\cdot y}{|x|^2} + O\left(\frac{|y|}{|x|}\right)^2\right]$$

$$= \frac{1}{|x|} + O\left(\frac{|y|}{|x|^2}\right)$$

$$\Rightarrow \frac{1}{|x-y|} \approx \frac{1}{|x|}$$

$$p'(x,t) = \int \frac{q(y, t-|x-y|/c)}{4\pi|x-y|}\, d^3y$$

Hence the solution (6.17) simplifies to

$$p'(x,t) = \frac{1}{4\pi|x|}\int q(y,\ t - \frac{|x|}{c} + \frac{x\cdot y}{|x|c})\, d^3y \quad \text{for } |x| \gg |y|$$

远场形式

Compact source

In addition, if the source is compact, i.e. source size $l \ll \lambda$ or $kl \ll 1$

Taylor expansion

$$f(x_0 + x) = f(x_0) + xf'(x_0) + \cdots$$

$$\Rightarrow \quad q\left(y, \underbrace{t-\frac{|x|}{c}}_{t_0}+\underbrace{\frac{x\cdot y}{|x|c}}_{t}\right)=q\left(y, t-\frac{|x|}{c}\right)+\frac{x\cdot y}{|x|c}\dot{q}\left(y, t-\frac{|x|}{c}\right)+\cdots$$

$\dot{q}/q \sim w$, $|y| \sim l$

$$=q\left(y, t-\frac{|x|}{c}\right)\left[1+O\left(\frac{wl}{c}\right)\right]$$

Hence for small wl/c

$$p'(x,t)=\frac{1}{4\pi|x|}\int q\left(y, t-\frac{|x|}{c}\right)d^3y=\frac{Q(t-|x|/c)}{4\pi|x|}$$

for $|x| \gg |y|$ and compact source

where $Q(t)=\int q(y,t)\,d^3y$

对于紧致的单极子声源的分布的远场，相当于一个在原点的单极子点源

- The far-field of a compact monopole source = that of monopole point source at the origin.
- N_0 - compact source $\longrightarrow$ retarded time dependent on y
  $\rightarrow$ possibility of significant cancellation

Example 6.1 A distributed monopole source of freq. w has its strength distributed uniformly over a spherical volume of radius a. By considering the far field, show that there is no sound radiated from the source whenever its size is such that $\tan(ka)=ka$, where $k=w/c$

Solution.

$$\text{strength}=\begin{cases} q_0 e^{i\omega t} & \text{in } |y|<a \\ 0 & \text{in } |y|>a \end{cases}$$

$$p'(x,t)=\int_a \frac{q_0 e^{i\omega(t-|x-y|/c)}}{4\pi|x-y|}d^3y$$

for x in far field $\longrightarrow$ $=\frac{1}{4\pi|x|}q_0 e^{i\omega(t-|x|/c)}\int_{|y|<a} e^{i\omega x\cdot y/|x|c}\,d^3y$

$\frac{x \cdot y}{|x|} = \sigma\cos\theta$ and $d^3y = 2\pi\sigma\sin\theta\sigma\, d\theta d\sigma$

$$p'(x,t) = \frac{q_0 e^{i\omega(t-|x|/c)}}{4\pi|x|} 2\pi \int_{\sigma=0}^{a} \sigma^2 \left[\int_0^{\pi} e^{ik\sigma\cos\theta} \sin\theta\, d\theta\right] d\sigma$$

$$= \frac{q_0 e^{i\omega(t-|x|/c)}}{2|x|} \int_{\sigma=0}^{a} \sigma^2 \left(-\frac{1}{ik\sigma}\right) \left[e^{ik\sigma\cos\theta}\right]_0^{\pi} d\sigma$$

$$= \frac{q_0 e^{i\omega(t-|x|/c)}}{k|x|} \int_0^a \sigma \sin(k\sigma)\, d\sigma$$

$$= \frac{q_0 e^{i\omega(t-|x|/c)}}{k|x|} \frac{1}{k} \left[\left[-\sigma\cos(k\sigma)\right]_0^a + \int_0^a \cos(k\sigma)\, d\sigma \right]$$

$$= \frac{q_0 e^{i\omega(t-|x|/c)}}{k^2|x|} \left[-a\cos(ka) + \frac{1}{k}\sin(ka)\right]$$

$p' = 0$ when $a\cos(ka) = \frac{1}{k}\sin(ka) \Rightarrow \tan(ka) = ka$

$p'(x,t) = 0$ only for x in the far field ? N_0

Spherically symmetric pressure field

$p' = A e^{i(\omega t - kr)}/r$ outside the source region $|x| \le a$

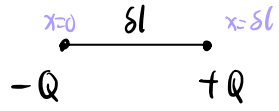
$\because A \equiv 0$ at frequency of $\tan(ka) = ka$

$\therefore p' = 0$ through $|x| \ge a$

- Source system is null. ← destructive interference
- Sound field inside source region does not vanish.
- Anti-sound : additional source → cancellation

6.3 The dipole source

6.3.1 A point dipole



Consider two point monopoles of strength Q and $-Q$ at a distance δl apart

Total strength $Q_d = -Q(t)\,[\delta(x) - \delta(x-\delta l)]$

Taylor expansion for small δl $\qquad \delta(x-\delta l) = \delta(x) - [\text{grad}\,\delta(x)]\cdot\delta l$

$$\delta(x-\delta l) = \delta(x) - \delta l_1 \frac{\partial}{\partial x_1}\delta(x) - \delta l_2 \frac{\partial}{\partial x_2}\delta(x) - \delta l_3 \frac{\partial}{\partial x_3}\delta(x) + O(\delta l)^2$$

$$\Rightarrow Q_d = -Q(t)\left[\delta l_1 \frac{\partial}{\partial x_1}\delta(x) + \delta l_2 \frac{\partial}{\partial x_2}\delta(x) + \delta l_3 \frac{\partial}{\partial x_3}\delta(x)\right]$$

$$= -\nabla\cdot[Q(t)\,\delta l\,\delta(x)]$$

$$= -\nabla\cdot[F(t)\,\delta(x)] \quad \text{as } \delta l \to 0$$

$\delta l\cdot(\text{grad}\,\delta(x)) = \text{div}(\delta l\,\delta(x)) - \delta(x)\,\text{div}\,\delta l$ (the last term $= 0$)

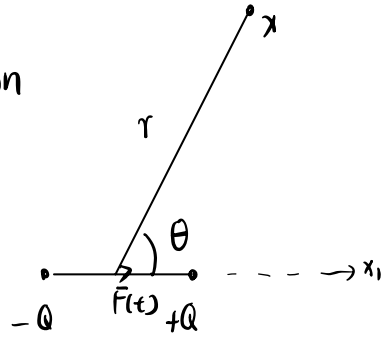
- The point dipole consist of two equal and opposite monopoles
- In the limit of large strength Q and small separation δl

If $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = -\nabla\cdot[F(t)\,\delta(x)]$

this source is a point dipole and $F(t)$ is called the dipole strength $Q(t)\delta l$

$F(t)$ ↓ force applied to the fluid

Solution



偶极子的指向性：从负指向正

Choose axes so that $F(t) = (F(t), 0, 0)$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = -\frac{\partial}{\partial x_1}[F(t)\,\delta(x)]$$

we already know

$$\frac{\partial}{\partial x_1} \longrightarrow \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{F(t-r/c)}{4\pi r} = F(t)\,\delta(x) \quad , r = |x|$$

$$\Rightarrow \quad \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{\partial}{\partial x_1}\left[\frac{F(t-r/c)}{4\pi r}\right] = \frac{\partial}{\partial x_1}\left[F(t)\,\delta(x)\right]$$

The solution is

$$p'(x,t) = -\frac{\partial}{\partial x_1}\left[\frac{F(t-r/c)}{4\pi r}\right]$$

Write $X = t - r/c \longrightarrow F(t-r/c) = F(X)$

$$\begin{cases} \frac{\partial F}{\partial r} = \frac{\partial X}{\partial r}\frac{dF}{dX} = -\frac{1}{c}\frac{dF}{dX} \\ \frac{\partial F}{\partial t} = \frac{\partial X}{\partial t}\frac{dF}{dX} = \frac{dF}{dX} \end{cases} \quad \therefore \quad \frac{\partial F}{\partial r} = -\frac{1}{c}\frac{\partial F}{\partial t} \; ※$$

$$p'(x,t) = -\frac{\partial r}{\partial x_1}\frac{d}{dr}\left[\frac{F(t-r/c)}{4\pi r}\right]$$

$$= \frac{\partial r}{\partial x_1}\left[\frac{1}{4\pi r^2}F(t-r/c) + \frac{1}{4\pi rc}\frac{\partial F}{\partial t}(t-r/c)\right]$$

$$r = (x_1^2 + x_2^2 + x_3^2)^{\frac{1}{2}}$$

$$\downarrow$$

$$\frac{\partial r}{\partial x_1} = \frac{x_1}{r} = \cos\theta \; ※$$

$$= \frac{\cos\theta}{4\pi}\left[\frac{1}{r^2}F(t-r/c) + \frac{1}{rc}\frac{\partial F}{\partial t}(t-r/c)\right]$$

near field $(kr \ll 1)$ $\sim O\left(\frac{F}{r^2}\right)$

far field $(kr \gg 1) \sim O\left(\frac{wF}{rc}\right)$

$$\frac{\frac{1}{lc}Fw}{F/\varepsilon^2} = \frac{kF/l}{F/\varepsilon^2}$$

Hence for a point dipole

远场最大 / 近场最大 $= \dfrac{\text{Max. } p' \text{ at } r = l \; (\gg k^{-1})}{\text{Max. } p' \text{ at } r = \varepsilon \; (\ll k^{-1})} = \dfrac{kF/l}{F/\varepsilon^2} = k\varepsilon \cdot \dfrac{\varepsilon}{l} \ll \dfrac{\varepsilon}{l}$

Recall for a point monopole $p'(x,t) = \dfrac{Q(t-r/c)}{4\pi r}$

$$\frac{\text{Max. } p' \text{ at } r=l \ (\gg k^{-1})}{\text{Max. } p' \text{ at } r=\varepsilon \ (\ll k^{-1})} = \frac{\varepsilon}{l}$$

近场压力脉动相同的条件下，dipole 幅射到远场的效率更低

- Dipoles are less effective than monopoles at turning near-field pressure into far-field sound by a factor of $k\varepsilon$, despite the intense near field

6.3.2 Dipole source distribution

Definition

If $$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) p' = -\nabla \cdot \mathbf{f}(\mathbf{x},t) = -\frac{\partial f_i}{\partial x_i}(\mathbf{x},t)$$

and $$\int f(\mathbf{x},t)\, dV = F(t) \neq 0$$

$\Rightarrow$ $f(\mathbf{x},t)$: a dipole distribution

A degeneration of a monopole distribution $q(\mathbf{x},t)$

Write $q(\mathbf{x},t) = -\nabla \cdot \mathbf{f}(\mathbf{x},t)$

$\downarrow$ $=0$ in sound field

$$Q(t) = \int_{\bar{V}} q(\mathbf{x},t)\, dV = -\int_{\bar{V}} \nabla \cdot \mathbf{f}\, dV = -\int_S f_n(\mathbf{x},t)\, dS = 0$$

- Volume $\bar{V}$ encloses source region, bounded by surface S on which $\mathbf{f}$ must vanish
- sum of source elements $\int_{\bar{V}} q\, dV = 0$ $\rightarrow$ destructive interference
- Non-zero net sound field $\leftarrow$ different retarded time

Solution

The solution to $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) S_i = f_i(\mathbf{x},t)$ is

$$\Rightarrow \quad S_i(\mathbf{x},t) = \int \frac{f_i(\mathbf{y}, t - |\mathbf{x}-\mathbf{y}|/c)}{4\pi|\mathbf{x}-\mathbf{y}|}\, d^3y$$

$$\Rightarrow \quad p'(x,t) = -\frac{\partial}{\partial x_i}\int \frac{f_i(y, t-|x-y|/c)}{4\pi|x-y|}\,d^3y$$

The far-field form

For $|x| >> |y|$ and the origin within source region

$$|x-y| \approx |x| - \frac{x\cdot y}{|x|} \qquad \frac{1}{|x-y|} \simeq \frac{1}{|x|}$$

Hence

$$p'(x,t) = -\frac{\partial}{\partial x_i}\left[\frac{1}{4\pi|x|}\int f_i\Big(y,\ \underbrace{t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}}_{\tau}\Big)\,d^3y\right]$$

$$\frac{\partial}{\partial x_i} = \frac{\partial \tau}{\partial x_i}\frac{\partial}{\partial \tau} = \frac{\partial |x|}{\partial x_i}\frac{\partial \tau}{\partial |x|}\frac{\partial}{\partial \tau}$$

$$= \frac{1}{4\pi|x|c}\frac{\partial|x|}{\partial x_i}\int \frac{\partial f_i}{\partial t}\Big(y, t-\frac{|x|}{c}-\frac{x\cdot y}{|x|c}\Big)\,d^3y + O\Big(\frac{1}{|x|^2}\Big)$$

$$|x| = (x_1^2+x_2^2+x_3^2)^{\frac{1}{2}}, \qquad \frac{\partial|x|}{\partial x_i} = \frac{x_i}{|x|}$$

$$\approx \frac{x_i}{4\pi|x|^2 c}\int \frac{\partial f_i}{\partial t}\Big(y, t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\Big)\,d^3y$$

Compact source

For $|x| >> |y|$ and a compact source region at the origin

$$p'(x,t) = \frac{x_i}{4\pi|x|^2 c}\frac{\partial}{\partial t}\int f_i\Big(y, t-\frac{|x|}{c}\Big)\,d^3y$$

$$= \frac{x\cdot F(t-|x|/c)}{4\pi|x|^2 c} \qquad \text{where} \quad F(t) = \int f(y,t)\,d^3y$$

Summary of Section 6.1–6.3

You should know:

Silence is the only homogeneous sound field in unbounded space since the field satisfies

$$\Big(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\Big)\frac{f(t-r/c)}{r} = 4\pi f(t)\,\delta(x)$$

An arbitary sound source in the form

$$q(x,t) = \int_V q(y,t)\,\delta(x-y)\,d^3y$$

generates a sound field

$$p'(x,t) = \int_V \frac{q(y,\, t-|x-y|/c)}{4\pi|x-y|}\,d^3y$$

A monopole point source of strength $Q(t)$ is the rate of change of mass addition rate

A monopole source distribution $q(x,t)$ generates a pressure field

$$p'(x,t) = \int \frac{q(y,\, t-|x-y|/c)}{4\pi|x-y|}\,d^3y$$

For x in the far field, this simplies to

$$p'(x,t) = \frac{1}{4\pi|x|}\int q\left(y,\, t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right)d^3y$$

and when the source region is compact and at the origin

$$p'(x,t) = \frac{Q(t-|x|/c)}{4\pi|x|} \quad \text{where} \quad Q(t)=\int q(y,t)\,d^3y \neq 0$$

A dipole point source of strength $F(t)$ at the origin generates a pressure field

$$p'(x,t) = -\frac{\partial}{\partial x_i}\left[\frac{F_i(t-|x|/c)}{4\pi|x|}\right] = \frac{x_i}{4\pi}\left[\frac{1}{|x|^3}F_i(t-|x|/c)+\frac{1}{|x|^2c}\frac{\partial F_i}{\partial t}(t-|x|/c)\right]$$

A dipole point source is the limit of two equal and opposite monopoles as their seperation δl becomes small and their strength $Q(t)$ large; $F(t)=Q(t)\delta l$ is the force applied to the field

A dipole source distribution $f(x,t)$ generates a pressure field

$$p'(x,t) = -\frac{\partial}{\partial x_i}\int \frac{f_i(y,\, t-|x-y|/c)}{4\pi|x-y|}\,d^3y$$

For x in the far field, this simplies to

$$p'(x,t) = \frac{x_i}{4\pi|x|^2c}\int \frac{\partial f_i}{\partial t}\left(y,\, t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right)d^3y$$

and when the source region is compact and at the origin

$$p'(x,t) = \frac{x \cdot \dot{F}(t-|x|/c)}{4\pi |x|^2 c} \quad \text{where} \quad F(t) = \int f(y,t)\, d^3y$$

6.4 Acoustic source process

Homogeneous wave equation

- conservation laws of mass and momentum
- weakly disturbed homogeneous medium in which $p = p(\rho)$

Source process

- forcing term in wave equation
- criteria leading to homogeneous wave equation violated

For example: fluid bounded by surface → sound generated at boundary surfaces

6.4.1 Combustion noise

过程不等熵, $p = p(\rho, s)$

sound source ← breakdown of state equation $p = p(\rho)$ ×

for example, heat added unsteady to a fluid

Combustion noise: one practical example of a monopole source distribution

Mass $\quad \frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot v = 0 \longrightarrow \frac{\partial}{\partial t}$

Momentum $\quad \rho_0 \frac{\partial v}{\partial t} + \nabla p' = 0 \longrightarrow \nabla \cdot$

neglect body force, inviscid

$$\frac{\partial^2 \rho'}{\partial t^2} - \nabla^2 p' = 0$$

In the case of combustion with heat addition

Entropy change $\rightarrow$ $\frac{\partial^2 \rho'}{\partial t^2} \neq \frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2}$

$\rho = \rho(p, s)$

$\Rightarrow \quad d\rho = \left.\frac{\partial \rho}{\partial p}\right|_s dp + \left.\frac{\partial \rho}{\partial s}\right|_p ds$ (全微分形式)

$= \frac{1}{c^2} dp + \left.\frac{\partial \rho}{\partial s}\right|_p ds$

For a perfect gas $s - s_0 = c_v \ln(p/p_0) - c_p \ln(\rho/\rho_0)$

$\downarrow \frac{\partial}{\partial \rho}$

$$\left.\frac{\partial \rho}{\partial s}\right|_p = -\frac{\rho}{c_p}$$

When viscous and heat conduction effects are neglected

Energy equation $\rho T \frac{ds}{dt} = w(x,t)$

w: rate of heat addition / unit volume

$$\frac{d\rho}{dt} = \frac{1}{c^2}\frac{dp}{dt} - \frac{\rho}{c_p}\frac{ds}{dt} = \frac{1}{c^2}\frac{dp}{dt} - \frac{1}{c_p T} w$$

$\downarrow \frac{1}{c_p T} = \frac{\gamma R}{c_p c^2}$, $\gamma = \frac{c_p}{c_v}$, $R = c_p - c_v$, $\frac{\gamma - 1}{c^2}$

$$\frac{d\rho}{dt} = \frac{1}{c^2}\frac{dp}{dt} - \frac{\gamma - 1}{c^2} w$$

linearising $\frac{\partial \rho'}{\partial t} = \frac{1}{c^2}\frac{\partial p'}{\partial t} - \frac{\gamma - 1}{c^2} w$

$\downarrow \frac{\partial^2 \rho'}{\partial t} - \nabla^2 p' = 0$

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = \frac{\gamma - 1}{c^2}\frac{\partial w}{\partial t}$$

distributed monopole

⇒ The solution is

$$p'(x,t) = \frac{\gamma-1}{4\pi c^2}\frac{\partial}{\partial t}\int \frac{w(y,\, t-|x-y|/c)}{|x-y|}\, d^3y$$

- Sound field ⟵ forced by rate of change in heat addition rate
- Steady heat is silent

热量注入率的变化率

- Sound of monopole source by unsteady heating

⟵ unsteady expansion of fluid at constant pressure

$$\frac{\partial}{\partial t}\Big|_p \quad \frac{\partial \rho'}{\partial t} = \frac{1}{c^2}\frac{\partial p'}{\partial t} - \frac{\gamma-1}{c^2}w$$

$$\Rightarrow \quad \frac{\gamma-1}{c^2}\frac{\partial w}{\partial t} = -\frac{\partial^2 p'}{\partial t^2}\Big|_p$$

monopole strength density

Mass continuity ⟶ $-\frac{\partial \rho'}{\partial t} = \rho_0 \nabla\cdot v$

$\nabla\cdot v$: divergence of velocity field

= rate of volumetric growth of unit material volume

⇒ $\frac{\gamma-1}{c^2}\frac{\partial w}{\partial t}$: ρ_0 x rate of change in the fractional rate of volumetric growth

The far field form

$$p'(x,t) = \frac{\gamma-1}{4\pi|x|c^2}\frac{\partial}{\partial t}\int w\left(y,\, t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right) d^3y$$

The far field of a compact flame

$$p'(x,t) = \frac{\gamma-1}{4\pi|x|c^2}\frac{\partial}{\partial t}\int w\left(y, t-\frac{|x|}{c}\right)d^3y$$

$$\propto \frac{\partial}{\partial t}\left(\text{rate of heat release with a time delay}\right)$$

Example 6.2 A time modulated laser beam supplies heat to a stationary gas at a rate per unit volume given by

$$w(x,t) = \frac{W_0}{l}H(x_1)\delta(x_2)\delta(x_3)(1-e^{i\omega t})e^{-x_1/l}$$

- The Heaviside function $H(x_1) = \begin{cases} 1 & \text{for } x_1 > 0 \\ 0 & \text{for } x_1 < 0 \end{cases}$
- l: the attenuation length due to thermal losses to the gas
- W_0: power output of the laser

Derive the far-field sound

The sound field $p'(x,t) = \frac{\gamma-1}{4\pi c^2}\frac{\partial}{\partial t}\int \frac{w(y, t-|x-y|/c)}{|x-y|}d^3y$

$$= \frac{\gamma-1}{4\pi c^2}(-i\omega)\frac{W_0}{l}\int_0^\infty \frac{e^{i\omega(t-|x-y|/c)}}{|x-(y,0,0)|}e^{-y_1/l}\,dy_1$$

For $|x|$ in the far field

$$\frac{1}{|x-y|} \approx \frac{1}{|x|}, \quad |x-y| \approx |x| - \frac{x\cdot y}{|x|c} = |x| - \frac{x_1 y_1}{|x|} = |x| - \cos\theta\, y_1$$

$$\Rightarrow \quad p'(x,t) = -\frac{i\omega(\gamma-1)}{4\pi|x|c^2}\frac{W_0}{l}e^{i(\omega t-k|x|)}\int_0^\infty e^{iky_1\cos\theta - y_1/l}\,dy_1$$

$$= -\frac{i\omega(\gamma-1)}{4\pi|x|c^2}\frac{W_0}{l}e^{i(\omega t-k|x|)}\left[\frac{e^{(ik\cos\theta - 1/l)y_1}}{ik\cos\theta - 1/l}\right]\Bigg|_0^\infty$$

$$\Downarrow \quad |e^{i\infty}| < 1, \quad e^{-\infty} \to 0$$

$$= -\frac{i\omega(\gamma-1)}{4\pi|x|c^2}\frac{W_0}{l}e^{i(\omega t-k|x|)}\frac{1}{1/l - ik\cos\theta}$$

$$= -\frac{i\omega(\gamma-1)}{4\pi|x|c^2}W_0 e^{i(\omega t-k|x|)}\frac{1}{1 - ik|\cos\theta}$$

6.4.2 Sound generated by linear mass creation & external forces.

Consider mass injected at a rate $m(x,t)$ / unit volume + a force of strength $f(x,t)$ / unit volume acting on the fluid.

Conservation of mass $\frac{\partial \rho}{\partial t} + \rho_0 \nabla \cdot v = m$

Momentum balance $\rho_0 \frac{\partial v}{\partial t} + \nabla p' = f$

$\frac{\partial}{\partial t}$, $\nabla\cdot$ $\rightarrow$ $\frac{\partial^2 \rho}{\partial t^2} - \nabla^2 p' = \frac{\partial m}{\partial t} - \nabla \cdot f$

Newly created mass $\longrightarrow$ a fraction β of unit volume with density ρ_m

The rest of fluid $\rightarrow$ the fraction $(1-\beta)$ of unit volume with density ρ_f

The total mass in unit volume

$$\rho = \beta \rho_m + (1-\beta)\rho_f$$

The volumetric mass creation rate

$$m = \frac{\partial}{\partial t}(\beta \rho_m)$$

Now $\rho \neq \rho(p)$, $p' \neq c^2 \rho'$

$$\Rightarrow \quad \frac{\partial \rho}{\partial t} = m + \frac{\partial \rho_f}{\partial t} - \frac{\partial}{\partial t}(\beta \rho_f) = m + \frac{1}{c^2}\frac{\partial p'}{\partial t} - \frac{\partial}{\partial t}(\beta \rho_0)$$

$$\Rightarrow \quad \frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 \rho}{\partial t^2} = \frac{\partial^2}{\partial t^2}(\beta \rho_0) - \frac{\partial m}{\partial t} \neq 0$$

$\Rightarrow$ Wave equation $\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = \rho_0 \frac{\partial^2 \beta}{\partial t^2} - \nabla \cdot f$

- $\rho_0 \frac{\partial^2 \beta}{\partial t^2}$: monopole source distribution
- $\nabla \cdot f$: dipole source distribution

- Mass creation rate m $\not\Rightarrow$ acoustic source
- Volume occupied by newly created mass $\longrightarrow$ Sound field

  mass created very densely ($\beta \to 0$) $\not\Rightarrow$ no disturbance

- steady creation of mass (volume) is silent

  Only unsteady displacement of fluid ($\frac{\partial^2 \beta}{\partial t^2} \neq 0$) → sound

- A very effective monopole source of strength $\rho_0 \partial^2\beta/\partial t^2$ per unit volume

  vs. less effective dipole source of strength f applied to unit volume

Using Eqs (6.16) and (6.30), the solution is

$$p'(x,t) = \rho_0 \frac{\partial^2}{\partial t^2} \int_V \frac{\beta(y, t-|x-y|/c)}{4\pi|x-y|} d^3y - \frac{\partial}{\partial x_i} \int_V \frac{f_i(y, t-|x-y|/c)}{4\pi|x-y|} d^3y$$

If 'volume' is is injected at the origin at a rate $\dot{\beta}(t)$

wave equation $\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = \underbrace{\rho_0 \ddot{\beta}(t)\, \delta(x)}$

↓

point monopole source

The solution $p'(x,t) = \frac{\rho_0 \ddot{\beta}(t - r/c)}{4\pi r}$ with $r = |x|$

↓

monopole source strength.

Example 6.3 ✡

紧致脉动球源在声学上等价于单极子点源

(i) Show that the pulsating compact sphere is acoustically equivalent to a point monopole of strength $\rho_0 \ddot{V}(t)$

↓

instantaneous volume of the sphere

(ii) Show that the oscillating compact sphere is acoustically equivalent to a point dipole of strength $F(t) + \rho_0 V \dot{U}(t)$

↘ sphere's acceleration

instantaneous force exerted on the fluid by the sphere

Solution:

(i) For a compact pulsating sphere

$$p'(r,t) = \frac{a^2}{r}\rho_0 \frac{\partial u_a(t-r/c)}{\partial t}$$

Since $\dot{V}(t) = 4\pi a^2 u_a(t)$

$$p'(r,t) = \frac{1}{4\pi r}\rho_0 \ddot{V}(t-r/c)$$

For a point monopole, $p'(x,t) = \frac{Q(t-r/c)}{4\pi r}$, $r = |x|$

$\Rightarrow$ The sound field is equivalent to that generated by a monopole of strength $\rho_0 \ddot{V}(t)$

(ii) For a compact oscillating sphere

$$p'(r,\theta,t) = -\frac{i\omega\rho_0 U a^3 \cos\theta}{2(1+ika)-(ka)^2}\frac{\partial}{\partial r}\left\{\frac{e^{i[\omega t - k(r-a)]}}{r}\right\}$$

$$\frac{\partial}{\partial x_1} = \frac{\partial r}{\partial x_1}\frac{\partial}{\partial r} = \cos\theta \frac{\partial}{\partial r}$$

$$p'(x,t) = -\frac{1}{2} i\omega\rho_0 U a^3 \frac{\partial}{\partial x_1}\left[\frac{e^{i(\omega t - kr)}}{r}\right]$$

For a point dipole

$$p'(x,t) = -\frac{\partial}{\partial x_1}\left[\frac{F(t-r/c)}{4\pi r}\right]$$

and $U(t) = (U(t), 0, 0)$, $F(t) = (F(t), 0, 0)$

This is equivalent to the pressure generated by a dipole of strength $2\pi\rho_0 a^3 U(t)$

The force exerted on a compact oscillating sphere is

$$-\frac{2}{3}\pi\rho_0 a^3 i\omega U e^{i\omega t} = -\frac{2}{3}\pi\rho_0 a^3 \dot{U}(t)$$

$$\Rightarrow F(t) + \underbrace{\rho_0 V \dot{U}(t)}_{\text{surface displacement}} = \left(\frac{2}{3}+\frac{4}{3}\right)\pi\rho_0 a^3 \dot{U}(t) = \underbrace{2\pi\rho_0 a^3 \dot{U}(t)}_{\text{dipole strength}}$$

6.5 Sound generation by flow — Lighthill's acoustic analogy

- To create mass and induce external

⟶ weak perturbation assumption ✗ ⟷ linearised equation

Real origins of a sound wave ?

Lighthill (1951) ⇒ theory of aerodynamic sound ⇐ emerging need to control jet noise

Acoustic analogy

exact statement of nature laws ⟺ acoustical approximations

⇓

sound source definition

Jet noise ← nonlinear flow perturbations ← nonlinear equation of motion

Consider a control volume V

Conservation of mass

$$\frac{\partial \rho}{\partial t}\delta x_1 \delta x_2 \delta x_3 = -\delta(\rho u_1)\delta x_2 \delta x_3 - \delta(\rho u_2)\delta x_1 \delta x_3 - \delta(\rho u_3)\delta x_1 \delta x_2$$

$$\downarrow \quad / \delta x_1 \delta x_2 \delta x_3$$

$$\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial x_1}(\rho u_1) - \frac{\partial}{\partial x_2}(\rho u_2) - \frac{\partial}{\partial x_3}(\rho u_3)$$

$$\downarrow$$

$$\boxed{\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = 0}$$

Conservation of momentum

τ_{ij} = viscous stress in i-direction on surface with normal in j-direction

Balance of momentum

$$\frac{\partial}{\partial t}(\rho u_1)\,\delta x_1\delta x_2\delta x_3 = -\delta p'\delta x_2\delta x_3 + \delta\tau_{11}\delta x_2\delta x_3 + \delta\tau_{12}\delta x_1\delta x_3 + \delta\tau_{13}\delta x_1\delta x_2$$
$$- \delta(\rho u_1^2)\delta x_2\delta x_3 - \delta(\rho u_1 u_2)\delta x_1\delta x_3 - \delta(\rho u_1 u_3)\delta x_1\delta x_2$$

$\downarrow$ $/\,\delta x_1\delta x_2\delta x_3$

$$\frac{\partial}{\partial t}(\rho u_1) = -\frac{\partial p'}{\partial x_1} + \frac{\partial \tau_{11}}{\partial x_1} + \frac{\partial \tau_{12}}{\partial x_2} + \frac{\partial \tau_{13}}{\partial x_3}$$
$$- \frac{\partial}{\partial x_1}(\rho u_1^2) - \frac{\partial}{\partial x_2}(\rho u_1 u_2) - \frac{\partial}{\partial x_3}(\rho u_1 u_3)$$

$\downarrow$

$$\frac{\partial}{\partial t}(\rho u_1) = -\frac{\partial p'}{\partial x_1} + \frac{\partial}{\partial x_j}(\tau_{1j} - \rho u_1 u_j) = -\frac{\partial}{\partial x_j}(\rho u_1 u_j + p'\delta_{1j} - \tau_{1j})$$

$\downarrow$

$$\frac{\partial}{\partial t}(\rho u_i) = -\frac{\partial}{\partial x_j}(\rho u_i u_j + p'\delta_{ij} - \tau_{ij})$$

$\rho u_i u_j$: Reynold's stress; $p'\delta_{ij} - \tau_{ij}$: P_{ij}

$\downarrow$

$$\frac{\partial}{\partial t}\rho u = -\text{grad}\, p' - \text{div}\,[\rho(u \otimes u) - \tau]$$

Combine $\Rightarrow$ $\frac{\partial^2 \rho}{\partial t^2} = \frac{\partial^2 \rho'}{\partial t^2} = \frac{\partial}{\partial x_i \partial x_j}(\rho u_i u_j + P_{ij})$

Subtract $c^2\nabla^2\rho' = c^2\frac{\partial^2 \rho'\delta_{ij}}{\partial x_i \partial x_j}$

$\Rightarrow$ $\frac{\partial^2 \rho'}{\partial t^2} - c^2\nabla^2\rho' = \frac{\partial^2}{\partial x_i \partial x_j}(\rho u_i u_j + P_{ij} - c^2\rho'\delta_{ij})$

Lighthill's stress tensor T_{ij}

$$T_{ij} = \rho u_i u_j + P_{ij} - c^2 \rho' \delta_{ij} \quad \text{with} \quad P_{ij} = p' \delta_{ij} - \tau_{ij}$$

$\Rightarrow$ Lighthill's equation

$$\frac{\partial^2 \rho'}{\partial t^2} - c^2 \nabla^2 \rho' = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

distributed quadrupole source

- No approximation is made in the derivation
- Nonlinear fluid motion in unbounded space

$\longrightarrow$ quadrupole distribution with strength T_{ij} in unit volume.

double divergence structure $\rightarrow$ double tendency to cancel

—— Dipole : divergence of monopole

—— Quadrupole : divergence of dipole or double divergence of monopole

Solution

The solution to $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) S = T_{ij}(x,t)$ is

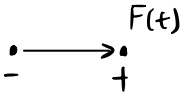
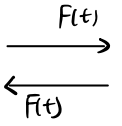
$$S(x,t) = \int \frac{T_{ij}(y, t - |x-y|/c)}{4\pi |x-y|} d^3y$$

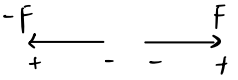
$$\Rightarrow \quad \rho'(x,t) = \frac{\partial^2}{\partial x_i \partial x_j} \int \frac{T_{ij}(y, t - |x-y|/c)}{4\pi c^2 |x-y|} d^3y$$

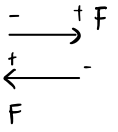
In linear invisid flow $\longrightarrow$ $T_{ij} = 0$

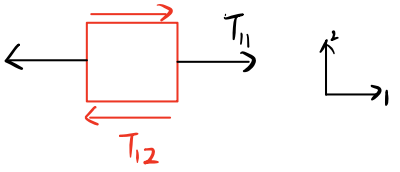
- $\rho u_i u_j \sim O^2(u)$, negligible
- $\tau_{ij} = 0 \rightarrow P_{ij} = p' \delta_{ij}$, $p' = c^2 \rho' \Rightarrow T_{ij} = \rho u_i u_j$

In turbulent flow $\longrightarrow T_{ij} \neq 0$

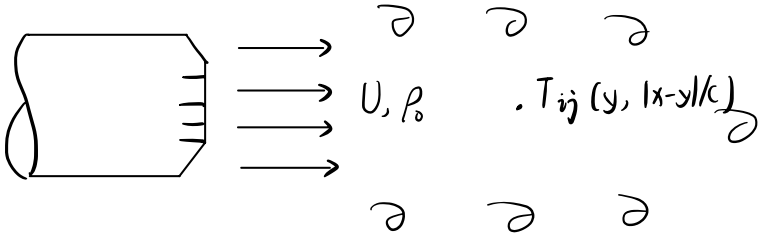
	Distributed source	Point-source	Interpretation
Monopole	$q(x,t)$	$Q(t)$	
Dipole	$-\nabla \cdot f(x,t)$	$F(t)$	
Quadrupole	$\frac{\partial^2 T_{ij}(x,t)}{\partial x_i \partial x_j}$	$T_{ij}(t)$	

e.g. T_{11} 

T_{12} 



6.5.1 Jet noise



Magnitude of jet noise by turbulent flow

← dimensional analysis ※

Consider a jet flow of velocity U, diameter D

- Characteristic frequency of flow fluctuations

$$\omega \sim \frac{U}{D}$$

- Time scale of sound field $\sim D/U$
- Wavelength $\lambda \sim \frac{c}{U} D = DM^{-1}$

- Compact ratio $kD = \omega D/c \sim U/c = M$

$\Rightarrow$ A low Mach number ($M \ll 1$) jet is compact ($D \ll \lambda$) $\longrightarrow$ retarded time $\tau \approx t - |x|/c$

In far field $\frac{\partial}{\partial x_i} = -\frac{x_i}{|x|c}\frac{\partial}{\partial t} \sim \frac{1}{c}\frac{U}{D} = \frac{M}{D}$ or $\frac{1}{\lambda}$

$\frac{\partial}{\partial x_i} = \frac{\partial |x|}{\partial x_i}\frac{\partial t}{\partial |x|}\frac{\partial}{\partial t} = -\frac{x_i}{|x|}\frac{1}{c}\frac{\partial}{\partial t}$

$T_{ij} \sim \rho_0 U^2$, $\int_V d^3y \sim D^3$

$\Rightarrow \rho' \sim \frac{M^2}{D^2}\frac{\rho_0 U^2}{|x| c^2} D^3 = \rho_0 M^4 \frac{D}{|x|}$

$\Rightarrow$ Lighthill's 8th-pow law

$\overline{\rho'^2} \sim \rho_0^2 M^8 \frac{D^2}{|x|^2}$ when $M \ll 1$

- extremely well confirmed experimentally
- low Mach number jets acoustically inefficient $\longleftarrow$ compact sources

低Ma数的射流噪声是低效的

Note:

- In jet $p' \sim \rho_0 U^2$ (hydrodynamic)
- In far field $p' \sim \rho_0 U^2 \times \frac{U^2}{c^2}\frac{D}{|x|}$ (acoustic)

流场噪声仅有很小一部分传到远场

High Mach number flows $\longrightarrow$ non-compact source

$\Rightarrow$ 8th-power scaling law X

For a supersonic jet, sound field structure $\longleftarrow$ $\tau = t - |x-y|/c$

In far field, $|x-y| \approx |x| - \frac{x\cdot y}{|x|} = r - y_r$

$$\frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} = \frac{\partial^2 r}{\partial x_i \partial x_j}\frac{\partial^2 T_{ij}}{\partial r^2} = \frac{\partial^2 T_{rr}}{\partial r^2}$$

$\frac{\partial^2 r}{\partial x_i \partial x_j} \rightarrow \frac{x_i x_j}{|x|^2}$

The solution simplies to

$$\rho'(x,t) = \frac{1}{4\pi c^2 r}\frac{\partial^2}{\partial r^2}\int T_{rr}(y, t-r/c+y_r/c)\, d^3y$$

$$= \frac{1}{4\pi c^2 r}\frac{\partial^2}{\partial r^2}\int T_{rr}(y,\tau)\, dy_r\, dS_r$$

- $T_{rr} = \frac{x_i x_j}{|x|^2} T_{ij}$: component of T_{ij} in observer's direction
- $y_r = x \cdot y/|x|$: component of y in direction of x v y
- S_r : surface of y_r = constant

change integration variable $dy_r = c\,d\tau$

$$\Rightarrow \quad \rho'(x,t) = \frac{1}{4\pi c r}\frac{\partial^2}{\partial r^2}\int T_{rr}(y,\tau)\, dS_r d\tau$$

- $y = (y_s, y_r) = (y_s, c(\tau - t + r))$

  ↓

  vector lying in S_r

y_r variation $\sim D/M$ $\qquad$ τ variation $\sim D/U$

↓

negligible for highly supersonic jets

Dimensional analysis

$T_{rr} \sim \rho_0 U^2$ $\qquad$ $\frac{\partial}{\partial r} \sim \frac{1}{D}$ $\qquad$ $\int_S dS_r \sim D^2$ $\qquad$ $\int_\tau d\tau \sim \frac{D}{U}$

$$\Rightarrow \quad \rho' \sim \frac{1}{Cr}\frac{1}{D^2}\rho_0 U^2 D^2 \frac{D}{U} = \rho_0 M \frac{D}{r}$$

$$\Rightarrow \quad \overline{\rho'^2} \sim \rho_0^2 M^2 \frac{D^2}{r^2}$$

Another effect : No. of eddies heard at $(x,t) \propto M$

$$\Rightarrow \quad \overline{\rho'^2} \sim U^2$$ in agreement with experiments of rocket noise

Example 6.4 Two jet engines have the same exit temperature and produce the same thrust but one has the twice the diameter of the other. Use LightHill's 8th-power law to estimate which engine is the lowest and how much.

Solution:

For equal thrust $\dot{m}V$ $\quad \rho_e U_1^2 D_1^2 = \rho_e U_2^2 D_2^2$

$\because \quad D_1 = 2D_2 \quad \therefore \quad U_2 = 2U_1$

$\Rightarrow$ Sound power for engine 1 $\sim k U_1^8 D_1^2$

Sound power for engine 2 $\sim k U_2^8 D_2^2 = k U_1^8 D_1^2 \times 2^6$

$\Rightarrow$ The smaller diameter engine is $10\log_{10}(2^6) = 18\ dB$ louder

The mass flow rate

$$\rho_e \frac{\pi D_1^2}{4} U_1 = \rho_e \frac{\pi D_2^2}{4} U_2 \times 2$$

$\Rightarrow$ Large bypass ratio engines

$\Rightarrow$ same thrust by jet Mach number ↓ mass flow rate ↑

在同等推力下，提高涵道比可以降低射流马赫数，进而降低射流噪声

significant noise reduction

For high speed jet, replace U^8 by U^3

$\Rightarrow$ ratio of sound power 2 to 1 $= \dfrac{U_2^3 D_2^2}{U_1^3 D_1^2} = \dfrac{D_1}{D_2} = 2$

Now $\Delta PWL = 10\log_{10}(2) = 3\ dB$

Summary of Section 6.4 and 6.5

Unsteady heat addition at a rate $w(x,t)$/unit volume is equivalent to a monopole source of strength

$\frac{\gamma-1}{c^2}\frac{\partial w}{\partial t}$: ρ_0 x rate of change in the fractional rate of volumetric growth.

Lighthill's acoustic analogy

$$\frac{\partial^2}{\partial t^2}\rho' - c^2\nabla^2\rho' = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

↓

distributed quadrupole source

where Lighthill's stress tensor

$$T_{ij} = \rho u_i u_j + P_{ij} - c^2\rho'\delta_{ij} \qquad \text{with} \quad P_{ij} = p'\delta_{ij} - \tau_{ij}$$

and the solution is

$$\rho'(x,t) = \frac{\partial^2}{\partial x_i \partial x_j}\int \frac{T_{ij}(y,\, t-|x-y|/c)}{4\pi c^2|x-y|}\, d^3y$$

A low Mach number jet is compact ($kD \sim M \ll 1$) and the sound power can be described by Lighthill's 8th-power law

$$\overline{\rho'^2} \sim \rho_0^2 M^8 \frac{D^2}{|x|^2} \qquad \text{when } M \ll 1$$

You should be able to derive scaling laws through dimensional analysis

6.6 The sound of foreign bodies

In Lighthill's theory, sound generation by

$$\frac{\partial^2}{\partial x_i \partial x_j}\int \frac{T_{ij}(y,\, t-|x-y|/c)}{|x-y|}\, d^3y$$

Foreign bodies in flow:

quadrupole distributed with the interior of foreign bodies ⟶ boundary sources.

$|x-y|$: Symmetric function of x and y

$$\frac{\partial}{\partial x_i}\left[\frac{T_{ij}(y,\, t-|x-y|/c)}{|x-y|}\right] = \frac{\partial T_{ij}/\partial y_i}{|x-y|} - \frac{\partial}{\partial y_i}\left[\frac{T_{ij}(y,\, t-|x-y|/c)}{|x-y|}\right]$$

$|x-y| = r$, $[\ ]$: function evaluated at $(y,\, t-|x-y|/c)$

Proof. left $= \frac{\partial}{\partial x_i}\left[\frac{T_{ij}}{r}\right] = \frac{1}{r}\left[\frac{\partial T_{ij}}{\partial t}\right]\left(-\frac{1}{c}\right)\frac{\partial r}{\partial x_i} - \frac{1}{r^2}\frac{\partial r}{\partial x_i}[T_{ij}]$

$$= \left(-\frac{1}{rc}\left[\frac{\partial T_{ij}}{\partial t}\right] - \frac{1}{r^2}[T_{ij}]\right)\frac{\partial r}{\partial x_i}$$

$\because \quad r = x - y \qquad i.e. \quad r_i = x_i - y_i$

$$\Rightarrow \quad \frac{\partial r}{\partial x_i} = \frac{\partial r_k}{\partial x_i}\frac{\partial r}{\partial r_k} = \delta_{ik}\frac{r_k}{r} = \frac{r_i}{r}$$

Similarly, $\frac{\partial}{\partial y_i}\left[\frac{T_{ij}}{r}\right] = \frac{1}{r}\frac{\partial T_{ij}}{\partial y_i} + \left(-\frac{1}{rc}\left[\frac{\partial T_{ij}}{\partial t}\right] - \frac{1}{r^2}[T_{ij}]\right)\frac{\partial r}{\partial y_i}$

$$\downarrow \quad \frac{\partial r}{\partial y_i} = \frac{\partial r_k}{\partial y_i}\frac{\partial r}{\partial r_k} = -\delta_{ik}\frac{r_k}{r} = -\frac{r_i}{r}$$

$$\frac{\partial}{\partial x_i}\left[\frac{T_{ij}}{r}\right] + \frac{\partial}{\partial y_i}\left[\frac{T_{ij}}{r}\right] = \frac{1}{r}\frac{\partial T_{ij}}{\partial y_i}$$

$T_{ij} = \rho u_i u_j + P_{ij} - c^2\rho'\delta_{ij}$ $\xrightarrow{\text{substitute}}$ $\frac{\partial}{\partial x_i}\left[\frac{T_{ij}(y,\, t-|x-y|/c)}{|x-y|}\right] = \frac{\partial T_{ij}}{|x-y|} - \frac{\partial}{\partial y_i}\left[\frac{T_{ij}(y,\, t-|x-y|/c)}{|x-y|}\right]$

$$\frac{\partial}{\partial x_i}\left[\frac{T_{ij}}{r}\right] = \frac{\partial}{\partial x_i}\left[\frac{\rho u_i u_j + P_{ij}}{r}\right] - c^2\frac{\partial}{\partial x_j}\left[\frac{\rho'}{r}\right]$$

$$= \left[\frac{1}{r}\frac{\partial}{\partial y_i}(\rho u_i u_j + P_{ij})\right] - \frac{\partial}{\partial y_i}\left[\frac{\rho u_{ij} + P_{ij}}{r}\right] - c^2\frac{\partial}{\partial x_j}\left[\frac{\rho'}{r}\right]$$

$$\downarrow$$

$$= -\frac{\partial}{\partial t}(\rho u_j)$$

$$= -\frac{\partial}{\partial y_i}\left[\frac{\rho u_i u_j + P_{ij}}{r}\right] - c^2\frac{\partial}{\partial x_j}\left[\frac{\rho'}{r}\right] - \left[\frac{1}{r}\frac{\partial}{\partial t}(\rho u_j)\right]$$

Differentiate w.r.t $x_j \Rightarrow$

$$\frac{\partial^2}{\partial x_i \partial x_j}\left[\frac{T_{ij}}{r}\right] = -\frac{\partial^2}{\partial y_i \partial x_j}\left[\frac{\rho u_i u_j + P_{ij}}{r}\right] - c^2 \frac{\partial^2}{\partial x_j \partial x_j}\left[\frac{\rho'}{r}\right] - \frac{\partial}{\partial x_j}\left[\frac{1}{r}\frac{\partial}{\partial t}(\rho u_j)\right]$$

Use $\frac{\partial}{\partial x_i}\left[\frac{T_{ij}(y, t-|x-y|/c)}{|x-y|}\right] = \frac{\partial T_{ij}}{|x-y|} - \frac{\partial}{\partial y_i}\left[\frac{T_{ij}(y, t-|x-y|/c)}{|x-y|}\right]$

$$\Rightarrow \quad \frac{\partial}{\partial x_j}\left[\frac{1}{r}\frac{\partial}{\partial t}(\rho u_j)\right] = \left[\frac{1}{r}\frac{\partial^2}{\partial t \partial y_i}(\rho u_j)\right] - \frac{\partial}{\partial y_i}\left[\frac{1}{r}\frac{\partial}{\partial t}(\rho u_j)\right]$$

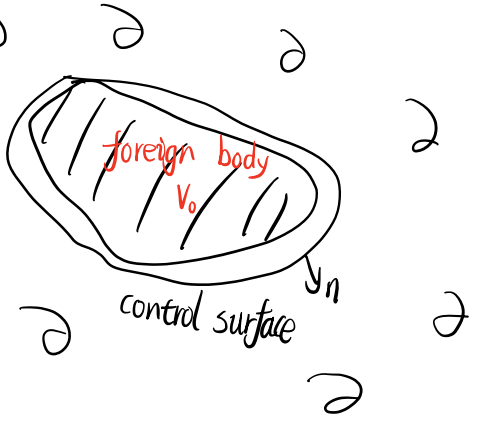
$$\downarrow \quad \frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial y_j}(\rho u_j)$$

$$= \left[-\frac{1}{r}\frac{\partial^2 \rho'}{\partial t^2}\right] - \frac{\partial}{\partial y_i}\left[\frac{1}{r}\frac{\partial}{\partial t}(\rho u_j)\right]$$

$$\Rightarrow \quad \frac{\partial^2}{\partial x_i \partial x_j}\left[\frac{T_{ij}}{r}\right] = -\frac{\partial^2}{\partial x_j \partial y_i}\left[\frac{\rho u_i u_j + P_{ij}}{r}\right] + \frac{\partial}{\partial y_i}\left[\frac{1}{r}\frac{\partial}{\partial t}(\rho u_j)\right]$$
$$+ \left(\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x_i^2}\right)\left[\frac{\rho'}{r}\right]$$

$\rho'(y, t-|x-y|/c)$ satisfies homogeneous wave equation in sound field x away from source region y

Integrate $\square$ over a stationary volume V_0 enclosing the foreign body + divergence theorem



$$\frac{\partial^2}{\partial x_i \partial x_j}\int_{V_0}\left[\frac{T_{ij}}{r}\right] dV = -\frac{\partial}{\partial x_j}\int_S n_i\left[\frac{\rho u_i u_j + P_{ij}}{r}\right] dS + \frac{\partial}{\partial t}\int_S \left[\frac{\rho u \cdot n}{r}\right] dS$$

dipole monopole

- Dipole strength /S = stress applied to S
- Monopole strength /S = mass flux across S

⇒ The exact solution of Lighthill's equation in a region V exterior to a stationary control surface S

$$\rho'(x,t) = \frac{1}{4\pi c^2}\frac{\partial^2}{\partial x_i \partial x_j}\int_V \left[\frac{T_{ij}}{r}\right] dV - \frac{1}{4\pi c^2}\frac{\partial}{\partial x_j}\int_S n_i \left[\frac{\rho u_i u_j + P_{ij}}{r}\right] dS + \frac{1}{4\pi c^2}\frac{\partial}{\partial t}\int_S \left[\frac{\rho u \cdot n}{r}\right] dS$$

- If foreign body stationary and rigid ⟶ $n_i(\rho u_i u_j) = 0$, $u \cdot n = 0$ on S
- The sound field by foreign body in inviscid fluid ($\tau_{ij} = 0$), $P_{ij} = p'\delta_{ij}$

$$-\frac{1}{4\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i \left[\frac{p'}{r}\right] dS$$

- For a compact body, τ variation ⟶ 0

$$-\frac{1}{4\pi c^2}\frac{\partial}{\partial x_i}\left[\frac{F_i}{r}\right] \quad \text{with} \quad F_i = \int_S n_i p' dS$$

↓ force applied by body to fluid

Sound by compact rigid bodies in turbulent flow

⇐ unsteady force acting on bodies $F \sim \rho_0 U^2 D^2$

Use $\partial/\partial x_i \sim M/D$ ⇒

$$\rho' \sim \rho_0 M^3 \frac{D}{r} \quad \text{and} \quad \overline{\rho'^2} \sim \rho_0^2 M^6 \frac{D^2}{r^2}$$

- Body forces generate sound more efficient than low Mach number free turbulence by a factor of M^{-2}

If foreign body pulsating in volume ⟶ efficient monopole

The sound for a compact body

$$\frac{1}{4\pi c^2}\frac{\partial}{\partial t}\left[\frac{Q}{r}\right] \quad \text{with} \quad Q=\int_S \rho u_n \, dS$$

Q: rate of mass flux out of S

Use $\frac{\partial}{\partial t} \sim U/D$, $Q \sim \rho_0 U D^2$

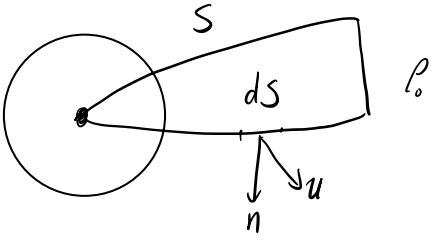
$$\Rightarrow \quad \rho' \sim \rho_0 M^2 \frac{D}{r} \quad \text{and} \quad \overline{\rho'^2} \sim \rho_0^2 M^4 \frac{D^2}{r^2}$$

- Sound generation ($M \ll 1$): mass (volume) flux >> body force >> turbulent flow by a factor of M^{-2} and M^{-4}

6.7 The effect of foreign bodies in linear motion

Simplied analysis for practical problems

Suppose fluid velocity u, unit normal n to surface S



- rate of mass flow at surface $= \rho_0 u \cdot n \, dS$
- force exerted on fluid at surface $= p n \, dS$
- ignore quadrupole source by turbulent flow

Recall Sec. 6.4.2 : Sound by mass creation & external force

Comparing with Eq(6.38), the sound field is

$$p'(x,t)=\frac{\partial}{\partial t}\int_S \frac{\rho_0 n \cdot u(y, t-|x-y|/c)}{4\pi|x-y|}\,dS - \frac{\partial}{\partial x_i}\int_S \frac{n_i p(y, t-|x-y|/c)}{4\pi|x-y|}\,dS$$

Assumption:
忽略由湍流产生的声源,
rigid body,
inviscid flow $\Rightarrow \tau = 0$
$\Downarrow$
$P_{ij} = p'\delta_{ij}$

$\frac{\partial}{\partial x_i} = \frac{\partial |x|}{\partial x_i}\frac{\partial \tau}{\partial |x|}\frac{\partial}{\partial \tau}$

linear motion
$\Rightarrow \rho u_i u_j \approx 0$
$p' = c^2 \rho'$
忽略二阶及以高小量

In far field,

$$p'(x,t)=\frac{1}{4\pi|x|}\frac{\partial}{\partial t}\int_S \rho_0 n\cdot u\left(y, t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right)dS + \frac{x_i}{4\pi|x|^2 c}\frac{\partial}{\partial t}\int_S n_i p\left(y, t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right)dS$$

Ffowcs Williams-Hawkings (FW-H) equation

- extremely useful ⟶ a wide range of sound generation problems
- far-field sound ⟵ motion of a foreign body + integral of pressure over the body surface
- the basis of predicting propeller and helicopter noise
  - —— surface motion known (rotating blades in forward flight)
  - —— surface pressure ⟵
    - * classical areodynamics (semi-emprical models)
    - * CFD simulations
  - —— CFD → far-field sound on LHS ?
    - * Sound: weak far-field product of much more intense near-field flows
    - * most CFD codes ⟶/ far-field sound by simply extending the calculation domain to large distances (dominated by numerical errors)
    - * CFD codes ⟶ direct calculation ✓ high order and very time consuming to run

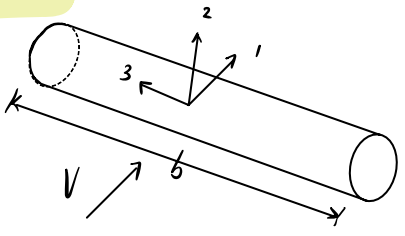
6.7.1 Rigid body at rest (Case 1 of FWH equation)

Consider a stationary body → $u \cdot n = 0$

For far field $|x| \gg |y|$ and compact body, FW-H equation reduce to

$$p'(x,t) = \frac{x_i}{4\pi |x|^2 c} \frac{\partial}{\partial t} \int_S n_i p\left(y, t - \frac{|x|}{c}\right) dS$$

Example 1: Aeolian tones



Aeolian tone : singing of wind blowing through wires

Suppose a cylinder of diameter D and length 2b

compact ($kb \ll 1$)

用于描述周期性非定常流动的无量纲数，通常f是涡脱落的频率

Steady wind of velocity V leads to vortex shedding

and an unsteady lift force

Strouhal number 斯特哈劳尔数

angle freq.

Frequency $\propto \frac{V}{D} = St\frac{2\pi V}{D}$ with $St = \frac{fD}{V} \sim 0.2$ for high Reynolds number

Note $kD = St\frac{2\pi V}{D}\frac{1}{c}D \sim St\,2\pi M \rightarrow$ Compactness means low Mach number

Let unsteady force/unit length $= f(y_3, t)$ in 2-direction

$\Rightarrow f_{rms} = C_L \frac{1}{2}\rho_0 V^2 D$ and $\frac{\partial f}{\partial t} = iSt\frac{2\pi V}{D} f$

单位展长的升力

lift coefficient

$= i\omega f \sim iSt\frac{2\pi V}{D} f$

Use $p'(x,t) = \frac{x_i}{4\pi|x|^2 c}\frac{\partial}{\partial t}\int_S n_i p\left(y, t-\frac{|x|}{c}\right) dS$

表面压力的面积分表现为电线受到的升力

$\Rightarrow p'(x,t) = \frac{x_2}{4\pi|x|^2 c} iSt\frac{2\pi V}{D}\int_{-b}^{b} f\left(y_3, t-\frac{|x|}{c}\right) dy_3$

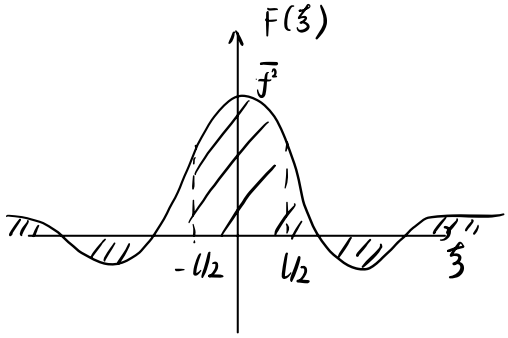
Definition :

Autocorrelation and integral length scale

$F(\xi)$ the autocorrelation of $f(y_3, t)$ is defined by

$$F(\xi) = \overline{f(y_3, t)\, f(y_3+\xi, t)}$$

- $\overline{(\,)}$: mean value ; $F(0) = \overline{f^2}$



The integral length scale is defined by $l\overline{f^2} = \int_0^\infty F(\xi)\, d\xi$

- l is a measure of the distance over which f and, in this case, vortex shedding is correlated $\longrightarrow$ correlation length

$$\overline{p'^2} = \left(\frac{x_2}{4\pi|x|^2 c}\right)^2 \left(St\frac{2\pi V}{D}\right)^2 \overline{\int_{-b}^{b} f\left(y_3, t-\frac{|x|}{c}\right) dy_3 \int_{-b}^{b} f\left(z_3, t-\frac{|x|}{c}\right) dz_3}$$

这里只是为了方便区分用了3另一积分

元素，本质是一样的

$$= \left(\frac{x_2 St V}{2|x|^2 c D}\right)^2 \int_{-b}^{b}\int_{-b-y_3}^{b-y_3} \underbrace{\overline{f(y_3, t-\tfrac{|x|}{c}) f(y_3+\xi, t-\tfrac{|x|}{c})}}_{F(\xi)} d\xi dy_3$$

$$= \left(\frac{x_2 St V}{2|x|^2 c D}\right)^2 \int_{-b}^{b}\int_{-b-y_3}^{b-y_3} F(\xi)\, d\xi dy_3$$

If $l \ll b$, $\int_{-b-y_3}^{b-y_3} F(\xi) d\xi \approx \int_{-\infty}^{+\infty} F(\xi) d\xi = l\overline{f^2}$

$$\Rightarrow \quad \overline{p'^2} = \left(\frac{x_2 St V}{2|x|^2 c D}\right)^2 \int_{-b}^{b} l\overline{f^2} dy_3 = \left(\frac{x_2 St V}{2|x|^2 c D}\right)^2 2bl\overline{f^2}$$

Use $f_{rms} = C_L \frac{1}{2} \rho_0 V^2 D$

$$\Rightarrow \quad \overline{p'^2} = \left(\frac{x_2 St V}{2|x|^2 c D}\right)^2 2bl \left(C_L \tfrac{1}{2} \rho_0 V^2 D\right)^2$$

$$= \frac{x_2^2}{8|x|^4 c^2} St^2 bl C_L^2 \rho^2 V^6$$

Applied unsteady force → dipole source → $\overline{p'^2} \sim V^6$

If the cylinder vibrates in response, the vortex shedding can 'lock on' and become coherent all along the cylinder

f is independent of y_3

这里的意思是，一旦 cylinder 振起来，correlation length 变为与 cylinder 有关即变为 $2b$

$2b\overline{f^2} = \int_{-\infty}^{+\infty} F(\xi) d\xi$

$$\therefore \quad \overline{p'^2} = \left(\frac{x_2 St V}{2|x|^2 c D}\right)^2 (2b)^2 \overline{f^2}$$

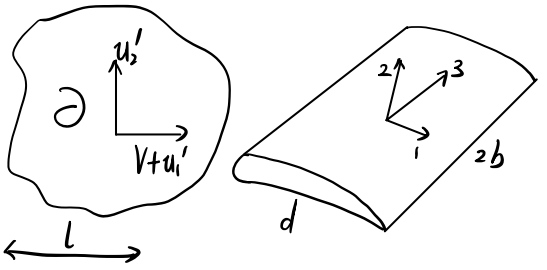
larger by a factor $2b/l$

Example 2: A thin strut in a turbulent flow (Case 2 of FWH equation)

Consider an aerofoil of chord d and span $2b$

† incoming turbulent eddy with velocity fluctuation

(u_1', u_2') and correlation length $l \gg d$



For an aerofoil shape, unsteady lift $\gg$ unsteady drag

Let $f(y_3, t)$ = unsteady force / unit length in 2-direction

$$\Rightarrow \quad \frac{\partial f}{\partial t} = i S_t \frac{2\pi V}{l} f$$

Use $p'(x,t) = \frac{x_i}{4\pi |x|^2 c} \frac{\partial}{\partial t} \int_S n_i p\left(y, t - \frac{|x|}{c}\right) dS$

$$\Rightarrow \quad \overline{p'^2} = \left(\frac{x_2 S_t V}{2|x|^2 c l}\right)^2 \int_{-b}^{b} f\left(y_3, t - \frac{|x|}{c}\right) dy_3 \int_{-b}^{b} f\left(z_3, t - \frac{|x|}{c}\right) dz_3$$

Angle of attack $\alpha = \frac{u_2'}{V}$ (very small)

垂直于来流方向的扰动会引起攻角的变化

$$f = \underbrace{D(\alpha)}_{\downarrow \text{drag}} \sin\alpha + L(\alpha)\cos\alpha$$

$$\approx D(0)\alpha + \left[L(0) + \alpha \frac{\partial L}{\partial \alpha}\right]$$

$$= L(0) + \left[D(0) + \frac{\partial L}{\partial \alpha}(0)\right]\alpha$$

$L(0)$: for symmetric strut, $L(0) = 0$ ← 对称翼型攻角为零时，升力为零

Write $D = \frac{1}{2}\rho_0 V^2 d\, C_D(\alpha)$, $L = \frac{1}{2}\rho_0 V^2 d\, C_L(\alpha)$

$$\Rightarrow \quad f = \frac{1}{2}\rho_0 V^2 d\,\alpha \left[C_D(0) + \frac{\partial C_L}{\partial \alpha}(0)\right]$$

$$= \frac{1}{2}\rho_0 u_2' V d \left[C_D(0) + \frac{\partial C_L}{\partial \alpha}(0)\right]$$

$$\Rightarrow \quad \overline{p'^2} = \left(\frac{x_2 S_t V}{2|x|^2 c l}\right)^2 \left(\frac{1}{2}\rho_0 V d \left[C_D(0) + \frac{\partial C_L}{\partial \alpha}(0)\right]\right)^2$$

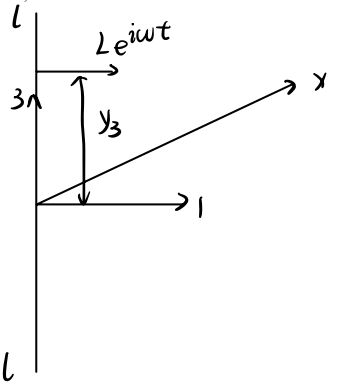
$$\cdot \int_{-b}^{b} \int_{-b-y_3}^{b-y_3} \overline{u_2'\left(y_3, t - \frac{|x|}{c}\right) u_2'\left(y_3 + \xi, t - \frac{|x|}{c}\right)}\, d\xi\, dy_3$$

$$= \left(\frac{x_2 S_t V}{2|x|^2 c l}\right)^2 \left(\frac{1}{2}\rho_0 V d \left[C_D(0) + \frac{\partial C_L}{\partial \alpha}(0)\right]\right)^2 2bl\, \overline{u_2'^2}$$

- well confirmed by experimental observations.

(Case 3 of FWH equation)

Example 6.5 The wind blowing past a long wire stretched from $(0,0,-l)$ to $(0,0,l)$ exerts a force in the x_1-direction on the wire of strength $Le^{i\omega t}$ per unit length of wire. Derive the far field sound. Comment on the limiting form when kl becomes small



Solution:

The source distribution is a collection of dipoles of strength $-Le^{i\omega t}dy_3$ in the 1-direction with $y=(0,0,y_3)$

$$p'(x,t)=\frac{\partial}{\partial x_1}\int_{-l}^{l}\frac{Le^{i(\omega t-k|x-y|)}}{4\pi|x-y|}dy_3$$

For x in far-field $|x-y|\approx|x|-\frac{x\cdot y}{|x|}=r-\frac{x_3y_3}{r}$, since $y_1=y_2=0$

$$p'(x,t)=\frac{\partial}{\partial x_1}\left[\frac{Le^{i(\omega t-kr)}}{4\pi r}\int_{-l}^{l}e^{ikx_3y_3/r}dy_3\right]$$

$$=\frac{\partial}{\partial x_1}\left\{\frac{Le^{i(\omega t-kr)}}{4\pi r}\left[\frac{e^{ikx_3y_3/r}}{ikx_3/r}\right]\Bigg|_{-l}^{l}\right\}$$

$$=\frac{\partial}{\partial x_1}\left[\frac{Le^{i(\omega t-kr)}}{4\pi r}\,\frac{2\sin(klx_3/r)}{kx_3/r}\right]$$

$$\frac{\partial}{\partial x_1}=\frac{\partial\tau}{\partial x_1}\frac{\partial}{\partial t}=\frac{\partial r}{\partial x_1}\frac{\partial\tau}{\partial r}\frac{\partial}{\partial t}=\frac{x_1}{r}\left(-\frac{1}{c}\right)i\omega$$

$$=-\frac{ikx}{4\pi r^2}2Ll\,\frac{\sin(kl(x_3/r))}{kl(x_3/r)}e^{i(\omega t-kr)}+\underbrace{O\left(\frac{1}{r^2}\right)}_{\downarrow\ \text{near field}}$$

If $kl\ll1$, $\frac{\sin(kl(x_3/r))}{kl(x_3/r)}\approx1$

and p' due to a dipole of strength $-2Ll e^{i\omega t}$ = total force on the fluid.

6.7.2 A compact body producing a net monopole

For a net monopole $\int_S n \cdot u(y,t)\, dS \neq 0$

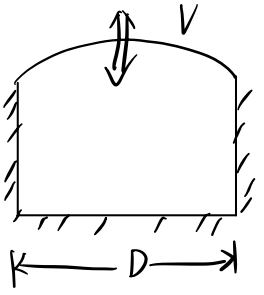
For $|x| \gg |y|$ and a compact body, Eq (6.58) reduce to

$$p'(x,t) = \frac{1}{4\pi|x|}\frac{\partial}{\partial t}\int_S \rho_0 n\cdot u\left(y, t-\frac{|x|}{c}\right) dS + \frac{x_i}{4\pi|x|^2 c}\frac{\partial}{\partial t}\int_S n_i p\left(y, t-\frac{|x|}{c}\right) dS$$

$$= \frac{\dot{Q}(t-|x|/c)}{4\pi|x|} + \frac{x\cdot \dot{F}(t-|x|/c)}{4\pi|x|^2 c}$$

where $Q(t) = \rho_0 \int_S n\cdot u\, dS$, $F(t) = \int_S n\, p\, dS$

Recap : monopoles more effective than dipoles (?) ⟶ prove



Relative orders of magnitude

Let ω : typical frequency

V : typical fluid velocity at surface

D : body dimension

⇒ Monopole $Q \sim \rho_0 V D^2$

$$\Rightarrow \frac{\dot{Q}}{4\pi|x|} \sim \frac{\rho_0 \omega V D^2}{|x|}$$

Dipole $-\nabla p = \rho_0 \frac{\partial u}{\partial t}$

$\downarrow$ $\sim p_s/D$ $\qquad$ $\downarrow$ $\sim \omega V$

$$\Rightarrow p_s \sim \rho_0 \omega V D \quad , \quad F \sim \rho_0 \omega V D^3$$

$$\Rightarrow \frac{x\cdot \dot{F}}{4\pi|x|^2 c} \sim \frac{\rho_0 \omega^2 V D^3}{|x| c} = \frac{\rho_0 \omega V D^2}{|x|} \quad (kD)$$

$\sim$ ↓ small since source is compact

$F \sim \rho_0 U^2 D^2$

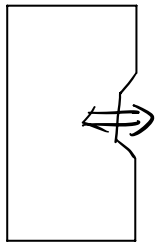
$\Rightarrow$ dipole sound < monopole sound by a factor kD, negligible for this compact body

Hence for a compact body producing a net monopole, the far-field pressure.

$$p'(x,t) = \frac{\dot{Q}(t-|x|/c)}{4\pi|x|} \quad \text{where} \quad Q(t) = \rho_0 \int_S n\cdot u\, dS$$

(Case 4 of FWH equation)

Example 3: An impenetrable compact body changing in volume.

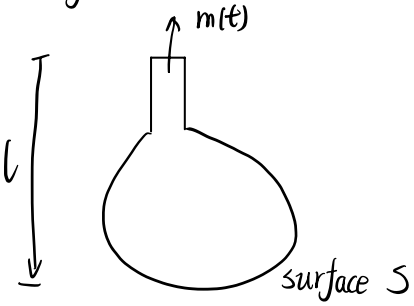


e.g. a loudspeaker $\int_S n\cdot u\, dS = \dot{V}(t)$

$$\Rightarrow \quad p'(x,t) = \rho_0 \frac{\ddot{V}(t-|x|/c)}{4\pi|x|}$$

Example 4: Unsteady flow across a control surface

(i) Helmholtz resonator (Case 5)



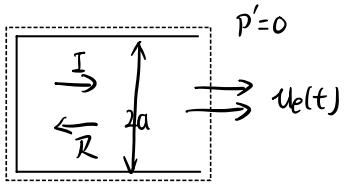
$kl \ll 1$ with l the length of Helmholtz resonator

Choose a control surface S across the neck

p' near neck $\longrightarrow$ large fluctuating $m(t)$ through neck $\rightarrow$ strong monopole source

$$p'(x,t) = \frac{\dot{m}(t-|x|/c)}{4\pi|x|} \quad \text{where} \quad m(t) = \rho_0 \int_S n\cdot u\, dS$$

(ii) Open end of organ pipes (Case 6)



$ka \ll 1$ with a the radius

For x in far field,

$$p'(x,t) = \frac{\rho_0 \pi a^2 \dot{u}_e(t-|x|/c)}{4\pi|x|}$$

Radiated sound power $= \dfrac{\overline{p'^2}}{\rho_0 c} 4\pi|x|^2 = \dfrac{\rho_0 \omega \pi a^2 u_e^2}{4c}$

B.C. at open end ($x=0$)

(1) $p' = 0 \Rightarrow I + R = 0$

(2) $u_e = \dfrac{I-R}{\rho_0 c} = \dfrac{2I}{\rho_0 c} \Rightarrow u_i = \dfrac{I}{\rho_0 c} = \dfrac{1}{2}u_e$

Incident sound power $= \overline{\left(\frac{1}{2}u_e\right)^2}\,\rho_0 c\pi a^2 = \frac{\rho_0 c\pi a^2 \overline{u_e^2}}{4}$

$$\Rightarrow \quad \frac{\text{Radiated sound power}}{\text{Incident sound power}} = (ka)^2 << 1$$

$\Rightarrow$ a good approximation to neglect the radiated sound of waves within a compact duct

(Case 6)

Example 6.8 A duct of length l and radius a has a closed end at $x=-l$ and an open end at $x=0$. Determine the far-field sound pressure radiated by the vibration of a tightly fixed piston in the duct near $x=-l$, which oscillates with velocity $u_0 e^{i\omega t}$. Assume that $ka<<1$

Solution:

Inside the duct

At $x=0$, $p' = (I e^{-ikx} + R e^{ikx}) e^{i\omega t} = 0 \Rightarrow R = -I$

At $x=l$, $u_0 e^{i\omega t} = \frac{1}{\rho_0 c}(I e^{-ikl} - R e^{ikl}) e^{i\omega t}$

$$= \frac{1}{\rho_0 c}(e^{-ikl} + e^{ikl}) e^{i\omega t}$$

$$= \frac{2I}{\rho_0 c}\cos(kl)\, e^{i\omega t}$$

$$\Rightarrow \quad I = -R = \frac{\rho_0 c u_0}{2\cos(kl)}$$

At $x=0$, $u_e = \frac{I-R}{\rho_0 c} e^{i\omega t} = \frac{u_0 e^{i\omega t}}{\cos(kl)}$

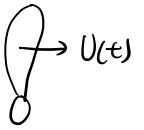
The far-field pressure

$$p'(x,t) = \frac{\rho_0 \pi a^2 i u_e (t - |x|/c)}{4\pi |x|} = \frac{\rho_0 \pi a^2 i\omega u_0 e^{i(\omega t - k|x|)}}{4\pi |x| \cos(kl)}$$

The duct enhances the sound field by a factor $[\cos(kl)]^{-1}$ compared to the piston placed at $x=0$

6.7.3 (Case 7) A moving rigid compact body

Suppose the body moving at $U(t)$



$\Rightarrow$ velocity normal to surface $U(t)\cdot n$

FWH equation

$$p'(x,t) = \frac{1}{4\pi|x|}\frac{\partial}{\partial t}\int_S \rho_0\, n\cdot U\left(t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right) dS + \frac{x_i}{4\pi|x|^2 c}\frac{\partial}{\partial t}\int_S n_i p\left(y, t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right) dS$$

For a compact body, Taylor expansion $\rightarrow$

$$U\left(t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right) = U\left(t-\frac{|x|}{c}\right) + \frac{x\cdot y}{|x|c}\dot{U}\left(t-\frac{|x|}{c}\right)+\cdots$$

$$= U(x_0) + x\dot{U}(x_0)+\cdots$$

Gauss (or divergence) theorem $\int_S n\cdot v\, dS = \int_V \nabla\cdot v\, dV = \int_V \frac{\partial v_i}{\partial y_i}\, dV$

$$\int_S n\cdot U\left(t-\frac{|x|}{c}+\frac{x\cdot y}{|x|c}\right) dS$$

$$= \int_S n\cdot U\left(t-\frac{|x|}{c}\right) dS + \int_S \frac{x\cdot y}{|x|c}\, n\cdot \dot{U}\left(t-\frac{|x|}{c}\right) dS$$

$$= \int_V \frac{\partial}{\partial y_i} U_i\left(t-\frac{|x|}{c}\right) d^3y + \int_V \frac{\partial}{\partial y_i}\left[\frac{x\cdot y}{|x|c}\dot{U}_i\left(t-\frac{|x|}{c}\right)\right] d^3y$$

($\int_V \frac{\partial}{\partial y_i} U_i\left(t-\frac{|x|}{c}\right) d^3y = 0$)

$\downarrow$ $\partial(x\cdot y)/\partial y_i = x_i$

$$= \int_V \frac{x_i}{|x|c}\dot{U}_i\left(t-\frac{|x|}{c}\right) d^3y$$

$$= \frac{x_i}{|x|c}\dot{U}_i\left(t-\frac{|x|}{c}\right) V$$

Substitude back to ..., $\rightarrow$ the sound field

$$p'(x,t) = \frac{\rho_0}{4\pi|x|}\frac{x\cdot\ddot{U}(t-|x|/c)\,V}{|x|c} + \frac{x\cdot\dot{F}(t-|x|/c)}{4\pi|x|^2 c}$$

$$= \frac{x \cdot \left[\rho_0 V \ddot{U}(t - |x|/c) + \dot{F}(t - |x|/c) \right]}{4\pi |x|^2 c}$$

$$\downarrow \quad \frac{\partial}{\partial x_i} = -\frac{1}{c} \frac{x_i}{|x|} \frac{\partial}{\partial t}$$

$$= \frac{-\nabla \cdot \left[\rho_0 V \dot{U}(t - |x|/c) + F(t - |x|/c) \right]}{4\pi |x|}$$

- a moving rigid compact body ⟶ dipole sound field

Dipole strength = $F(t) + \rho_0 V \dot{U}(t)$

- $F(t)$: force exerted on fluid by body
- $\rho_0 V \dot{U}(t)$: rate of dissipated momentum

- primarily due to the unsteady force the body exerts on fluid
  + additional term due to the inertia of displaced fluid.

- Example 6.3 → the oscillating compact sphere acoustically point dipole of strength $F(t) + \rho_0 V \dot{U}(t)$

⇒ True for the unsteady motion of any compact rigid body

- Intense forces of short duration are very noisy

Note: If the only force on the body is due to the fluid

Force on body $= -F(t) = m\dot{U}(t)$

Net dipole strength $= (-m + \rho_0 V)\dot{U}(t)$

↓ mass of the body

- Displaced mass term is negligible if $m \gg \rho_0 V$

In Sec. 6.7.1, for wires or struts $m \gg \rho_0 V$

→ Result of compact stationary bodies ✓

if the cylinder or strut vibrates in response to the unsteady force.

- If the body is neutrally buoyant (fish or submarine in water)

$m = \rho_0 V$ → no net dipole 浮力 = 重力

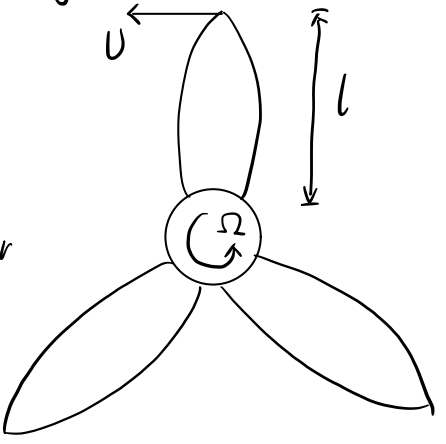
(Case 7)

Example 5. Propeller noise

Consider a propeller or fan with n blades of length l, angular velocity Ω

the speed $U = \Omega l$

sound of frequency $\omega = n\Omega$ 要乘上葉片數



compactness ratio $= \frac{n\Omega l}{c} = nM \ll 1$ for compact propeller

Helmholtz number $= \frac{\omega}{c} l$

↓

top Mach number

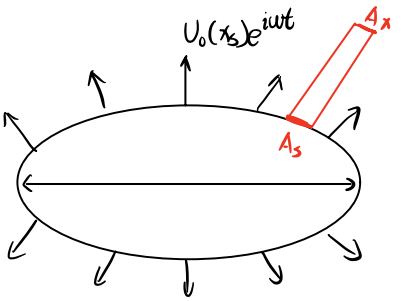
$\bar{F} = \bar{f} l \sim (C_D \rho_0 U^2 l) l$

Unsteady force $F \sim C_D \rho_0 U^2 l^2$

Far-field sound $p' \sim \frac{C_D}{|x| c} \left(\frac{nU}{l}\right) \rho_0 U^2 l^2 = C_D n \rho_0 \frac{U^3}{c} \frac{l}{r}$

- For fixed force, $l \uparrow$ and $U \downarrow$ ⇒ noise ↓
- $\frac{\partial F}{\partial t} \uparrow$ if blades cuts through the wakes of fixed struts or an upstream blade row ⇒ radiated $p' \uparrow$

6.7.4 Ray theory for high-frequency sound



Example 6 : A vibrating impentrable convex surface

If the surface vibrates in the form $n \cdot U = U_0(x_s) e^{i\omega t}$

- $U_0(x_s)$ varies gradually over the surface
- $kl >> 1$ ⟶ large surface compared to wavelength
- Ray theory ⟶ Sound travels as if in straight lines ⊥ Surface

  & $p' = \rho_0 c u$ as in plane wave

On the surface $p'(x_s, t) = \rho_0 c U_0(x_s) e^{i\omega t}$

Recall in a ray $\frac{\overline{p'^2} A}{\rho_0 c}$ = constant

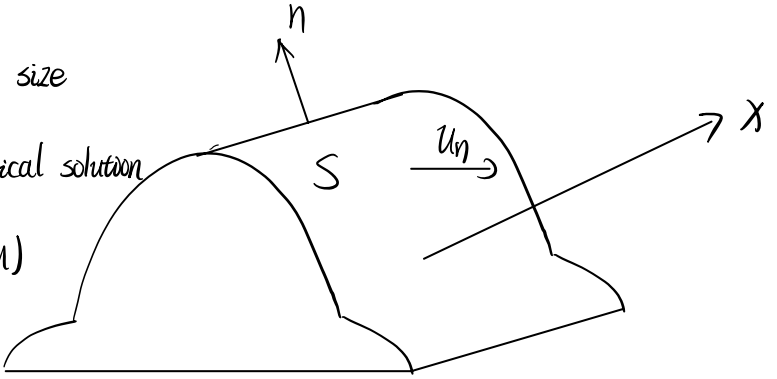
联立得

At a position x away from the surface

$$p'(x,t) = \sqrt{\frac{A_s}{A_x}}\, \rho_0 c\, U_0(x_s)\, e^{i(\omega t - kd)} \quad \text{as } \omega \to \infty$$

- e^{-ikd} : the time taken for the sound from the surface to x
- Area ratio A_x/A_s : geometrical spread of a ray tube
- $\sqrt{A_s/A_x}$: energy flow down each tube is conserved.

6.7.5 A vibrating body of arbitray size



For complicated geometries ⟶ numerical solution

eg. Boundary Element Method (BEM)

Suppose on the surface

$$u \cdot n = \hat{u}_n e^{i\omega t}$$

For a general position x, FWH equation ⇒

$$4\pi H(x) \hat{p}(x) = \rho_0 i\omega \int_S \frac{\hat{u}_n e^{-ik|x-y|}}{|x-y|} dS - \int_S \hat{p}(y) n_i \frac{\partial}{\partial x_i} \frac{e^{-ik|x-y|}}{|x-y|} dS$$

where Heaviside function $H(x) = \begin{cases} 1 & \text{with fluid} \\ 1/2 & \text{on } S \\ 0 & \text{within } S \end{cases}$

For x on surface

$$\Rightarrow \quad 2\pi \hat{p}(x) = \rho_0 i\omega \int_S \frac{u_n e^{-ik|x-y|}}{|x-y|} dS - \int_S \hat{p}(y) \frac{\partial}{\partial x_i} \left(\frac{e^{-ik|x-y|}}{|x-y|} \right) dS$$

An integral equation to be solved numerically

(i) Consider surface devided into N elements, each sufficiently small for $\hat{p}_j$ and $\hat{u}_{nj}$, $j = 1, \cdots, N$

(ii) Discretization $\longrightarrow N$ equations for N unknowns $\hat{p}_j$ in terms of $\hat{u}_{nj}$

(iii) Solve the matrix equation for $\hat{p}_j$

(iv) Far-field sound $\hat{p}(x)$ $\longleftarrow$ discretized form of the integral in Eq. 6.66

Summary of Sec. 6.6 and 6.7

The far field sound due to "bodies" of typical size D in a fluid of low Mach number with typical velocity U. The frequency of the flutuation in flow will scale on U/D. In a low Mach number flow, $kD \sim M \ll 1$ and the body is compact.

(i) When there is a net monopole (e.g. the 'body' change in size)

$$p'(x,t) = \frac{\dot{Q}(t - |x|/c)}{4\pi |x|} \quad , \text{where } Q(t) = \rho_0 \int_S n \cdot u \, dS$$

$$\dot{Q} = \rho_0 \ddot{V} \sim \rho_0 \omega^2 D^3 \sim \rho_0 U^2 D$$

$$\Rightarrow \quad \overline{\rho'^2} = \frac{\overline{p'^2}}{c^4} \sim \rho_0^2 M^4 \frac{D^2}{|x|^2}$$

These are efficient noise sources and are to be avoided if a quiet flow is required.

(ii) When a stationary / moving solid body exerts a force on the fluid the sound source is dipole

$$p'(x,t) = \frac{x \cdot [\dot{F}(t-|x|/c) + \rho_0 V \ddot{U}(t-|x|/c)]}{4\pi |x|^2 c}$$

$$|\dot{F}| \sim \rho_0 \omega U^2 D^2 \sim \rho_0 U^3 D \quad \Rightarrow \quad \overline{\rho'^2} = \frac{\overline{p'^2}}{c^4} \sim \rho_0 M^6 \frac{D^2}{|x|^2}$$

a factor M^2 smaller than the sound due to the monopole, but a factor M^2 larger than the sound due an unsteady flow in the absence of any bodies (i.e. jet noise) as described by Lighthill's 8th-power law

7. Sound generated near Surfaces of Discontinuity

7.1 Reciprocity of source and field

Simplication of harmonic time dependence $\frac{\partial}{\partial t} = i\omega$

The wave operator

$$\left(\frac{\partial^2}{\partial t} - c^2\nabla^2\right)\phi = (-\omega^2 - c^2\nabla^2)\phi \longleftarrow \frac{1}{c^2}$$

Time harmonic sound waves conform with the Helmholtz equation

$$k^2\phi + \nabla^2\phi = 0$$

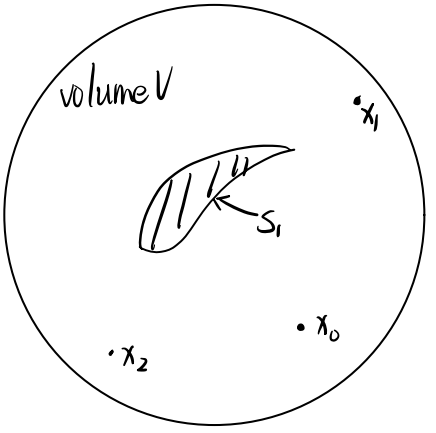
↓ non-zero if there is a source

Consider $\phi_1 e^{i\omega t}$ by a point monopole at x_0 and a point quadrupole at x_1

ϕ_1 is a solution of the inhomogeneous Helmholtz equation

$$k^2\phi_1 + \nabla^2\phi_1 = Q\delta(x - x_0) + T_{ij}\frac{\partial^2\delta(x - x_1)}{\partial x_i \partial x_j}$$

monopole strength　　quadrupole strength



Consider a second field $\phi_2 e^{i\omega t}$ by a unit strength point monopole at x_2

$$k^2\phi_2 + \nabla^2\phi_2 = \delta(x - x_2)$$

The two fields $\phi_1 e^{i\omega t}$ and $\phi_2 e^{i\omega t}$ in a space of volume V bounded by a surface S, S is made up of two distinct sections

① a surface at infinity S_∞ ⟶ outgoing waves with $\phi \propto \partial\phi/\partial n$

∵ $\phi(r - ct)$: harmonic in time → harmonic in r

② a surface S_1 → B.C.s to be specified

$\phi_2 \times \square - \phi_1 \times \square$

$$\Rightarrow \quad \phi_2 \nabla^2 \phi_1 - \phi_1 \nabla^2 \phi_2 = \phi_2 Q \delta(x - x_0) + \phi_2 T_{ij} \frac{\partial^2 \delta(x - x_1)}{\partial x_i \partial x_j} - \phi_1 \delta(x - x_2)$$

$$\downarrow$$

$$\frac{\partial}{\partial x_i}\left(\phi_2 \frac{\partial \phi_1}{\partial x_i} - \phi \frac{\partial \phi_2}{\partial x_i}\right)$$

Integrate over volume V bounded by S_1 and S_∞ + divergence theorem

$$\int_{S_1 + S_\infty} \left(\phi_2 \frac{\partial \phi_1}{\partial n} - \phi_1 \frac{\partial \phi_2}{\partial n}\right) dS$$

$$= \int_V \left[\phi_2 Q \delta(x - x_0) + \phi_2 T_{ij} \frac{\partial^2 \delta(x - x_0)}{\partial x_i \partial x_j} - \phi_1 \delta(x - x_2)\right] dV$$

$$= Q \phi_2(x_0) + T_{ij} \frac{\partial^2 \phi_2(x_1)}{\partial x_i \partial x_j} - \phi_1(x_2)$$

The L.H.S. $= 0$ in all cases we consider

- The integrand on $S_\infty = 0$ $\longleftarrow$ $\phi \propto \partial\phi/\partial n$ in far radiational field (plane waves)

  i.e. $p' = \rho_0 c v_n = -i\omega \rho_0 \phi = \rho_0 c \frac{\partial \phi}{\partial n}$

- The integrand on $S_1 = 0$ if $\phi \propto \partial\phi/\partial n$ for a surface impedance condition

  Or $\partial\phi/\partial n = 0$ for a rigid surface

  $\phi = 0$ for a free surface of water

$\Rightarrow$ The reciprocal theorem

$$\phi_1(x_2) = Q \phi_2(x_0) + T_{ij} \frac{\partial^2 \phi_2(x_1)}{\partial x_i \partial x_j}$$

- The field at x_2 due to a point monopole at x_0 + a point quadrupole at x_1

= the field at x_0 + double divergence of the field at x_1 due to a monopole at x_2

- In the case $T_{ij} = 0$ $\Rightarrow$

  The field at x_2 due to a source at x_0 $=$ the field at x_0 due to a source at x_2

- $Q = 0$

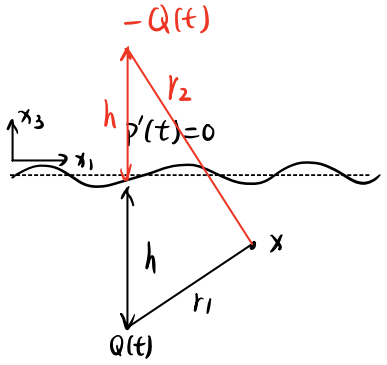
The field at x_2 due to a quadrupole at x_1 $=$ the double gradient of the field at x_1 due to a source at x_2

- The field near a wave scattering body due to a distant source $>>$ much simpler

  the distant field due to a complex source near the body

7.2 Sound sources near a plane surface of discontinuity

7.2.1 A source beneath a plane free surface



Statement of problem

We want to find $p'(x,t)$ in $x_3 < 0$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) p'(x,t) = Q(t)\,\delta(x_1)\delta(x_2)\,\delta(x_3+h)$$

with B.C. $p'(x,t) = 0$ on the surface $S: x_3 = 0$

↓ free surface or pressure release surface

For example, a limp, light surface $\mathcal{Z}=0$ and $\mathcal{R} = -I$

B.C. $\Rightarrow$ Source is asymmetric about surface S, i.e. 反对称

$$Q(x_1, x_2, x_3) = -Q(x_1, x_2, -x_3)$$

$\longrightarrow$ asymmetric sound field i.e. $p'(x_1, x_2, x_3) = -p'(x_1, x_2, -x_3)$

$\longrightarrow$ $p'(x_1, x_2, 0) = 0$ on the plane of asymmetry $x_3 = 0$

$\therefore$ free surface $=$ image system of sources with negative image strength

$$\Longrightarrow \quad p'(x,t) = \frac{Q(t - r_1/c)}{4\pi r_1} - \frac{Q(t - r_2/c)}{4\pi r_2}$$

Prove by checking: required condition satisfied ?

(1) On $x_3 = 0$, $r_1 = r_2 \Rightarrow p'(x,t) = 0 \Rightarrow$ B.C. satisfied

(2) Monopole solution in Eq. (6.13) $\longrightarrow$ the solution satisfied

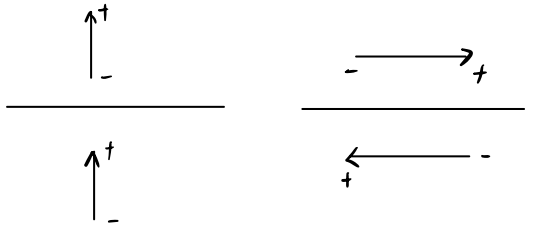
$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) p'(x,t) = Q(t)\,\delta(x_1)\delta(x_2)\delta(x_3+h) - Q(t)\,\delta(x_1)\delta(x_2)\,\delta(x_3-h)$$

$= Q(t)\,\delta(x_1)\,\delta(x_2)\,\delta(x_3+h)$ in $x_3<0$

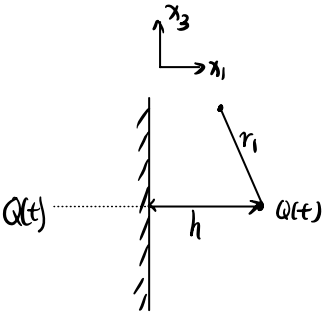
The solution satisfies the required homogeneous wave equation in the fluid $x_3<0$

Note:

The images of high order multipoles ← images of their constituent monopoles



7.2.2 A rigid plane surface



B.C.: $u_1 = 0$ or $\frac{\partial p'}{\partial x_1} = 0$ on the surface $x_1 = 0$

⇒ Image source (Symmetric) about the surface 对称

$$Q(x_1, x_2, x_3) = Q(-x_1, x_2, x_3)$$

⇒ Symmetric sound field, i.e. $p'(x_1, x_2, x_3) = p'(-x_1, x_2, x_3)$

⇒ $$\left.\frac{\partial p'}{\partial x_1}\right|_{x_1=0} = \frac{\partial p'}{\partial x_1}(x_1=0, x_2, x_3) - \frac{\partial p'}{\partial x_1}(-x_1=0, x_2, x_3)$$

⇒ rigid surface = image system of sources

The solution is $$p'(x,t) = \frac{Q(t-r_1/c)}{4\pi r_1} + \frac{Q(t-r_2/c)}{4\pi r_2}$$

Note:

(i) If $h=0$, $p'(x,t) = \frac{Q(t-r/c)}{2\pi r}$

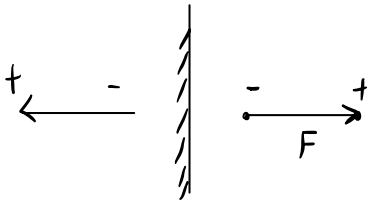
⟶ The adjacent wall doubles p' by the same source in unbounded space

intensity I ↑ by a factor of 4, S ↓ by 1/2

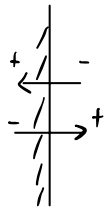
⇓

power output is only doubled

(ii)

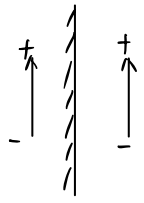


if $h = 0$



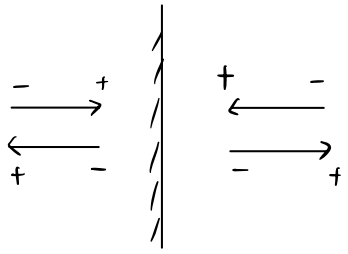
cancel $\Rightarrow$ no sound

(iii) $h = 0$

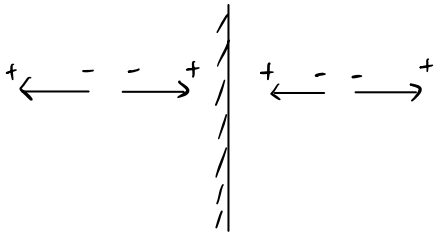


$\Rightarrow$ double

(iv) The field quadrupole sources near a hard surface



$\Rightarrow$ cancel



$\Rightarrow$ double

Recall in ch.6

- Jet noise is a quadrupole source
- In a low Mach number flow, far-field p' ↑ by M^{-1} when a rigid compact body is placed in the jet
- If a very large plane surface is placed near the jet.

  $\longrightarrow$ p' ↑ by at most a factor of 2

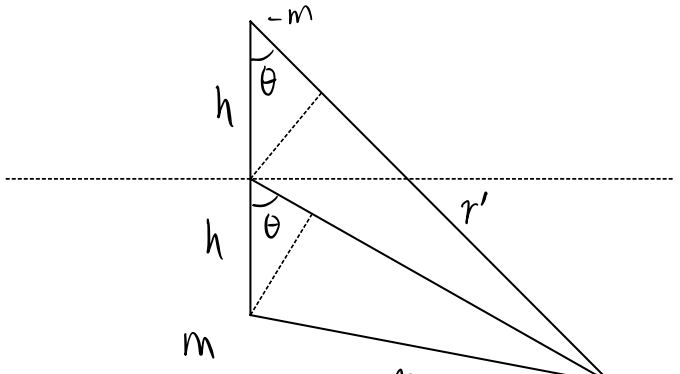
Example 7.1 A very small spherically symmetric acoustic transmitter is operated at a radian frequency ω at depth h below the free ocean surface. Show that the radiated sound power is

given by

$$\bar{p} = \frac{(m\omega)^2 \rho_0}{4\pi c}\left[1 - \frac{\sin(2kh)}{2kh}\right]$$

where m^2 is the mean square of volume outflow from the source.

To what depth should such a transmitter be lowered in order to achieve 50% of its deep water radiation efficiency at a frequency of 100 Hz ?



Solution :

The image is a source of strength $-m$ at $x_3 = -h$

$$r = \left[x_1^2 + x_2^2 + (x_3 - h)^2\right]^{1/2}$$

$$\approx |x| - \frac{x_3 h}{|x|} = |x| - h\cos\theta$$

$$\Rightarrow r' = |x| + h\cos\theta$$

$$|\vec{x} - \vec{y}| \approx |\vec{x}| - \frac{\vec{x}\cdot\vec{y}}{|\vec{x}|}$$

$$\frac{x \cdot y}{|x|} = \begin{cases} h\cos\theta & \text{for } m \\ h\cos(\pi - \theta) & \text{for } -m \end{cases}$$

Volume outflow rate $\dot{V} = \sqrt{2}\, m \longleftarrow m^2 = \overline{\dot{V}^2} = \frac{1}{2}\dot{V}^2$

$$\Rightarrow Q = \rho_0 \dot{V} = \rho_0 \sqrt{2}\, \dot{m} = i\omega \rho_0 \sqrt{2}\, m$$

$$\Rightarrow p'(x,t) = \frac{i\omega \rho_0 \sqrt{2} m}{4\pi}\left[\frac{e^{i(\omega t - kr)}}{r} - \frac{e^{i(\omega t - kr')}}{r'}\right]$$

$$\approx \frac{i\omega \rho_0 \sqrt{2} m}{4\pi |x|} e^{i(\omega t - k|x|)}\left(e^{ikh\cos\theta} - e^{-ikh\cos\theta}\right)$$

$$= -\frac{\omega \rho_0 \sqrt{2} m}{2\pi |x|} e^{i(\omega t - k|x|)} \sin(kh\cos\theta)$$

Radiated sound power $\longleftarrow$ energy flux through a large hemisphere centred on origin

$$\bar{P} = \int_0^{\frac{\pi}{2}} \frac{\overline{p'^2}}{\rho_0 c} \cdot 2\pi R \sin\theta\, R\, d\theta$$

$$= \left(\frac{\omega \rho_0 m}{2\pi}\right)^2 \frac{1}{\rho_0 c} \int_0^{\frac{\pi}{2}} 2\pi \sin\theta \sin^2(kh\cos\theta)\, d\theta$$

$$= \frac{\rho_0 \omega^2 m^2}{4\pi c} \int_0^{\frac{\pi}{2}} \left[1 - \cos(2kh\cos\theta)\right] \sin\theta \, d\theta$$

$$= \frac{\rho_0 \omega^2 m^2}{4\pi c} \left[-\cos\theta + \frac{\sin(2kh\cos\theta)}{2kh}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{\rho_0 \omega^2 m^2}{4\pi c} \left[1 - \frac{\sin(2kh)}{2kh}\right]$$

In deep water $h \to \infty$, $\bar{P} = \frac{\rho_0 \omega^2 m^2}{4\pi c}$

$\bar{P}$ has half of this value when $1 - \frac{\sin(2kh)}{2kh} = \frac{1}{2}$

$\Rightarrow$ Numerical solution of $\sin x = x/2 \Rightarrow x \sim 1.9$

and hence $kh \approx 0.95$, $h \sim 2.2\,m$ for $100\,Hz$ and $c = 1450\,m/s$

Note:

emitted sound power ↓↓ when source is close to the surface ($kh \ll 1$)

$\longrightarrow$ the nearby image changes the impedance near the source

$\longrightarrow$ altering the work it does on the near

The same idea $\longrightarrow$ 'anti-sound' to inhibit sound generation by source

Example 7.2 Due to a misbalance in its components a compact machine exerts a fluctuating force $Fe^{i\omega t}$ on the surrounding air. If the sound is not reflected from any boundaries, determine the rms pressure that would be radiated to a point x in the far field. This machine is placed on a hard floor and the centre of action of the force can be considered to be a distance $L \ll c/\omega$ above the floor. Using the method of images, determine the rms pressure radiated to a point x in the far field.

(i) F in a direction parallel to the floor.

(ii) F is perpendicular to the floor.

Solution:

Suppose F acts in the 1-direction, then in the absence of boundaries

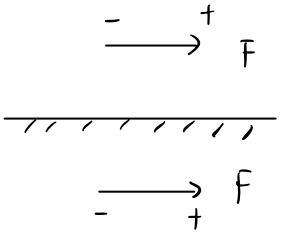
$$p'(x,t) = -\frac{\partial}{\partial x}\left[\frac{Fe^{i(\omega t - kr)}}{4\pi r}\right] = \frac{ik\cos\theta}{4\pi r} F e^{i(\omega t - kr)}$$

In the far field $r = |x| \Rightarrow p'_{rms} = \frac{k\cos\theta}{4\sqrt{2}\pi r} F$

(i) when F is parallel to the floor ⟶ the image dipole is parallel to F

In the far field, $p'(x,t) = 2 \times \frac{ik\cos\theta}{4\pi r} F e^{i(\omega t - kr)}$

$$\Rightarrow p'_{rms} = \frac{k\cos\theta}{2\sqrt{2}\pi r} F$$

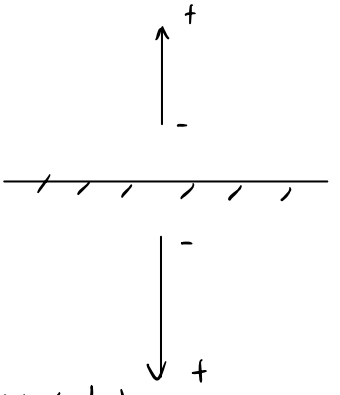


(ii) When F is perpendicular to the floor ⟶

the image dipole is equal & opposite to F ⟶

two opposite dipoles separated by a distance 2L

i.e. quadrupole



for small L and in the far field

$$p'(x,t) = 2L\frac{\partial^2}{\partial x_1^2}\left[\frac{Fe^{i(\omega t - kr)}}{4\pi r}\right] = -\frac{2Lk^2\cos^2\theta}{4\pi r} F e^{i(\omega t - kr)}$$

Taylor expansion as for dipole source $\Rightarrow p'_{rms} = \frac{Lk^2\cos^2\theta}{2\sqrt{2}\pi r} F$

7.3 Kichhoff's theorem for plane surfaces.

Kichhoff's theorem ⟶

- a source free linear acoustic medium bounded by surface
- sound ⟷ surface conditions

$$\frac{\partial^2}{\partial x_i \partial x_j}\int_{V_0}\left[\frac{T_{ij}}{r}\right]dV = -\frac{\partial}{\partial x_j}\int_S n_i\left[\frac{\rho u_i u_j + P_{ij}}{r}\right]dS + \frac{\partial}{\partial t}\int_S\left[\frac{\rho u\cdot n}{r}\right]dS$$

dipole monopole

provided x is never in V_0

Linear condition : $T_{ij} = \rho_0 v_i v_j \sim 0$, $P_{ij} = p'\delta_{ij}$, $\rho v_i = \rho_0 u_i$

$$\Rightarrow \quad 0 = -\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS + \rho_0\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS$$

provided x and y are never coincident.

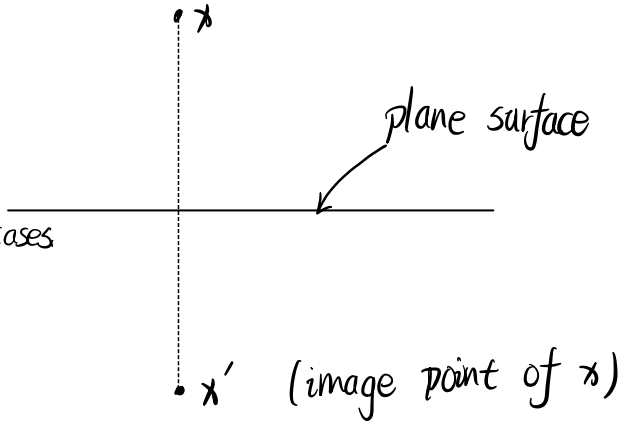
Under linearity constraints, Lighthill's acoustic analogy → a bounded region

$$\Rightarrow \quad \rho'(x,t) = -\frac{1}{4\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS + \frac{\rho_0}{4\pi c^2}\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS$$

Kirchoff's theorem

- Sound field at x ⟵ p', u at the fixed control surface S
- Acoustic medium is linear and source free.
- [ ] : retarded time $t - |x-y|/c$
- unit vector n → observation point x

Plane control surface



If x' @ the image point of x, for the two cases

- $t - |x-y|/c$ is the same
- n_i and $\frac{\partial}{\partial x_i}$ are opposite

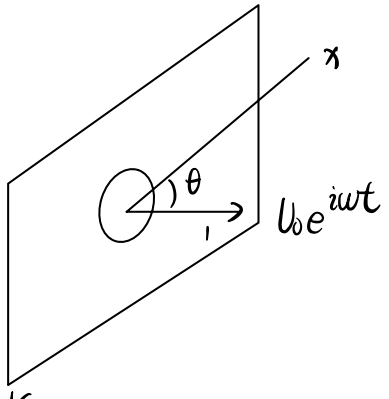
$$0 = -\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS + \rho_0\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS$$

Combine $p'(x,t) = -\frac{1}{4\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS + \frac{\rho_0}{4\pi c^2}\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS$

$$p'(x,t) = \frac{\rho_0}{2\pi c^2}\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS = -\frac{1}{2\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS$$

7.3.1 A baffled piston on a plane surface

Consider a circular piston of radius a



- flush mounted in an infinite hard wall

In free space

$$p'(x,t) = \frac{1}{4\pi}\frac{\partial}{\partial t}\int_{S_0}\left[\frac{\rho_0 u\cdot n}{r}\right]dS - \frac{1}{4\pi}\frac{\partial}{\partial x_i}\int_{S_0}\left[\frac{pn_i}{r}\right]dS \qquad , \quad r = |x-y|$$

monopole — dipole normal to surface

- S_0 : piston surface on $x_1 = 0$
- $[\ ]$: to be evaluated at time $t - r/c$
- Sources on an infinite surface

$\longrightarrow$ monopole doubled by image , dipole cancelled by image

$$\Rightarrow \quad p'(x,t) = \frac{1}{2\pi}\frac{\partial}{\partial t}\int_{S_0}\frac{\rho_0 U_0}{r}e^{i\omega(t-r/c)}dS$$

Alternatively, use $p'(x,t) = \frac{\rho}{2\pi c^2}\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS = -\frac{1}{2\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS$ for a plane control.

For large $|x|$, $r \approx |x| - x\cdot y/|x|$, this integral simplifies to

$$p'(x,t) = \frac{i\omega\rho_0 U_0}{2\pi|x|}e^{i(\omega t - k|x|)}\int_{S_0}e^{ikx\cdot y/|x|}dS$$

Polar coordinates

$$y_1 = 0 \quad , \quad y_2 = \sigma\cos\mu \quad , \quad y_3 = \sigma\sin\mu$$

$$x_1 = R\cos\theta \qquad x_2 = R\sin\theta\cos\phi \qquad x_3 = R\sin\theta\sin\phi$$

$$|x| = R \qquad x\cdot y = R\sin\theta\,\sigma\cos(\phi-\mu)$$

$$\Rightarrow \quad p'(x,t) = \frac{i\omega\rho_0 U_0}{2\pi R} e^{i(\omega t - kR)} \int_0^a \sigma\, d\sigma \int_0^{2\pi} e^{ik\sigma \sin\theta \cos(\phi-\mu)}\, d\mu$$

Now $\quad J_0(z) = J_0(-z) = \frac{1}{2\pi}\int_0^{2\pi} e^{iz\cos\theta}\, d\theta$

$$\Rightarrow \quad p'(x,t) = \frac{i\omega\rho_0 U_0}{R} e^{i(\omega t - kR)} \int_0^a \sigma J_0(k\sigma\sin\theta)\, d\sigma$$

Use $\quad z J_1(z) = \int_0^z z' J_0(z')\, dz'$

$$\Rightarrow \quad \int_0^a \sigma J_0(k\sigma\sin\theta)\, d\sigma = \int_0^{ka\sin\theta} \frac{z'}{ka\sin\theta} J_0(z') \frac{dz'}{ka\sin\theta}, \qquad (z' = k\sigma\sin\theta)$$

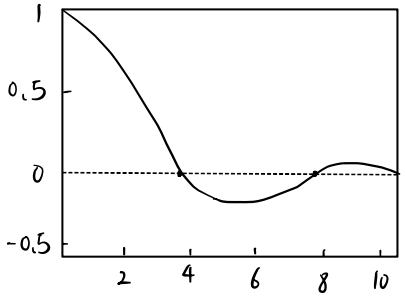
$$= \frac{1}{(ka\sin\theta)^2} ka\sin\theta\, J_1(ka\sin\theta)$$

$$= a^2 \frac{J_1(ka\sin\theta)}{ka\sin\theta}$$

$\Rightarrow$ The far field pressure

$$p'(x,t) = \frac{i\omega\rho_0 U_0 a^2}{2R} e^{i(\omega t - kR)} \frac{2J_1(ka\sin\theta)}{ka\sin\theta}$$

$\frac{2J_1(z)}{z}$ $\longleftarrow$ Tables of Bessel function



If $ka \ll 1$,

$$\frac{2J_1(ka\sin\theta)}{ka\sin\theta} \sim 1 \quad \text{for all angles } \theta$$

$$\Rightarrow \quad p'(x,t) = \frac{i\omega\rho_0 U_0 a^2}{2R} e^{i(\omega t - kR)} = 2\frac{\rho_0}{4\pi R}\dot{Q}(t - R/c)$$

where $\dot{Q}(t) = i\omega\pi a^2 U_0 e^{i\omega t}$ = rate of change of volume displaced by the vibrating piston

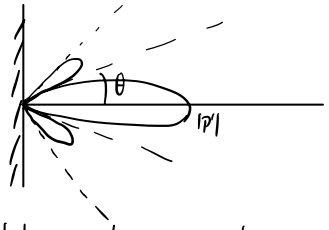
- In the compact limit, vibrating piston radiates sound omnidimensionally like a point source
- The strength of the field = 2× the same volume displacement that would generate in unbounded space
- Constructive interference with the image source ⟵ the rigid baffle surface.

If $ka \gg 1$ (non-compact) $\longrightarrow \bar{p}' = \bar{p}'(\theta)$

- On the axis through the piston centre, $\theta = 0$, $\frac{2J_1(ka\sin\theta)}{ka\sin\theta} = 1$

  $\bar{p}' = p'$ at the same radial distance from a compact piston
- $\bar{p}' = 0$ at an angle θ_1 when $ka\sin\theta_1 = 3.83$ $\longrightarrow$ major lobe $|\theta| < \theta_1$
- $\bar{p}' = 0$ at an angle θ_2 when $ka\sin\theta_2 = 7.02$

  $\longrightarrow$ second lope $\theta_1 < |\theta| < \theta_2$

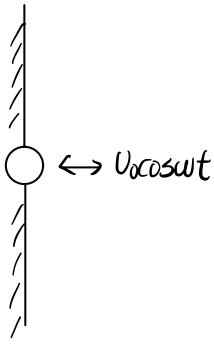


- If the piston is grossly non-compact $\longrightarrow$ a large number of side lobes with a similar peak intensity than the preceding lobe.

Example 7.3 A rigid compact sphere of radius a oscillates with speed $U_0\cos\omega t$ and small displacement amplitude, in a circular hole of the same radius in a large rigid screen.

Determine the acoustic power radiated into

What sound power would be radiated by the oscillating sphere in the unbounded space?



Solution:

This is a piston on a plane. For compact sphere, $|x-y| \approx |x|$

$$p'(x,t) = \frac{\rho_0 \pi a^2}{2\pi |x|} \frac{d}{dt}\left(U_0 \cos[\omega(t-|x|/c)]\right)$$

$$= -\frac{\rho_0 \pi a^2}{2\pi |x|} \omega U_0 \sin[\omega(t-|x|/c)]$$

Radiated sound power $= \frac{1}{2}\left(\frac{\rho_0 \pi a^2 \omega U_0}{2\pi |x|}\right)^2 \frac{1}{\rho_0 c} \times 2\pi R^2 = \frac{\pi}{4}\rho_0 c (ka)^2 a^2 U_0^2$

In unbound space, radiated sound power $= \frac{\pi}{6}\rho_0 c (ka)^4 a^2 U_0^2$

The plane increases the power output by a factor $\frac{3}{2}(ka)^{-2}$

$\longrightarrow$ the advantages of mounting a loudspeaker in a cabinet

- cancels the sound field of a normal dipole ?
- doubles the power output of a monopole ?

Source mechanism changed by the plane : dipole → monopole

Example 7.4 A square piston with sides of length a is mounted flush in an infinite hard wall and vibrates with a velocity $U_0 e^{i\omega t}$. Determine the far-field radiated pressure comment on the form of the radiation pattern in the limits of (i) $wa/c \ll 1$ (ii) $wa/c \gg 1$.

Solution :

For large $|x| = R$, $p'(x,t) = \frac{i\omega \rho_0 U_0}{2\pi R} e^{i(\omega t - kR)} \int_{-a/2}^{a/2} e^{ikx_2 y_2/R}\, dy_2 \int_{-a/2}^{a/2} e^{ikx_3 y_3/R}\, dy_3$

$$= \frac{i\omega \rho_0 U_0}{2\pi R} e^{i(\omega t - kR)} \left[\frac{e^{ikx_2 y_2/R}}{ikx_2/R}\right]\Bigg|_{-a/2}^{a/2} \left[\frac{e^{ikx_3 y_3/R}}{ikx_3/R}\right]\Bigg|_{-a/2}^{a/2}$$

$$= \frac{i\omega \rho_0 U_0}{2\pi R} e^{i(\omega t - kR)} \frac{2i \sin\left(\frac{kax_2}{2R}\right) 2i \sin\left(\frac{kax_3}{2R}\right)}{(ikx_2/R)\ (ikx_3/R)}$$

$$= \frac{i\omega \rho_0 U_0 a^2}{2\pi R} e^{i(\omega t - kR)} \frac{\sin\left(\frac{kax_2}{2R}\right) \sin\left(\frac{kax_3}{2R}\right)}{\frac{kax_2}{2R} \cdot \frac{kax_3}{2R}}$$

For $ka \ll 1$, $p'(x,t) \approx \frac{i\omega \rho_0 U_0 a^2}{2\pi R} e^{i(\omega t - kR)}$

an omni-directional pattern with source strength dependent on total area a^2

For $ka \gg 1$,

$$p'(x,t) = 0 \quad \text{when} \quad \sin\left(\frac{ka x_2}{2R}\right), \quad x_2 \neq 0$$

$$\sin\left(\frac{ka x_3}{2R}\right), \quad x_3 \neq 0$$

i.e. $\frac{x_2}{R} = \frac{2n\pi}{ka}$ or $\frac{x_3}{R} = \frac{2m\pi}{ka}$ with m and n non-zero integers

$\longrightarrow$ a pattern of lobes

7.4 Sound scattered by a spherial bubble in turbulence

Sound of turbulent water flows $\xleftarrow[\text{heard}]{}$ enerained gas bubbles.

Example: babbling break, domestic water flows

Reciprocal relations $\longrightarrow$ Simple description of the sound of a single spherical bubble

B.C. on the bubble surface: $\phi \times \partial\phi/\partial n \Rightarrow$

The reciprocal theorem is applicable

Taking one frequency component,

Lighthill's equation $\frac{\partial^2 \rho'}{\partial t^2} - c^2 \nabla^2 \rho' = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$

$$\Rightarrow \quad k^2 \rho' + \nabla^2 \rho' = -\frac{1}{c^2} \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

Sound induced by a quadrupole distribution

$\phi e^{i\omega t}$: the hypothetical field generated by a point source at x_0

$$k^2\phi + \nabla^2\phi = \delta(x - x_0)$$

The reciprocal theorem :

source point for the reciprocal problem = observation point for all real problem

$$\Rightarrow \quad \underbrace{\rho'(x_0)\, e^{i\omega t}}_{\text{real scattered field at } x_0} = -\frac{T_{ij}}{c^2} \underbrace{\frac{\partial^2 \phi(x)}{\partial x_i \partial x_j}}_{\text{reciprocal field at real quadrupole } x} e^{i\omega t}$$

Without loss of generality, position the bubble at ω-ordinate origin

$\Rightarrow$ the incident field in its vicinity due to the distant reciprocal source at x_0

The solution of $k^2\phi + \nabla^2\phi = \delta(x - x_0)$

$$\phi(x) e^{i\omega t} = -\frac{1}{4\pi |x_0|} e^{i\omega(t - |x_0|/c)}$$

For a bubble of radius a, the scattered field (see Eq.(6.23) in the textbook)

$$-\frac{a}{r} e^{i\omega(r-a)/c} \left[1 - \frac{\omega_0^2}{\omega^2}\left(1 + \frac{i\omega a}{c}\right)\right]^{-1} \times \underbrace{\text{incident field}}$$

r : distance from bubble centre

$\therefore$ The reciprocal field in the vicinity of the bubble

$$\phi(x) e^{i\omega t} = -\frac{1}{4\pi |x_0|} \left\{ \underbrace{e^{i\omega(t - |x_0| - x|/c)}}_{\text{incident field}} - \underbrace{\frac{a}{r} e^{-i\omega(r-a)/c} e^{i\omega(t-|x_0|/c)} \left[1 - \frac{\omega_0^2}{\omega^2}\left(1 + \frac{i\omega a}{c}\right)\right]^{-1}}_{\text{scattered field}} \right\}$$

where the spatial variation of the incident field over the bubble surface is ignored

In reciprocal theorem, $\partial\phi/\partial x_i$ includes two parts

(i) phase variation $\quad \dfrac{\partial \phi}{\partial x_i} \sim \dfrac{\omega}{rc}$

(ii) the factor $\frac{\partial \phi}{\partial x_i} \sim \frac{1}{r^2}$

2nd term > 1st term by a factor $\frac{1}{kr}$ $\longrightarrow$ vary large when $kr \ll 1$

Hence, the far sound field scattered by bubble

$$\rho'(x_0)\, e^{i\omega t} = -\frac{e^{i\omega(t-|x-x_0|/c)}}{4\pi |x_0| c^2} \frac{T_{ij}}{a^2} \frac{\partial^2}{\partial x_i \partial x_j}\left(\frac{a^3}{r}\right)\left[1-\frac{\omega_0^2}{\omega^2}\left(1+\frac{i\omega a}{c}\right)\right]^{-1}$$

$\left(r \sim a \longrightarrow e^{i\omega(r-a)/c} \approx 1 \right)$

Sound generated by turbulence in the absence of the bubble.

Use the first term in $\sim$ (incident field only)

$$\rho'(x_0)\, e^{i\omega t}\Big|_{\text{no bubble}} = -\frac{e^{i\omega(t-|x_0|/c)}}{4\pi |x_0| c^2} \frac{\omega^2}{c^2} T_{ij} \underbrace{\hat{x}_{0i}\hat{x}_{0j}}_{\downarrow}$$

unit vector in the direction of x_0

$\because \frac{\partial^2}{\partial x_i \partial x_j}\left(\frac{a^3}{r}\right) \sim O(1)$

$$\therefore \quad \frac{\rho'(x_0)\big|_{\text{bubble}}}{\rho'(x_0)\big|_{\text{no bubble}}} \sim (ka)^{-2}\left[1-\frac{\omega_0^2}{\omega^2}(1+ika)\right]^{-1}$$

very large $\rightarrow$ Bubble are very important in the acoustics of unsteady water flow

Sound intensity ↑ by a factor of order

$$(ka)^{-4}\left|1-\frac{\omega_0^2}{\omega^2}(1+ika)\right|^{-2}$$

i.e. $\left|(ka)^2-(k_0 a)^2(1+ika)\right|^{-2}$ where $k_0 = \omega_0/c$

- The typical Helmholtz number at bubble resonance $(k_0 a) = 0.014$

  $\Rightarrow$ at $\omega = \omega_0$, amplification factor of $(k_0 a)^{-6} \sim 10^{11}$ or 110 dB
- well below resonance, amplification $\sim (k_0 a)^{-4} \approx 70$ dB

Summary of Chapter 7

The reciprocal theorem

$$\phi_1(x_2) = Q\phi_2(x_0) + T_{ij}\frac{\partial^2\phi(x_1)}{\partial x_i \partial x_j}$$

- A monopole source is reflected in a plane free surface as an equal and opposite source.
- A monopole source is reflected in a plane rigid surface as an equal source.
- The reflection of dipoles and quadrupoles can be worked out by determining the images of their constituent monopoles.

kirchhoff's theorem

$$p'(x,t) = -\frac{1}{4\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS + \frac{\rho_0}{4\pi c^2}\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS$$

For a piston on a finite plane surface.

$$p'(x,t) = \frac{\rho_0}{2\pi}\frac{\partial}{\partial t}\int_S \frac{n\cdot u(y, t-|x-y|/c)}{|x-y|}\,dS$$

even when the piston is noncompact and be able to use this expression to calculate the far-field sound.

Bubble significantly amplify the sound of turbulent flow by scattering

The final

5 question

基本声源描述：脉动、振动

不连续界面：反射，折射

Ray theorem （声速变化，ray trace）

Sound source：scales analysis ☆

images